**061982 - Advanced Motorsport Engineering**
**Mechanical Engineering**
**Automotive and Motorsport Engineering, M.Sc.**

# Laboratories Summary


**Group:**
Cassano Marco
Meli Gianluca
Nardon Pietro


July 10, 2025

# Contents

# 1 Lab.00 - Basic equations

## 1.1 Ex.1

Considering a point mass vehicle with characteristics collected in (Tab. 1.1), the request was to find the maximum speed achievable by the vehicle when cornering with a radius of $R_1 = 150\,m$ and $R_2 = 30\,m$.

| | |
|---|---|
| $F_{down}$ | $2500\,N$ |
| $\mu$ | $0.75$ |
| $m$ | $500\,kg$ |

Table 1.1: Data Ex.1

To compute the velocity it was needed to scale $F_{down}$, as in (Tab. 1.1) it was computed at $v_{ref} = 50\,m/s$. Therefore, the constant part of the force was computed as:

$$\frac{1}{2}\rho c_z A = \frac{F_z}{v_{ref}^2} = 1\,\frac{Ns^2}{m^2} \tag{1.1}$$

Using the point mass equation while cornering and the definition of downforce, it is possible to derive (Eq. 1.2).

$$m\frac{v^2}{R} = \mu F_z = \mu(mg + \frac{1}{2}\rho c_z A v^2) \tag{1.2}$$

Then, rearranging, the velocity could be computed in (Eq. 1.3).

$$v = \sqrt{\frac{\mu m g}{\frac{m}{R} - \mu \cdot \frac{1}{2}\rho c_z A}} \tag{1.3}$$

The results are collected in (Tab. 1.3).

| | |
|---|---|
| $v(R_1)$ | $37.73\,m/s$ |
| $v(R_2)$ | $15.20\,m/s$ |

Table 1.2: Results Ex.1

When not considering the effect of downforce, at the denominator of (Eq. 1.3) only $m/R$ will remain, therefore the resulting velocities are:

| | |
|---|---|
| $v(R_1)$ | $33.22\,m/s$ |
| $v(R_2)$ | $14.86\,m/s$ |

Table 1.3: Results Ex.1

## 1.2 Ex.2

Aim of this exercise was to compute the minimum time in which a vehicle, represented by a point mass, could travel the trajectory shown in (Fig. 1.1). The driver would keep a constant velocity while cornering, then in the straights it would accelerate and brake as much as possible. Therefore the maximum acceleration achievable is limited by grip and is equal to $a_{max} = \mu g$, both when braking and accelerating.

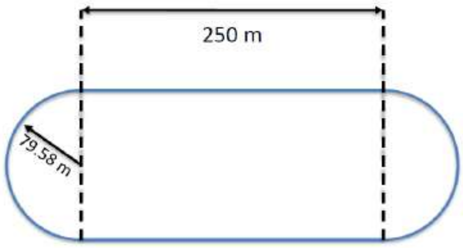



Figure 1.1: Trajectory

At first the vehicle's speed while cornering $v_c$ was computed starting from the cornering equation (Eq. 1.4), where $\mu = 1$, $R = 79.58\,m$.

$$m\frac{v_c^2}{R} = \mu m g \quad \longrightarrow \quad v_c = \sqrt{\mu g R} = 27.94\,m/s \tag{1.4}$$

As said before, since the driver has to accelerate and brake as much as possible, the acceleration of the vehicle is equal when accelerating and braking (apart from a minus sign), therefore the time and space needed by the driver to accelerate outside of the corner and decelerate to enter the next one is equal, since the speed at the corner exit and entry has to be equal. Therefore the driver would accelerate from the exit of the corner up to half of the straight and then it would brake starting from half of the straight up to beginning of the next corner.

The time needed can be computed using the uniformly accelerated motion equation, where $l = 250\,m$

$$\frac{1}{2}a_{max}t^2 + v_c t - \frac{l}{2} = 0 \tag{1.5}$$

Whose solution reads:

$$t_{half} = \frac{-v_c \pm \sqrt{v_c^2 + 4 \cdot \frac{1}{2}a_{max}\frac{l}{2}}}{2 \cdot \frac{1}{2}a_{max}} = 2.946\,s \tag{1.6}$$

The time needed to complete one corner is:

$$t_c = \frac{\pi R}{v_c} = 8.94\,s \tag{1.7}$$

Therefore, the total time needed to travel the trajectory is:

$$t_{tot} = 2 \cdot t_c + 4 \cdot t_{half} = 29.66\,s \tag{1.8}$$

## 1.3 Ex.3

For this third exercise, we were requested to find the rolling resistance and the drag area of a vehicle starting from a dataset of velocity vs time points related to a coast down maneuver.

In particular, the following equation (longitudinal forces equilibrium) has been exploited.

$$ma_x = F_{engine} - F_{drag} - F_{roll} - F_{grade}$$

Being $F_{engine} = 0$ and $F_{grade} = 0$, the equation can be written as

$$ma_x = -\frac{1}{2}\rho c_d S v^2 - f_{roll}(mg + \frac{1}{2}\rho c_z S v^2)$$

which is a second order polynomial in the variable $v$, where the unknowns are $f_{roll}$ and $c_d S$. In particular:

$$ma_x = -\frac{1}{2}\rho(c_d + c_z f_{roll})Sv^2 - f_{roll}mg$$

The values of $ma_x$ are derived from the dataset and through the `polyfit` Matlab function the polynomial coefficients whose best fit the dataset are found. The following results have been obtained:

$$p_2 = -\frac{1}{2}\rho(c_d + c_z f_{roll})S = -0.8698$$

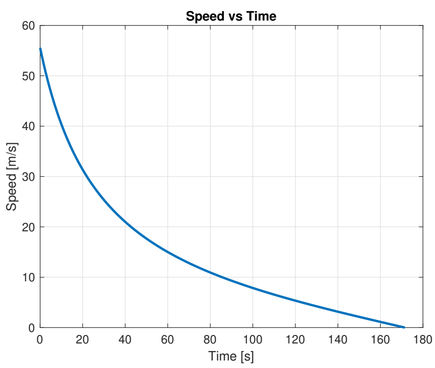
$$p_1 \simeq 0$$

$$p_0 = -f_{roll}mg = -135.3780$$

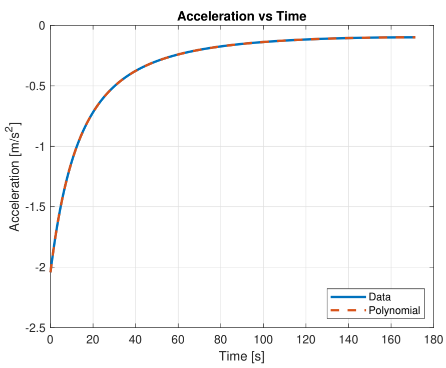
From which

$$f_{roll} = 0.01$$

$$c_d S = 1.4296\, m^2$$




(a) Provided dataset plot




(b) Dataset vs polynomial acceleration

Figure 1.2: Velocity and acceleration

## 1.4 Ex.4

The request of this exercise was to determine the vertical load $F_z$ acting on front and rear tires of a car. The car data are reported in (Tab. 1.4)

| | |
|---|---|
| $m$ | $1300\,kg$ |
| $Wheelbase$ | $2.65\,m$ |
| $v$ | $200\,km/h$ |
| $a_x$ | $-17\,m/s^2$ |

| | |
|---|---|
| $h_{COG}$ | $0.4\,m$ |
| $c_zA$ | $3.2\,m^2$ |
| $WD$ | $45\%$ |
| $AB$ | $52\%$ |

Table 1.4: Data Ex.4

At first the downforce was computed as:

$$F_{down} = \frac{1}{2}\rho c_z A v^2 = 5925.92\,N \tag{1.9}$$

Then, using the aerodynamic balance, it was possible to compute the downforce at the front and at the rear.

$$AB = \frac{F^F_{down}}{F_{down}} \quad \longrightarrow \quad \begin{cases} F^F_{down} = 3081.48\,N \\ F^R_{down} = 2844.44\,N \end{cases} \tag{1.10}$$

Using the weight distribution instead, it was possible to compute the static vertical loads.

$$WD = \frac{F^F_{z,static}}{F^{tot}_{z,static}} \quad \longrightarrow \quad \begin{cases} F^F_{z,static} = 5738.85\,N \\ F^R_{z,static} = 7014.15\,N \end{cases} \tag{1.11}$$

The load transfer due to deceleration was computed as:

$$\Delta F_z = \frac{m a_x h_{COG}}{Wheelbase} = -3335.85\,N \tag{1.12}$$

So the total vertical forces acting on front and rear tires can be computed as:

$$\begin{cases} F^F_{z,tot} = F^F_{z,static} + |\Delta F_z| + F^F_{down} = 12156.18\,N \\ F^R_{z,tot} = F^R_{z,static} - |\Delta F_z| + F^R_{down} = 6522.74\,N \end{cases} \tag{1.13}$$

## 1.5 Ex.5

Aim of the exercise is to compute, from a constant speed test, the following quantities:

1. Aerodynamic balance: $BAL$
2. Aerodynamic efficiency: $EFF$
3. $c_z S$
4. $c_d S$

The springs compression (zeroed on the static position) can be exploited to compute the aerodynamic downforce on each axle, which is the only component of the vertical force apart from the weight. Thus:

$$F^F_{down} = 2 \cdot \frac{k_{front} \Delta l_{front}}{R^F_m} = 2 \cdot \frac{350 \, \frac{N}{mm} \cdot 6.4 \, mm}{1.82} = 2461.5 \, N$$

$$F^R_{down} = 2 \cdot \frac{k_{rear} \Delta l_{rear}}{R^R_m} = 2 \cdot \frac{200 \, \frac{N}{mm} \cdot 11.5 \, mm}{1.3} = 3538.5 \, N$$

Which directly brings to the balance:

$$BAL = \frac{F^F_{down}}{F^F_{down} + F^R_{down}} = 0.4103$$

For the aerodynamic efficiency the longitudinal force equilibrium is exploited:

$$ma_x = F_{engine} - F_{drag} - F_{roll} - F_{grade} = 0$$

Where

$$F_{engine} = \frac{\tau T}{R_{wheel}} = \frac{4.45 \cdot 350 \, Nm}{0.345 \, m} = 4514.5 \, N$$

$$F_{roll} = f_{roll}(mg + F_{down}) = 0.01 \cdot (1380 \, kg \cdot 9,81 \, m/s^2 + 6000 \, N) = 195.38 \, N$$

$$F_{grade} = 0 \, N$$

Thus

$$F_{drag} = F_{engine} - F_{roll} = 4319.1 \, N$$

Which means that

$$EFF = \frac{F_{down}}{F_{drag}} = \frac{6000 \, N}{4319.1 \, N} = 1.3892$$

Finally

$$c_z S = \frac{2F_{down}}{\rho v^2} = \frac{2 \cdot 6000 \, N}{1.225 \, kg/m^3 \cdot (68.33 \, m/s)^2} = 2.0981 \, m^2$$

$$c_d S = \frac{2F_{drag}}{\rho v^2} = \frac{2 \cdot 4319.1 \, N}{1.225 \, kg/m^3 \cdot (68.33 \, m/s)^2} = 1.5103 \, m^2$$

## 2 Lab.01 - Tyres

Focus of this laboratory is to observe the effects of some different set-ups on the vehicle performance, with a particular focus on the relationship between the tyre behavior depending on its temperature evolution during 5 laps. Finally, an estimation of the tyre friction law, starting from a data set, is carried out.

### 2.1 Exercise 1

#### 2.1.1 Toe angle - Cold track

Considering a night race, in which the track, ambient and tyre temperatures are low, different front and rear toe angles combinations were tested to study their influence on performance.

The following data are provided.

| $T_{air}$ | $T_{track}$ | $t_{sim,end}$ | $T_{0,tyre}$ |
|---|---|---|---|
| $15°C$ | $20°C$ | $250\ s$ | $25°C$ |

Table 2.1: Exercise 1.1 Provided Data

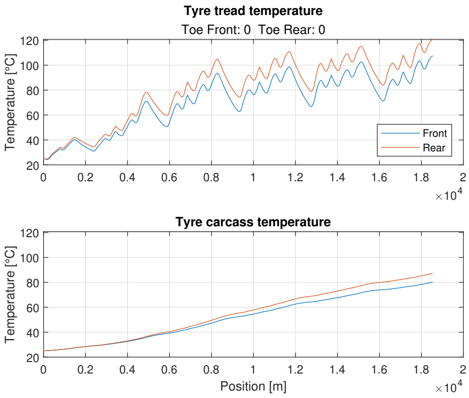
The following combinations are tested.

| $Toe_F$ **[deg]** | $Toe_R$ **[deg]** | **1° Lap t. [s]** | **Last Lap t. [s]** | **Avg Lap t. [s]** | $\mu_{avg,F}$ | $\mu_{avg,R}$ |
|---|---|---|---|---|---|---|
| 0 | 0 | 52.54 | 44.29 | 44.82 | 1.0648 | 1.0753 |
| 0.35 | 0 | 51.73 | 44.51 | 44.90 | 1.0704 | 1.0778 |
| 0.65 | 0 | 50.76 | 44.52 | 44.56 | 1.0809 | 1.0828 |
| 1 | 0 | 51.16 | 44.33 | 44.39 | 1.0836 | 1.0837 |
| 0 | 0.35 | 52.25 | 44.36 | 44.93 | 1.0673 | 1.0784 |
| 0.35 | 0.35 | 50.80 | 44.32 | 44.48 | 1.0803 | 1.0817 |
| 0.65 | 0.35 | 51.35 | 44.36 | 44.87 | 1.0819 | 1.0838 |
| 1 | 0.35 | 50.83 | 44.47 | 44.42 | 1.0853 | 1.0857 |
| 0 | 0.65 | 51.71 | 45.90 | 45.23 | 1.0692 | 1.0814 |
| 0.35 | 0.65 | 50.83 | 44.55 | 44.62 | 1.0806 | 1.0822 |
| 0.65 | 0.65 | 51.29 | 44.45 | 44.58 | 1.0810 | 1.0807 |
| 1 | 0.65 | 52.49 | 45.17 | 44.70 | 1.0839 | 1.0818 |
| 0 | 1 | 51.21 | 44.97 | 44.71 | 1.0779 | 1.0806 |
| 0.35 | 1 | 50.80 | 44.55 | 44.74 | 1.0797 | 1.0795 |
| 0.65 | 1 | 50.63 | 44.50 | 44.86 | 1.0841 | 1.0805 |
| 1 | 1 | 50.44 | 45.05 | 44.73 | 1.0867 | 1.0806 |

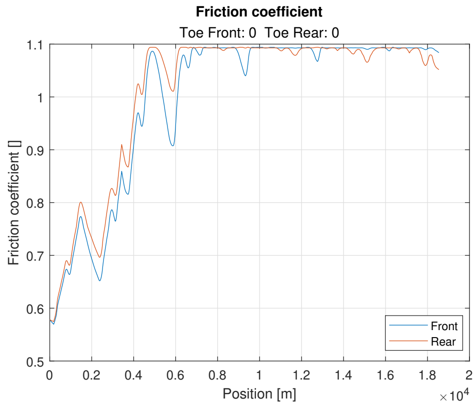
Table 2.2: Toe combinations cold track

The main getaway found by analyzing (Tab. 2.2) and the graphs is that the front tyres' temperature is particularly critical, as being the track very cold and the car being RWD is harder for the front tyres to heat up to the correct temperature, therefore a higher toe angle at the front will be required to increase it.

As a reference the neutral case with $Toe_F = 0$ and $Toe_R = 0$ is presented.

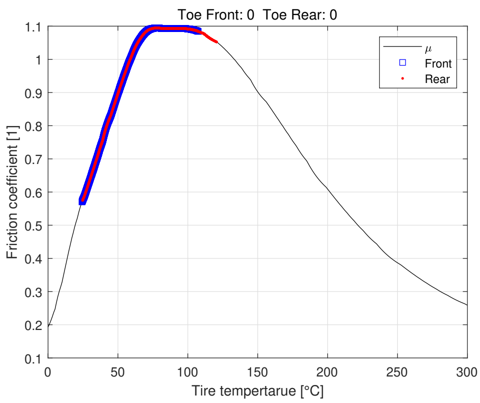



(a) Temperature vs Position

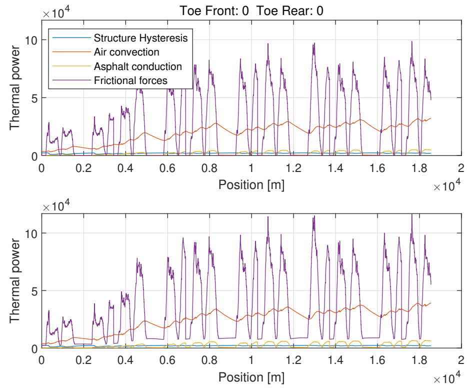



(b) Friction coefficient vs Position

Figure 2.1: $Toe_F = 0$ and $Toe_R = 0$




(a) Friction coefficient vs Temperature




(b) Power flows - Front and Rear

Figure 2.2: $Toe_F = 0$ and $Toe_R = 0$

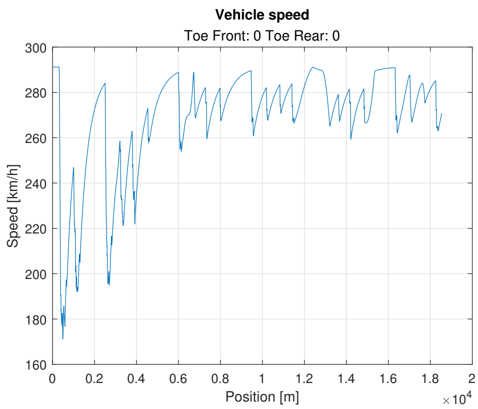
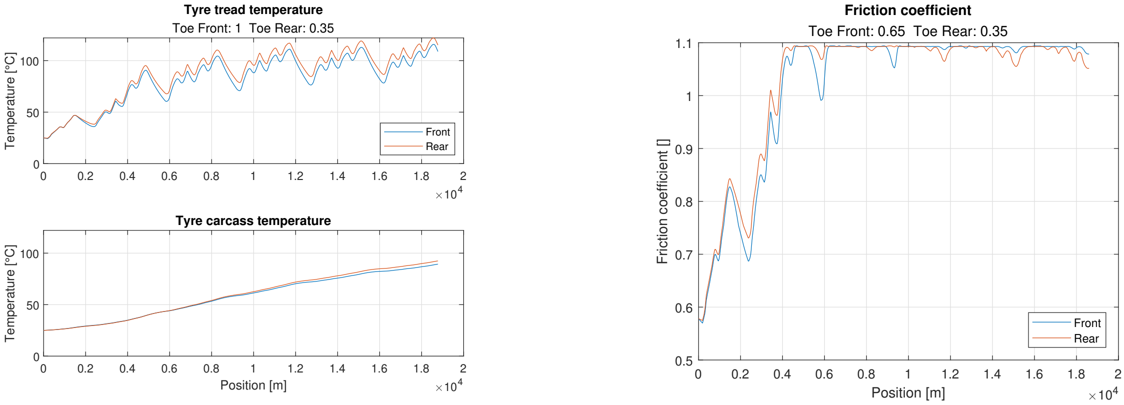



Figure 2.3: $Toe_F = 0$ and $Toe_R = 0$ - Speed vs Position

As described before the front tyres tend to be colder than the rear tyres, which in the end also overheat leading to a grip loss. In general grip for the first two laps is very low.

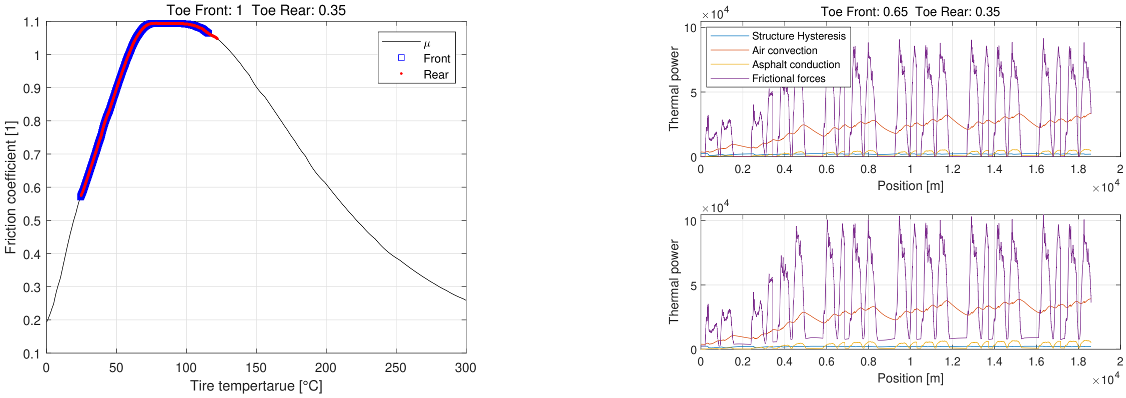
The first case analyzed is $Toe_F = 1$ and $Toe_R = 0.35$.




(a) Temperature vs Position

(b) Friction coefficient vs Position

Figure 2.4: $Toe_F = 1$ and $Toe_R = 0.35$




(a) Friction coefficient vs Temperature

(b) Power flows - Front and Rear

Figure 2.5: $Toe_F = 1$ and $Toe_R = 0.35$

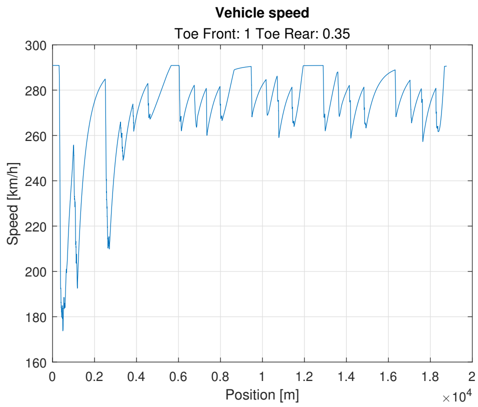
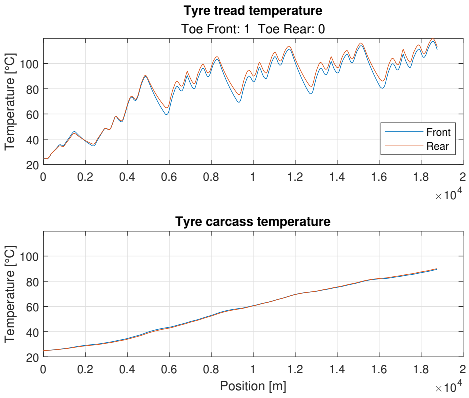


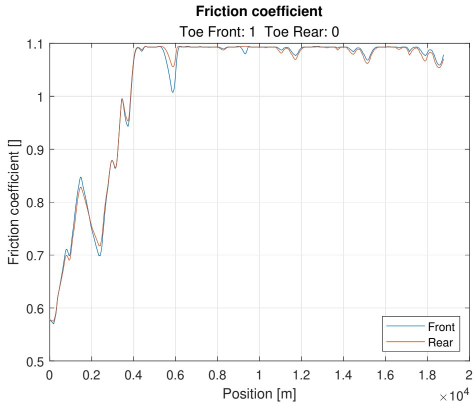

Figure 2.6: $Toe_F = 1$ and $Toe_R = 0.35$ - Speed vs Position

With this combination of toe angles the highest $\mu_{avg}$ are reached, meaning that the tyres work in the best temperature window for these few laps but, especially from the last lap, the rear tyres start to overheat and the rear grip starts to drop, this could be a problem in longer stints.

The second case analyzed is $Toe_F = 1$ and $Toe_R = 0$.

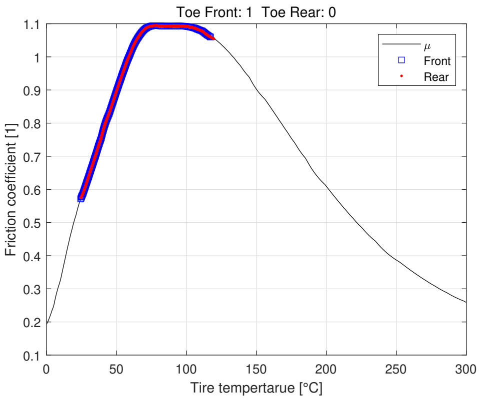



(a) Temperature vs Position

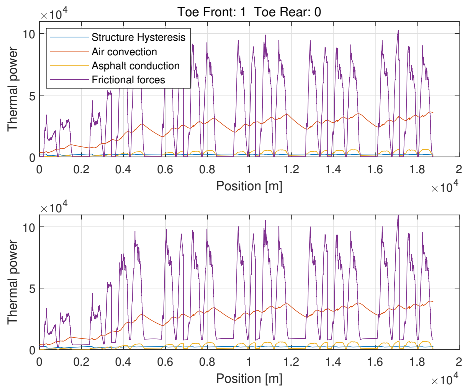



(b) Friction coefficient vs Position

Figure 2.7: $Toe_F = 1$ and $Toe_R = 0$




(a) Friction coefficient vs Temperature




(b) Power flows - Front and Rear

Figure 2.8: $Toe_F = 1$ and $Toe_R = 0$

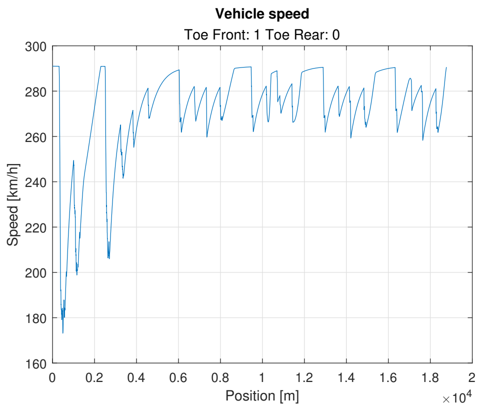



Figure 2.9: $Toe_F = 1$ and $Toe_R = 0$ - Speed vs Position

In this toe angles combination the best average lap time is achieved, both front and rear tyres experience the same temperature and grip variation, leading to a stable, and ultimately, predicable car along longer stints. The speed profile is much more uniform during all the laps analysed, unlike the previous case.

In general, the best toe angle combination is the one that allows for a fast heating up of the tyres, especially of the front tyres, getting them to work in their optimal temperature window. When both front and rear tyres experience the same trend in temperature and grip during the various lap, the average lap time is lowest. But if considering longer stints, a careful choice must be carried out to avoid overheating and grip losses in the long term.

### 2.1.2 Toe angle - Hot track

Let's now consider a hot summer day race, in which the track, ambient and tyre temperatures are higher with respect to the previous case. The same front and rear toe angles combinations are tested.

The following data are provided.

| $T_{air}$ | $T_{track}$ | $t_{sim,end}$ | $T_{0,tyre}$ |
|---|---|---|---|
| $30°C$ | $50°C$ | $250\ s$ | $70°C$ |

Table 2.3: Exercise 1.1 Provided Data

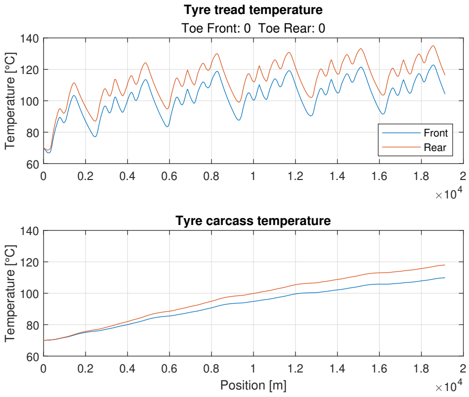
The following combinations are tested.

| $Toe_F$ **[deg]** | $Toe_R$ **[deg]** | **1° Lap t. [s]** | **Last Lap t. [s]** | **Avg Lap t. [s]** | $\mu_{avg,F}$ | $\mu_{avg,R}$ |
|---|---|---|---|---|---|---|
| 0 | 0 | 43.94 | 45.13 | 45.02 | 1.0817 | 1.0567 |
| 0.35 | 0 | 44.24 | 45.13 | 44.91 | 1.0770 | 1.0570 |
| 0.65 | 0 | 43.90 | 44.66 | 44.68 | 1.0722 | 1.0567 |
| 1 | 0 | 44.27 | 45.23 | 45.38 | 1.0668 | 1.0591 |
| 0 | 0.35 | 44.09 | 45.17 | 44.87 | 1.0826 | 1.0488 |
| 0.35 | 0.35 | 44.20 | 45.31 | 44.94 | 1.0789 | 1.0507 |
| 0.65 | 0.35 | 44.42 | 45.00 | 44.93 | 1.0730 | 1.0490 |
| 1 | 0.35 | 44.63 | 45.07 | 44.98 | 1.0680 | 1.0508 |
| 0 | 0.65 | 43.80 | 45.23 | 44.90 | 1.0839 | 1.0425 |
| 0.35 | 0.65 | 44.19 | 45.26 | 44.86 | 1.0784 | 1.0405 |
| 0.65 | 0.65 | 44.01 | 45.70 | 45.33 | 1.0753 | 1.0441 |
| 1 | 0.65 | 44.20 | 45.28 | 45.21 | 1.0678 | 1.0422 |
| 0 | 1 | 44.30 | 45.19 | 45.08 | 1.0840 | 1.0315 |
| 0.35 | 1 | 44.19 | 45.56 | 45.24 | 1.0796 | 1.0320 |
| 0.65 | 1 | 44.20 | 45.53 | 45.23 | 1.0749 | 1.0316 |
| 1 | 1 | 44.35 | 45.49 | 45.30 | 1.0697 | 1.0334 |

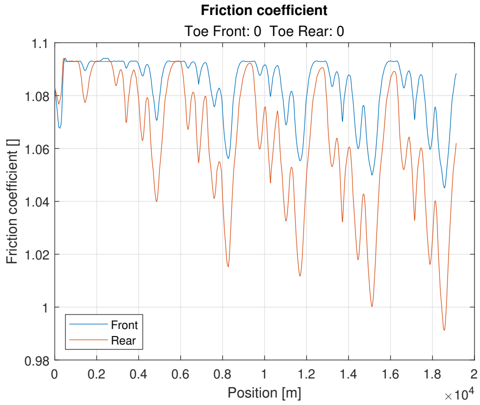
Table 2.4: Toe combinations hot track

Analyzing (Tab. 2.4), straightaway is clear that the higher initial tyre temperature $T_{0,tyre}$ has a great impact on the car's performance during the first lap. The tyre is already way closer to its optimal temperature, therefore has high grip available, resulting in a first lap that is usually even faster than the average.
Instead, the overall average lap time is slightly lower than the cold track case, this is due to an extreme overheating, when running on the hot track, especially of the rear tyre. This results in a drop in terms of friction coefficient, which worsens every lap, leading to a $\mu_{avg,R}$ on the hot track way worse than $\mu_{avg,R}$ on the cold track. Moreover, also the difference between $\mu_{avg,F}$ and $\mu_{avg,R}$ is much more visible on the hot track, creating an unhealthy grip difference, increasing unpredictability and worsening lap times.

As a reference the neutral case with $Toe_F = 0$ and $Toe_R = 0$ is presented.

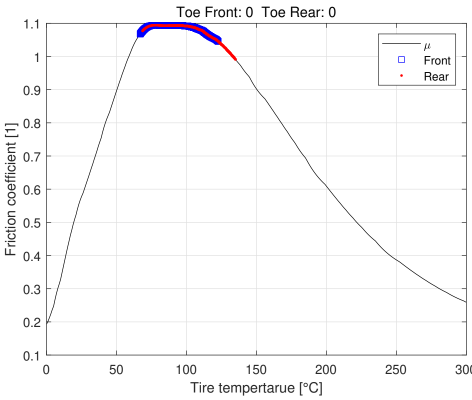



(a) Temperature vs Position

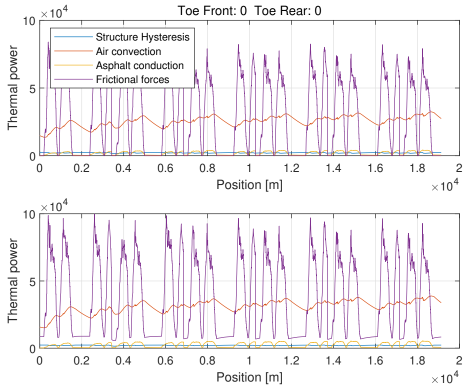



(b) Friction coefficient vs Position

Figure 2.10: $Toe_F = 0$ and $Toe_R = 0$




(a) Friction coefficient vs Temperature




(b) Power flows - Front and Rear

Figure 2.11: $Toe_F = 0$ and $Toe_R = 0$

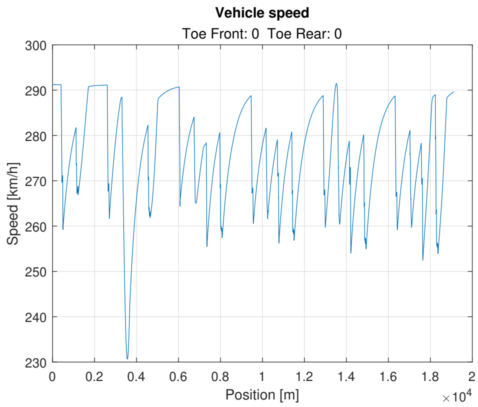
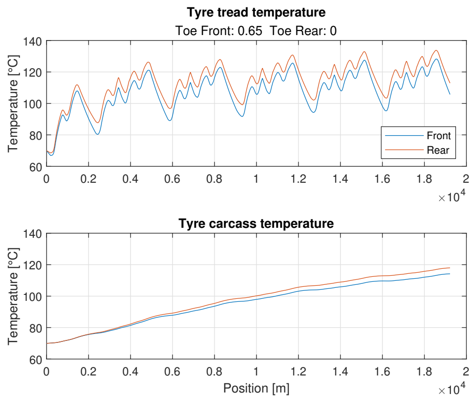


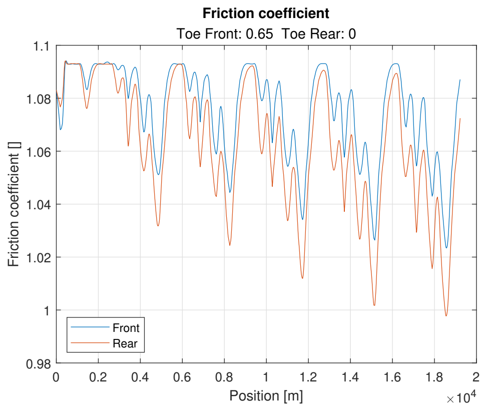

Figure 2.12: $Toe_F = 0$ and $Toe_R = 0$ - Speed vs Position

Even if in the first lap the grip is way higher than in the cold track, it drops more drastically, leading to a grip loss visible also in the velocity distribution, where it is shown that the driver is never able to fully exploit all the vehicle's power.

The best case analyzed is $Toe_F = 0.65$ and $Toe_R = 0$.

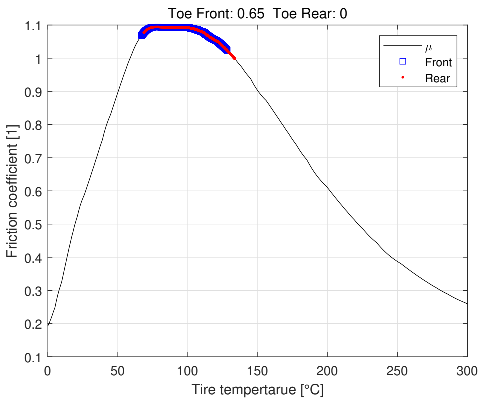



(a) Temperature vs Position

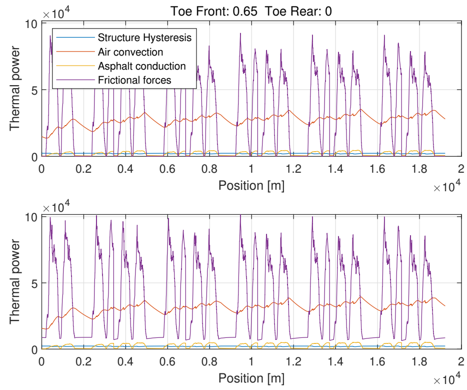



(b) Friction coefficient vs Position

Figure 2.13: $Toe_F = 0.65$ and $Toe_R = 0$




(a) Friction coefficient vs Temperature




(b) Power flows - Front and Rear

Figure 2.14: $Toe_F = 0.65$ and $Toe_R = 0$

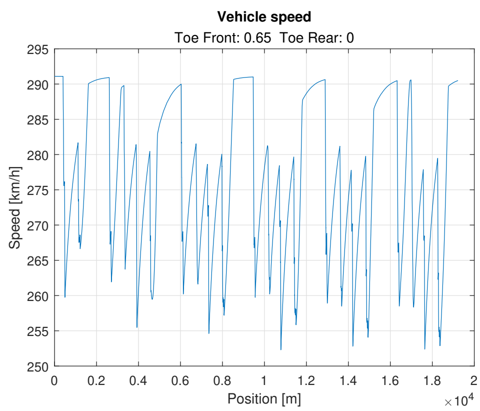



Figure 2.15: $Toe_F = 0.65$ and $Toe_R = 0$ - Speed vs Position

Using the same reasoning of the previous track, is possible to notice that the best average lap times are achieved when both front and rear tyres experience the same temperature and grip trend, leading to a stable car, able to perform well during the stint.

If just one setup has to be chosen, which optimizes both cold and hot track condition (for example during a 24h race), probably the best choice would be a tradeoff between the best cases of $Toe_F = 1$ and $Toe_R = 0$ for the cold track and $Toe_F = 0.65$ and $Toe_R = 0$. In this way the heating of the front tyres is easier and faster, without excessively overheating the rear tyres. This will allow for a stable car, without excessive grip drops, that will keep good lap times even in long stints.

A very small rear toe angle could be implemented to ease the heating up process when in cold track conditions and thus be beneficial to have a fast outlap.

### 2.1.3 Weight Distribution and Aerodynamic Balance - Hot track

Still considering the hot track conditions, let's now investigate the effects of some different values of weight distribution and aerodynamic balance, defined as front/total ratio. The following combinations have been tested.

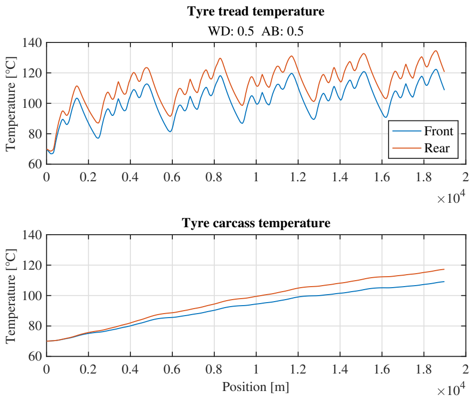
| **WD** | **AB** | **1° Lap t. [s]** | **Last Lap t. [s]** | **Avg Lap t. [s]** | $\mu_{avg,F}$ | $\mu_{avg,R}$ |
|---|---|---|---|---|---|---|
| 0.35 | 0.35 | 44.21 | 45.39 | 45.18 | 1.0930 | 1.0051 |
| 0.50 | 0.35 | 44.15 | 45.11 | 44.93 | 1.0574 | 1.0757 |
| 0.65 | 0.35 | 44.27 | 46.14 | 45.91 | 0.9587 | 1.0931 |
| 0.35 | 0.50 | 44.22 | 45.45 | 45.57 | 1.0910 | 0.9740 |
| 0.50 | 0.50 | 44.02 | 44.89 | 45.48 | 1.0827 | 1.0589 |
| 0.65 | 0.50 | 44.02 | 46.14 | 45.37 | 1.0039 | 1.0929 |
| 0.35 | 0.65 | 45.15 | 46.28 | 46.22 | 1.0782 | 0.9459 |
| 0.50 | 0.65 | 44.30 | 45.06 | 45.01 | 1.0915 | 1.0261 |
| 0.65 | 0.65 | 43.78 | 45.26 | 44.93 | 1.0357 | 1.0891 |

Table 2.5: Weight Distribution (front/total) and Aerodynamic Balance (front/total) combinations

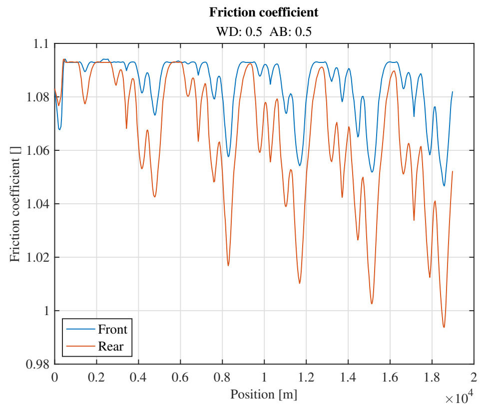
As previously observed, in neutral conditions, the temperature of the rear tyres are higher than the front ones. As expected then, the best lap times are reached when avoiding to heat up too much the rear tyres in favor of the front axle, so to contemporarily bring into the correct window all the four wheels more quickly without exceeding the limits in the last laps. For this reason, the best combination, in terms of average lap time, is $WD = 0.5$ and $AB = 0.35$: a larger normal force on the rear axle corresponds to a larger capability to accelerate and a quicker heat up of the front tyres thanks to more slip. This solution has been preferred with respect to the one with the same average time ($WD = 0.65$ and $AB = 0.65$) because the latter is strongly pushed towards the front axle, with some drawback better explained with the diagrams in the following pages.

Roughly, as will be better shown, moving the weight distribution and the aero balance in the same direction has opposite effects: more weight means hotter tyres, more downforce means colder tyres.

The neutral-balance diagrams ($WD = 0.5$ and $AB = 0.5$) are shown below as a reference.

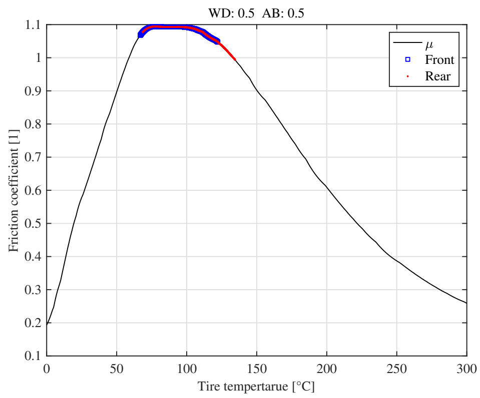



(a) Temperature vs Position

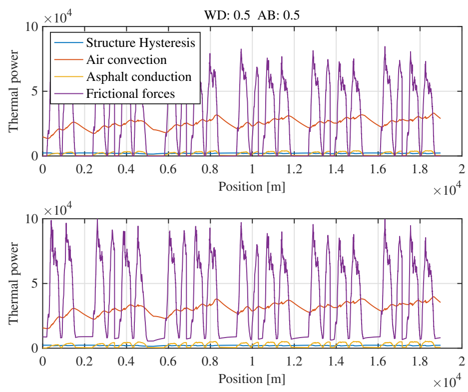



(b) Friction coefficient vs Position

Figure 2.16: $WD = 0.5$ and $AB = 0.5$




(a) Friction coefficient vs Temperature




(b) Power flows - Front and Rear

Figure 2.17: $WD = 0.5$ and $AB = 0.5$

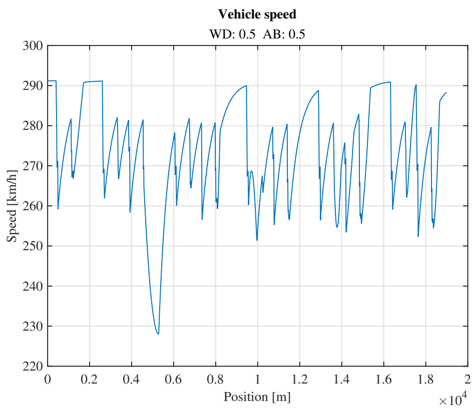
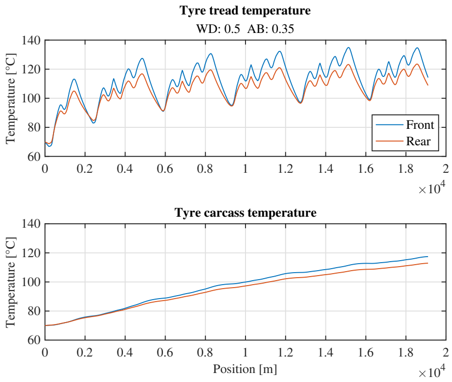


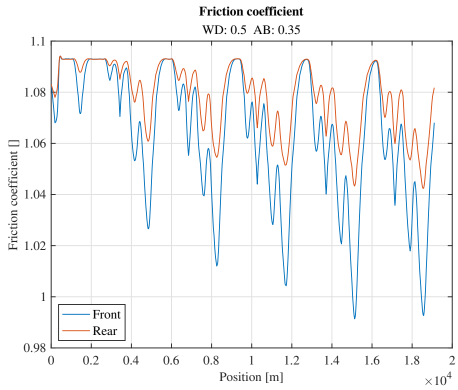

Figure 2.18: $WD = 0.5$ and $AB = 0.5$ - Speed vs Position

A large drop in the speed diagram has been experienced between 5000 $m$ and 6000 $m$, position at which the car should accelerate. This could be directly connected with the temperature increase and friction coefficient reduction which happen at the rear tyres between 3000 $m$ and 5000 $m$.

Let’s now analyze the best case ($WD = 0.5$ and $AB = 0.35$).

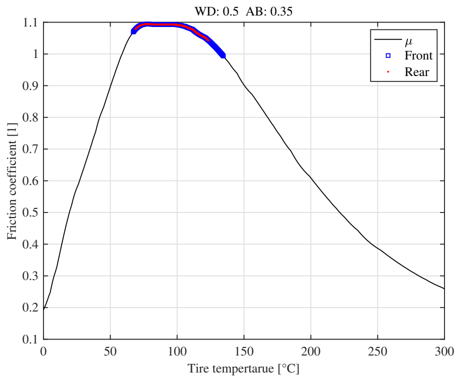



(a) Temperature vs Position

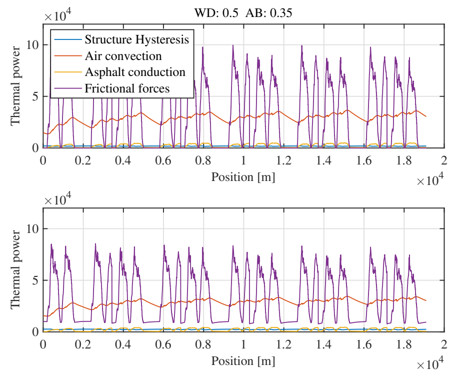



(b) Friction coefficient vs Position

Figure 2.19: $WD = 0.5$ and $AB = 0.35$




(a) Friction coefficient vs Temperature




(b) Power flows - Front and Rear

Figure 2.20: $WD = 0.5$ and $AB = 0.35$

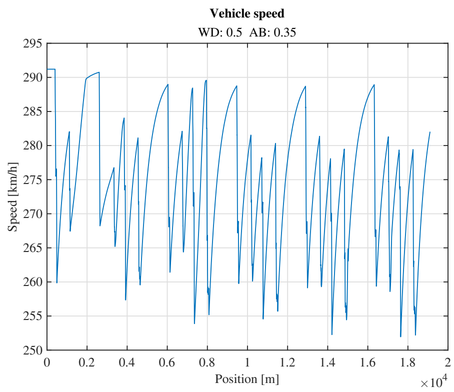
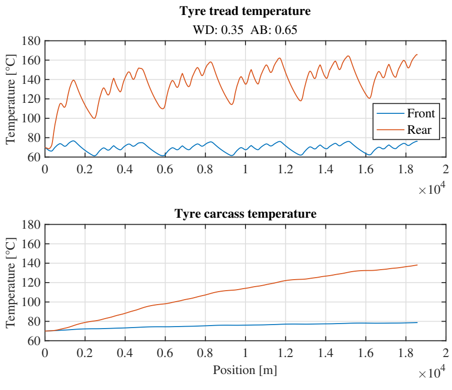


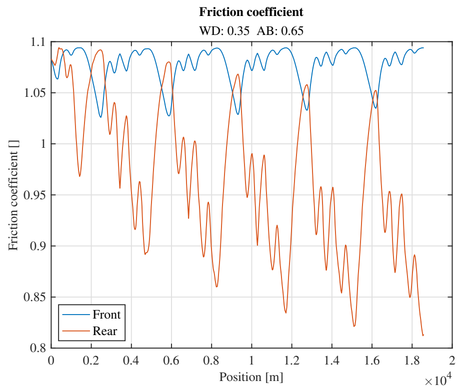

Figure 2.21: $WD = 0.5$ and $AB = 0.35$ - Speed vs Position

Larger temperatures at the front axle with respect to rear axle are experienced, with a consequent larger grip at the rear tyres. As expected this causes a faster temperature increase at the front tyres in the first lap and the capability to develop larger longitudinal accelerations with consequent larger speed.

In the worst situation instead ($WD = 0.35$ and $AB = 0.65$) presents the diagrams below.

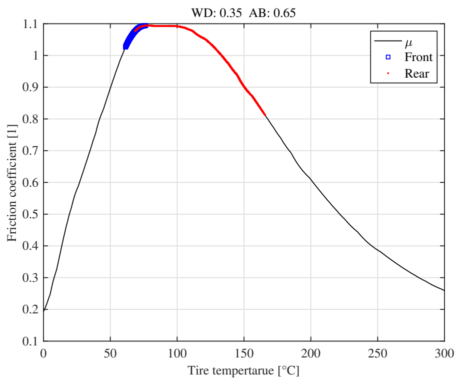



(a) Temperature vs Position

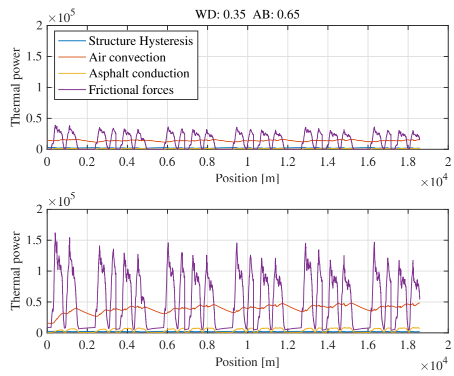



(b) Friction coefficient vs Position

Figure 2.22: $WD = 0.35$ and $AB = 0.65$




(a) Friction coefficient vs Temperature




(b) Power flows - Front and Rear

Figure 2.23: $WD = 0.35$ and $AB = 0.65$

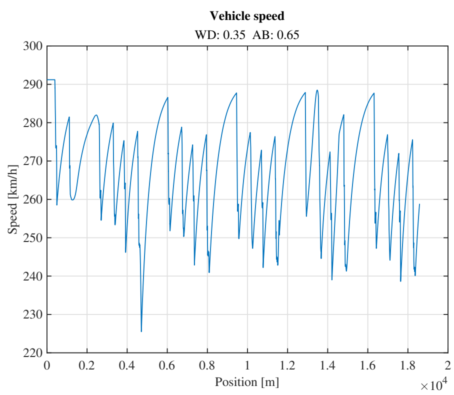
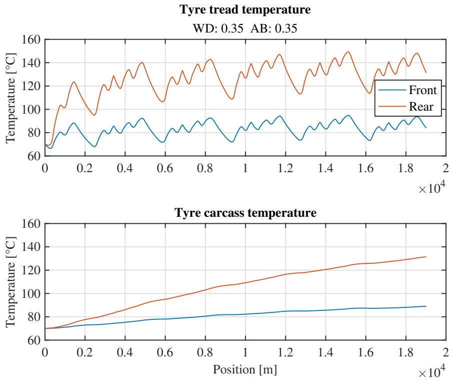


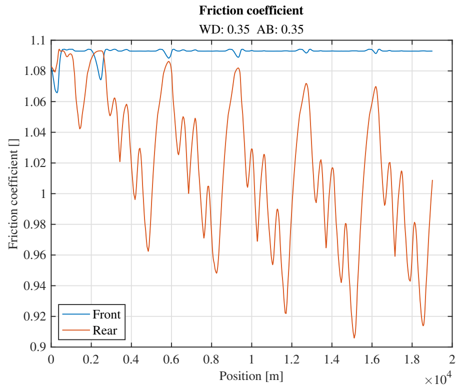

Figure 2.24: $WD = 0.35$ and $AB = 0.65$ - Speed vs Position

The aero-balance to the front causes lower slip in (high-speed) cornering, thus low thermal power due to friction. The weight distribution to the rear causes more oversteer (larger slip while cornering). These generate a too low temperature at the front tyres and a too large temperature at the rear, overheating them, thus the car becomes less stable, worsening the performance.

With both the balances to the rear axle ($WD = 0.35$ and $AB = 0.35$) the following results are obtained.

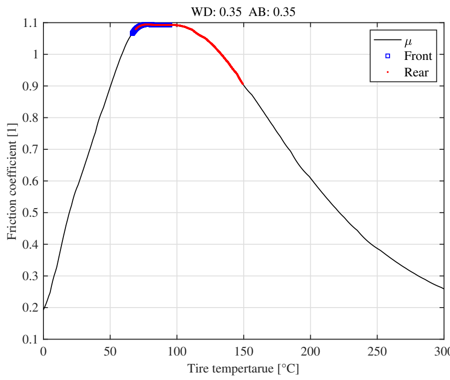



(a) Temperature vs Position

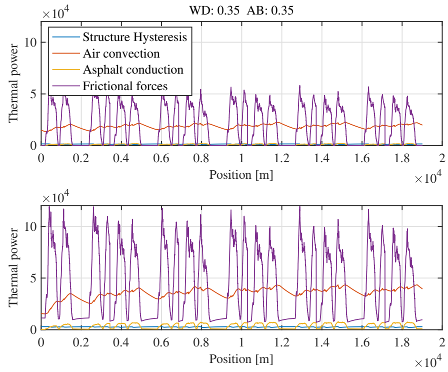



(b) Friction coefficient vs Position

Figure 2.25: $WD = 0.35$ and $AB = 0.35$




(a) Friction coefficient vs Temperature




(b) Power flows - Front and Rear

Figure 2.26: $WD = 0.35$ and $AB = 0.35$

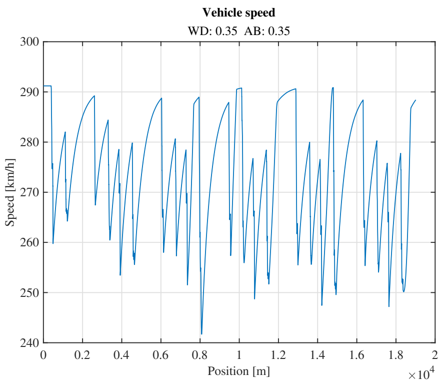
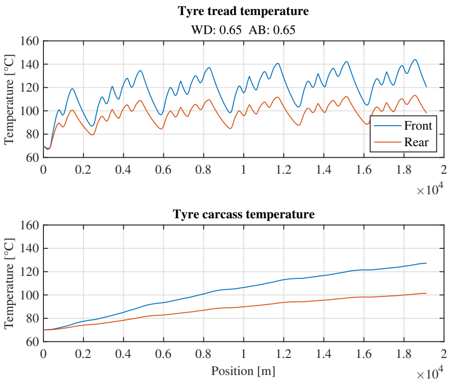


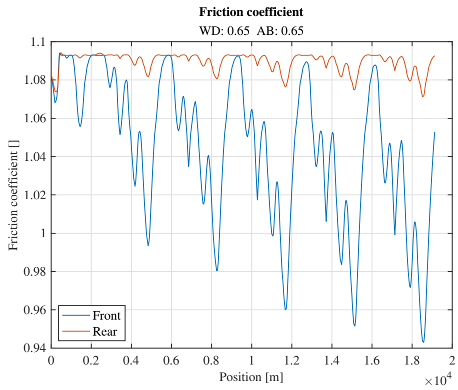

Figure 2.27: $WD = 0.35$ and $AB = 0.35$ - Speed vs Position

This case is similar to the previous one, with the aerodynamic balance trying to compensate the effects of the COG close to the rear axle, resulting in better front axle (low temperatures but not so bad in terms of friction coefficient, which is instead the best average front one within all the cases) and better rear one (still slightly too hot).

Finally, both balances to the front are analyzed ($WD = 0.65$ and $AB = 0.65$)

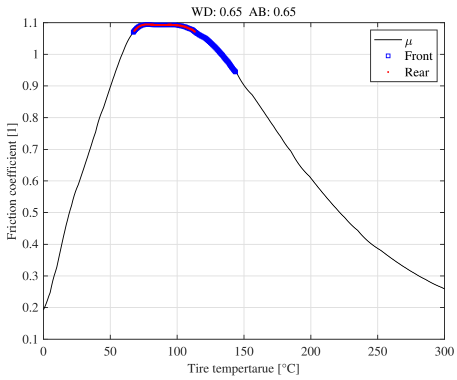



(a) Temperature vs Position

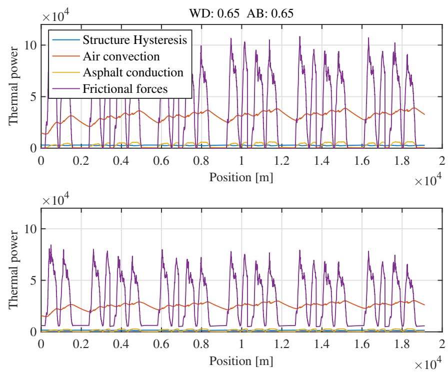



(b) Friction coefficient vs Position

Figure 2.28: $WD = 0.65$ and $AB = 0.65$




(a) Friction coefficient vs Temperature




(b) Power flows - Front and Rear

Figure 2.29: $WD = 0.65$ and $AB = 0.65$

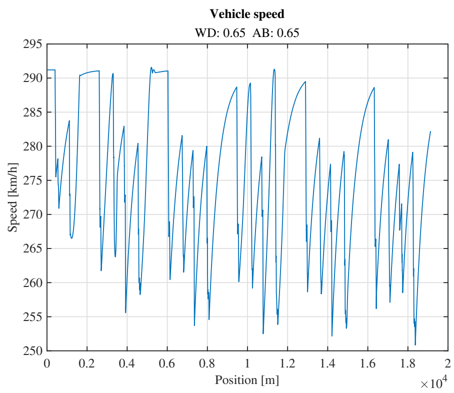
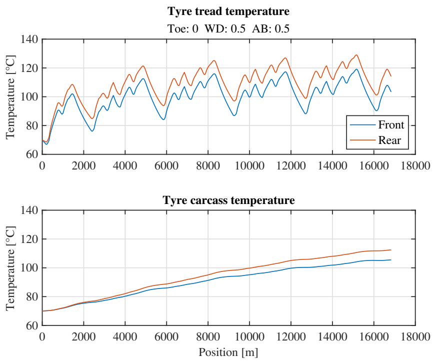



Figure 2.30: $WD = 0.65$ and $AB = 0.65$ - Speed vs Position

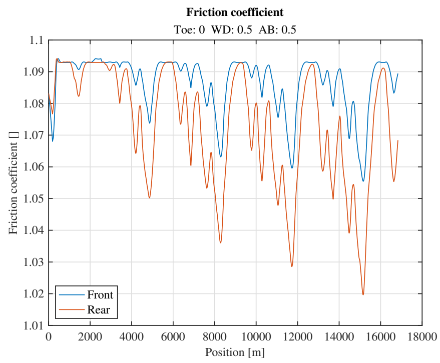
The reasons why this solution has not been chosen as the best one are: larger discrepancy between front and rear tyres temperature; very good rear temperature but worse front one, with bad development which could cause slower laps while going on with the race; set-up strongly pushed toward the front axle which could be not the best solution in a deeper analysis.

### 2.1.4 Different track

We now want to perform the same analysis on a different track and compare the results.
The second track is characterized by a longer lap, 4 laps in total instead of 5, almost the same peak speed, lower-speed intermediate peaks and slower corners (lower minimum speeds).
As a reference, the diagrams of neutral set-up (0 toe angles and 0.5 for both balances) are reported below.

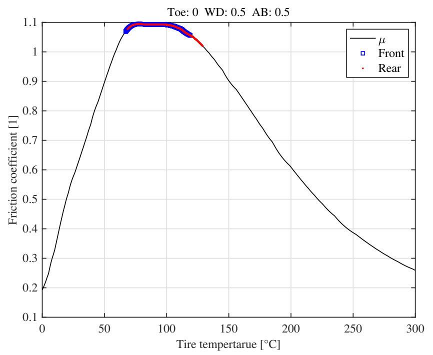



(a) Temperature vs Position

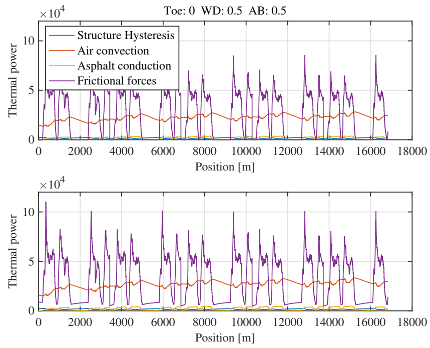



(b) Friction coefficient vs Position

Figure 2.31: Neutral set-up




(a) Friction coefficient vs Temperature




(b) Power flows - Front and Rear

Figure 2.32: Neutral set-up

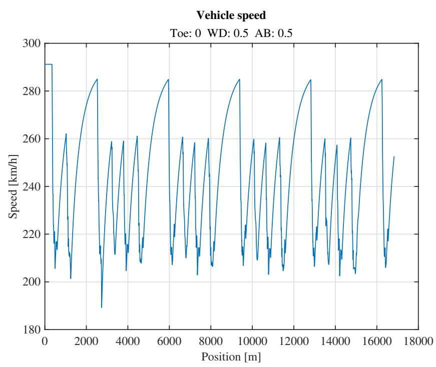



Figure 2.33: Neutral set-up - Speed vs Position

Comparing the neutral set-ups on the two different tracks, it can be noticed that this second solution provides a slightly better result, with a good temperature window both at front and rear, a lower temperature difference between the two, lower thermal power due to friction especially at the rear axle, lower temperatures especially at the end of the stint and globally a larger average friction coefficient.

**Toe effects.** The results for different toe angles with neutral AB and WD are reported.

| $Toe_F$ **[deg]** | $Toe_R$ **[deg]** | **1° Lap t. [s]** | **Last Lap t. [s]** | **Avg Lap t. [s]** | $\mu_{avg,F}$ | $\mu_{avg,R}$ |
|---|---|---|---|---|---|---|
| 0 | 0 | 49.75 | 51.28 | 51.18 | 1.0855 | 1.0681 |
| 0.35 | 0 | 49.59 | 51.35 | 51.19 | 1.0796 | 1.0646 |
| 0.65 | 0 | 49.67 | 51.12 | 51.16 | 1.0769 | 1.0670 |
| 1 | 0 | 49.33 | 51.50 | 51.52 | 1.0748 | 1.0704 |
| 0 | 0.35 | 49.89 | 51.64 | 51.40 | 1.0860 | 1.0622 |
| 0.35 | 0.35 | 49.86 | 51.48 | 50.98 | 1.0819 | 1.0609 |
| 0.65 | 0.35 | 49.69 | 51.41 | 51.23 | 1.0782 | 1.0612 |
| 1 | 0.35 | 49.85 | 51.52 | 51.39 | 1.0739 | 1.0623 |
| 0 | 0.65 | 49.91 | 51.16 | 50.95 | 1.0861 | 1.0551 |
| 0.35 | 0.65 | 50.31 | 51.44 | 51.33 | 1.0828 | 1.0554 |
| 0.65 | 0.65 | 49.81 | 51.61 | 51.35 | 1.0790 | 1.0554 |
| 1 | 0.65 | 49.94 | 51.50 | 51.40 | 1.0737 | 1.0554 |
| 0 | 1 | 49.66 | 51.52 | 51.30 | 1.0863 | 1.0466 |
| 0.35 | 1 | 49.92 | 51.59 | 51.49 | 1.0840 | 1.0493 |
| 0.65 | 1 | 49.97 | 51.60 | 51.47 | 1.0796 | 1.0472 |
| 1 | 1 | 49.98 | 51.66 | 51.86 | 1.0760 | 1.0498 |

Table 2.6: Toe combinations second track

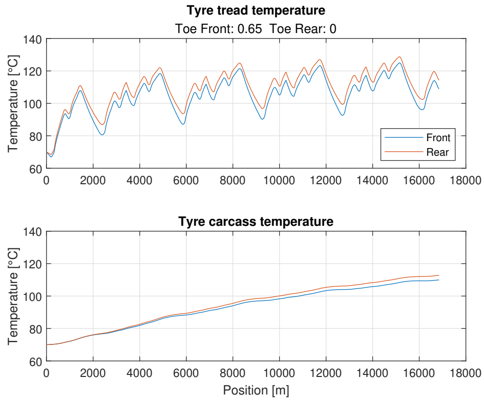
The results in (Tab. 2.6) show that the different setups do not have a significant influence on the average lap time, therefore to improve the lap time other changes will probably be more efficient.

Being this track with many high speed corners, temperature management at the front axle is crucial, so the setup should focus on maintain the front tyres at their optimal temperature. Due to the many consecutive high speed corners the tyres will heat up considerably, then in the very long straight they will cool down.

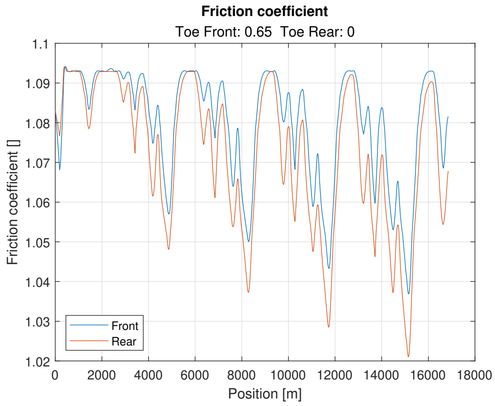
The combination of toe angles that results in the lowest lap time is $Toe_F = 0$ and $Toe_R = 0.65$, which tends to keep the front tyres as cold as possible, neglecting the grip loss at the rear, inducing oversteering.

But, reasoning about real life scenarios, and considering that the average lap time is basically the same for each combination, the more sensible solution would be to maintain a setup similar to the one used in the previous track, meaning $Toe_F = 0.65$ and $Toe_R = 0$.
This setup is more general for a RWD car, and can be used in different tracks as it allows to have a similar trend in temperature and grip for both front and rear tyres. This develops into an almost neutral car, which is usually preferred by drivers, avoiding excessive under/over-steering. This is a reasonable trade-off in real life: a driver with a faster but unpredictable car is more prone to make costly mistakes, rather than a driver with a slightly slower, but predictable car.

The former best case analyzed $Toe_F = 0.65$ and $Toe_R = 0$.

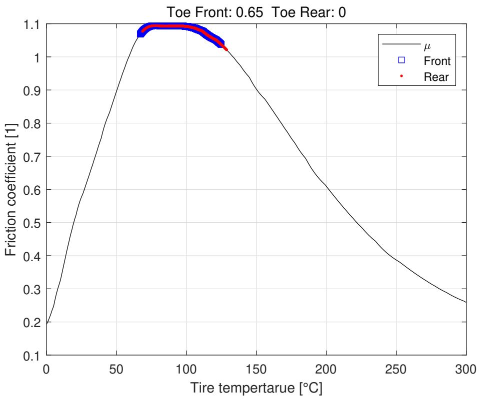



(a) Temperature vs Position

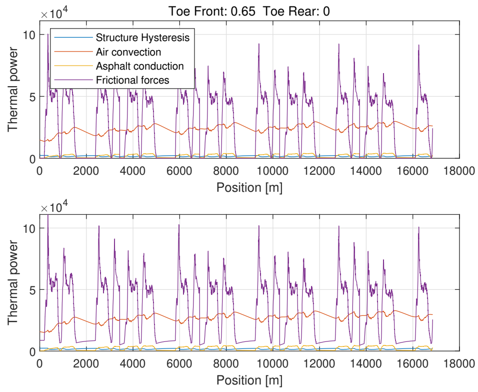



(b) Friction coefficient vs Position

Figure 2.34: $Toe_F = 0.65$ and $Toe_R = 0$




(a) Friction coefficient vs Temperature




(b) Power flows - Front and Rear

Figure 2.35: $Toe_F = 0.65$ and $Toe_R = 0$

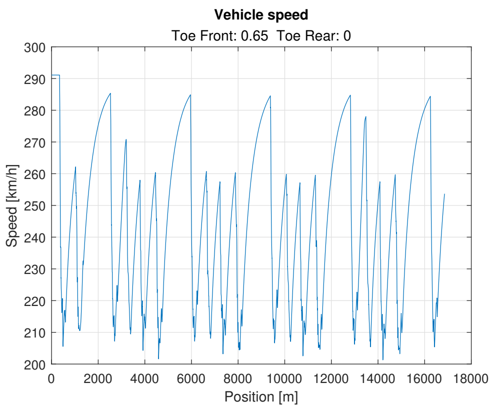
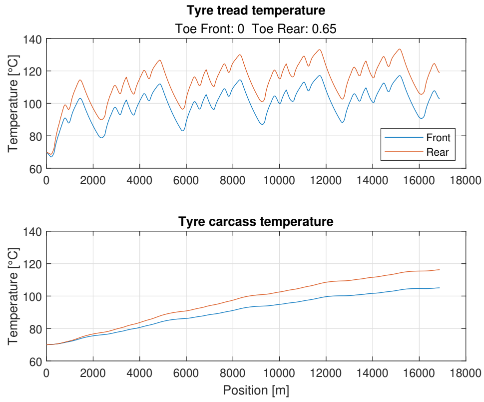


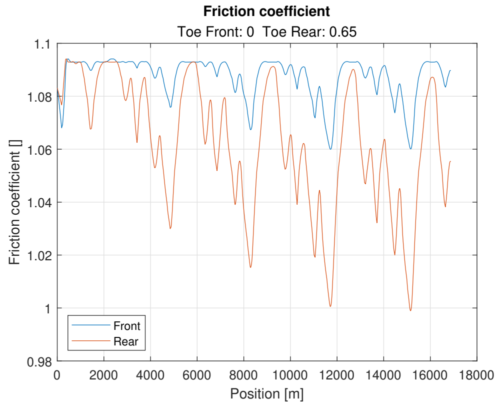

Figure 2.36: $Toe_F = 0.65$ and $Toe_R = 0$ - Speed vs Position

In this case the setup is again showing at the front and rear that tyres have similar trends. but their grip is lower than in the previous track due to overheating in the track's section with many turns. This lowers the lap time.

The former best case analyzed $Toe_F = 0$ and $Toe_R = 0.65$.

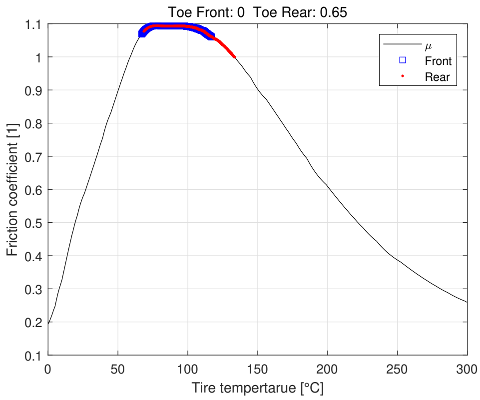



(a) Temperature vs Position

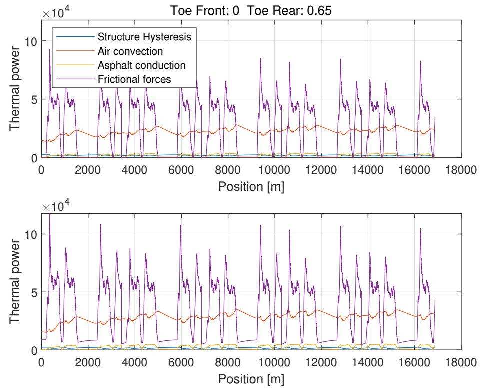



(b) Friction coefficient vs Position

Figure 2.37: $Toe_F = 0$ and $Toe_R = 0.65$




(a) Friction coefficient vs Temperature




(b) Power flows - Front and Rear

Figure 2.38: $Toe_F = 0$ and $Toe_R = 0.65$

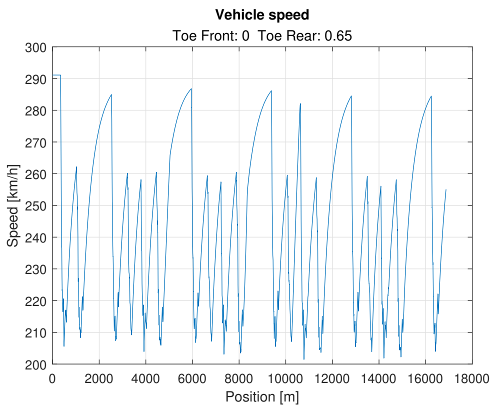



Figure 2.39: $Toe_F = 0$ and $Toe_R = 0.65$ - Speed vs Position

This combination is the best one in terms of lap time, having lower temperature at the front tyres prevents them from overheating, this allows to have always high grip at the front, which helps in this track with many corners, but extreme oversteer may be a problem for the driver.

**WD and AB effects.** The new results are reported in the following table.

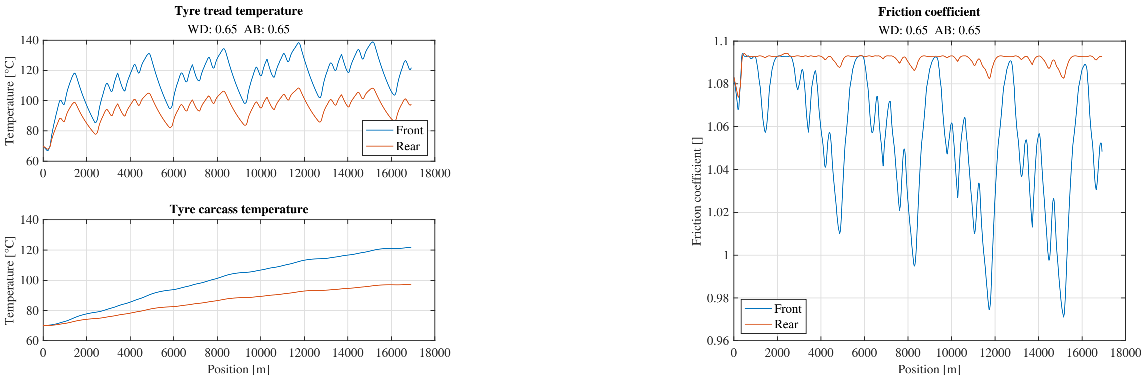
| WD | AB | 1° Lap t. [s] | Last Lap t. [s] | Avg Lap t. [s] | $\mu_{avg,F}$ | $\mu_{avg,R}$ |
|---|---|---|---|---|---|---|
| 0.35 | 0.35 | 50.01 | 51.72 | 51.52 | 1.0930 | 1.0248 |
| 0.50 | 0.35 | 49.56 | 51.27 | 51.19 | 1.0688 | 1.0806 |
| 0.65 | 0.35 | 50.21 | 52.15 | 51.96 | 0.9947 | 1.0931 |
| 0.35 | 0.50 | 50.19 | 51.88 | 51.77 | 1.0910 | 1.0047 |
| 0.50 | 0.50 | 50.24 | 51.21 | 51.09 | 1.0852 | 1.0675 |
| 0.65 | 0.50 | 49.52 | 51.51 | 51.50 | 1.0268 | 1.0930 |
| 0.35 | 0.65 | 50.08 | 52.95 | 52.45 | 1.0831 | 0.9784 |
| 0.50 | 0.65 | 49.34 | 51.55 | 51.32 | 1.0917 | 1.0467 |
| 0.65 | 0.65 | 49.28 | 50.96 | 51.03 | 1.0466 | 1.0914 |

Table 2.7: Weight Distribution (front/total) and Aerodynamic Balance (front/total) combinations

It can be observed as the new best average lap time (and first lap time too) is the one which was the non-chosen solution of the previous analysis, confirming that, in different conditions, it could be a feasible set-up, but the choice is highly dependent on the track layout.
It is also a reasonable result since an aero-balance towards the front can be helpful in turns at lower speed, which are more frequent in this track than in the previous one.
The solution of the previous analysis is now still one of the best, but slightly worsening the average time with respect to the neutral set-up.

In the following, the most interesting and representative diagrams are reported.

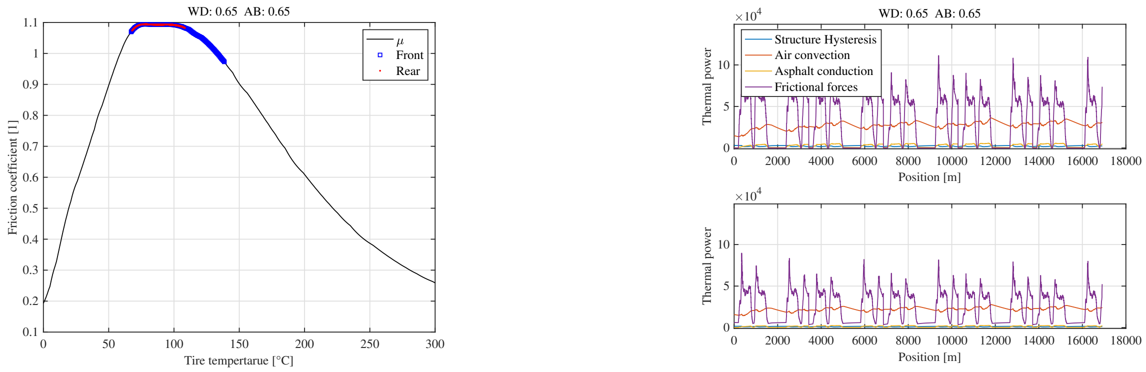
The best performance is reached with $WD = 0.65$ and $AB = 0.65$.




(a) Temperature vs Position

(b) Friction coefficient vs Position

Figure 2.40: $WD = 0.65$ and $AB = 0.65$




(a) Friction coefficient vs Temperature

(b) Power flows - Front and Rear

Figure 2.41: $WD = 0.65$ and $AB = 0.65$

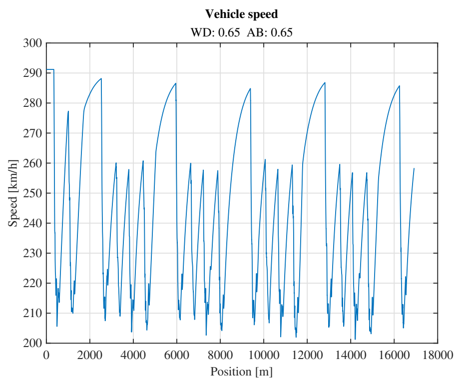
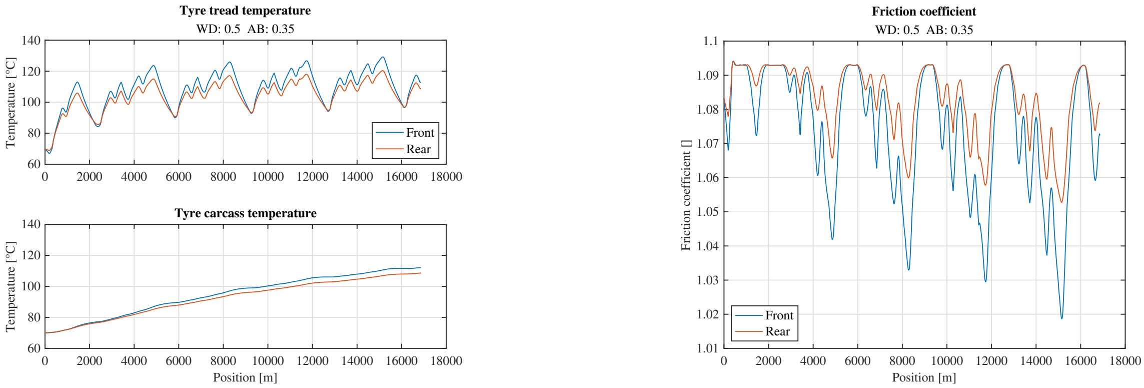



Figure 2.42: $WD = 0.65$ and $AB = 0.65$ - Speed vs Position

Differently from the other track, the temperatures here are globally lower, which removes the main problem of this set-up in the other solution. The other concepts are substantially the same, resulting in very good temperatures at the rear tyres and an acceptable range at the front ones with the tendency of the friction coefficient to decrease while increasing the number of laps. Lower frictional thermal power is experienced at both axles.

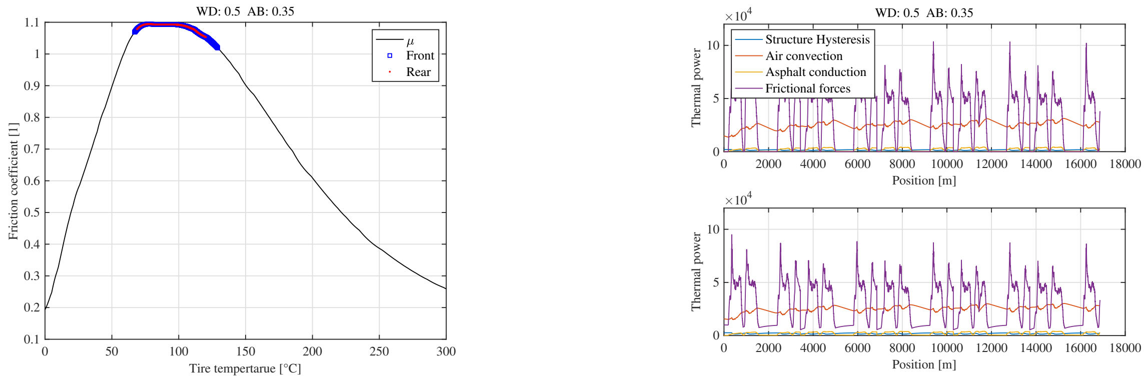
Let's analyze also $WD = 0.5$ and $AB = 0.35$ (previous best solution).




(a) Temperature vs Position

(b) Friction coefficient vs Position

Figure 2.43: $WD = 0.5$ and $AB = 0.35$




(a) Friction coefficient vs Temperature

(b) Power flows - Front and Rear

Figure 2.44: $WD = 0.5$ and $AB = 0.35$

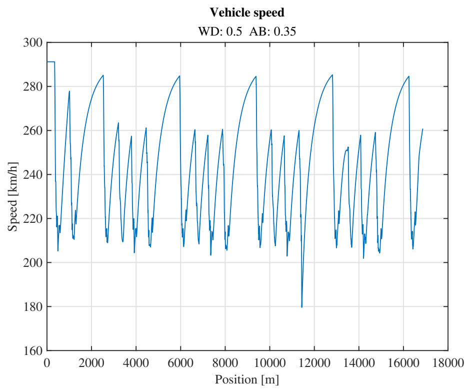



Figure 2.45: $WD = 0.5$ and $AB = 0.35$ - Speed vs Position

Even if the lap time is not the best, this set-up definitely has the best temperature ranges. The trend with respect to the previous track are the same already stated. The comparison with the best set-up of this track suggests lower and more stable temperatures for this case, making this a possible set-up for longer stints.

### 2.1.5 Tyre warming up without acting on the toe angles

The general observed trends suggest that the tyres can be warmed up more quickly by increasing the percentage of weight on the specific axle or by reducing the vertical load coming from the aerodynamic downforce.
It has in fact been observed that by moving the weight balance towards one axle (i.e. by moving the center of gravity towards it) the effects were a quicker warming up in the first lap and possibly a larger grip decay in the last laps due to too high temperatures.
At the same time, opposite effects has been experienced while moving the aerodynamic balance towards the same axle (i.e. by moving the center of pressure towards it): increasing the downforce on the tyre resulted in a slower temperature increase, due to smaller slip.

These effects can be explained by remembering that, as seen, the largest contribution of thermal power comes from the frictional effects: moving the COG towards an axle makes it to slip more in turns and consequently to heat up more; on the contrary, a larger downforce reduces the slip on the tyres, thus reducing the frictional thermal power. So finally reducing the AB on the tyres can help in the warm up.
The increase/reduction of frictional thermal power could be counterbalanced by a reduction/increase of the hysteretic losses, which however represent only a marginal effect on the heat production in the tyres.

Clearly everything needs to be considered as a trade-off: an example could be the fact that a lower downforce on the rear tyres causes a quicker warm up which could put the tyres in the correct window in terms of friction coefficient, but at the same time reduces the vertical load on the tyres themselves and so the capability to transmit longitudinal and lateral forces.

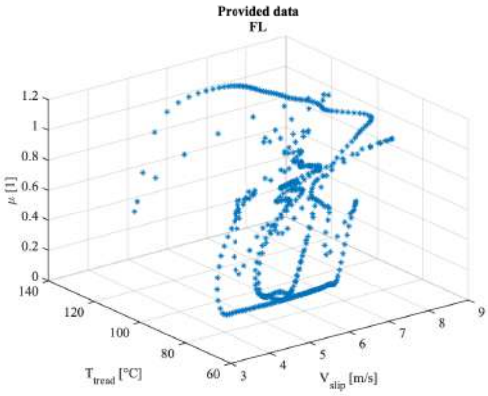
Other ways to warm up tyres without changing the setup of the car could be:

- Follow a wave-like trajectory to create high lateral forces, helping especially the front tyres.
- Accelerate harshly to force the rear tyres to slip, heating them up due to friction.
- Brake hard from high speed in order to create heat in the brakes, which in turn heat up the tyre carcass from the inside.
- Brake hard while turning, to generate slip on the outside of the tyre, helping especially the front tyres.
- Deforming the tyre by taking wide turns at high speed to exploit the heat generated by hysteresis.

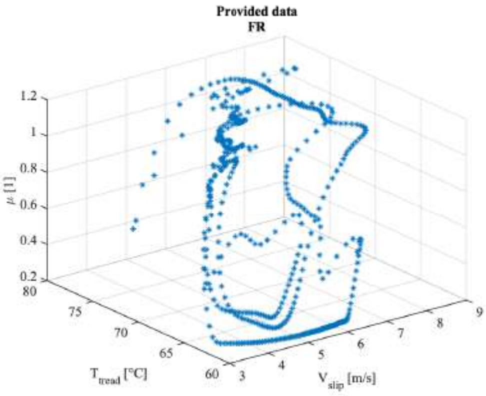
## 2.2 Exercise 2

The request of this second exercise is to try to extract, from a given data set, the friction law of the tyres. In particular, the provided data regard the longitudinal, lateral and vertical forces, the slip velocity and the tread temperature for all the four tyres during a lap.
The given quantities are plotted here below, considering that:

$$\mu = \frac{\sqrt{F_x^2 + F_y^2}}{Fz}$$

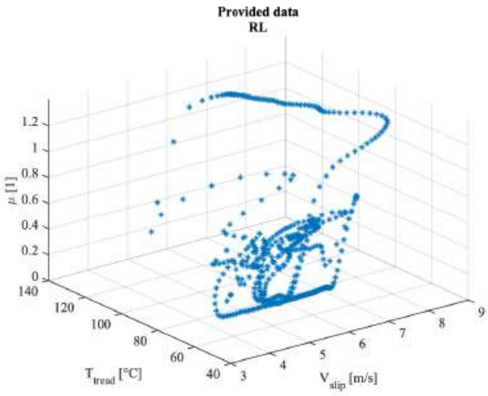



(a) Front Left Tyre

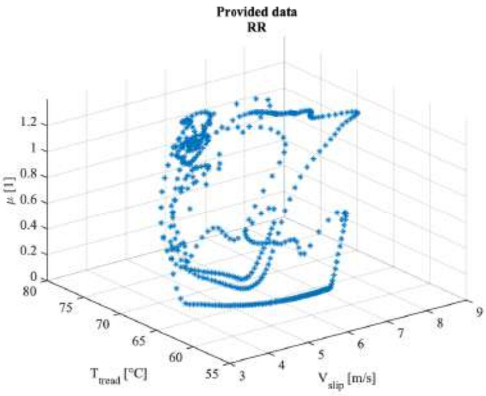



(b) Front Right Tyre

Figure 2.46: Friction coefficient front




(a) Rear Left Tyre




(b) Rear Right Tyre

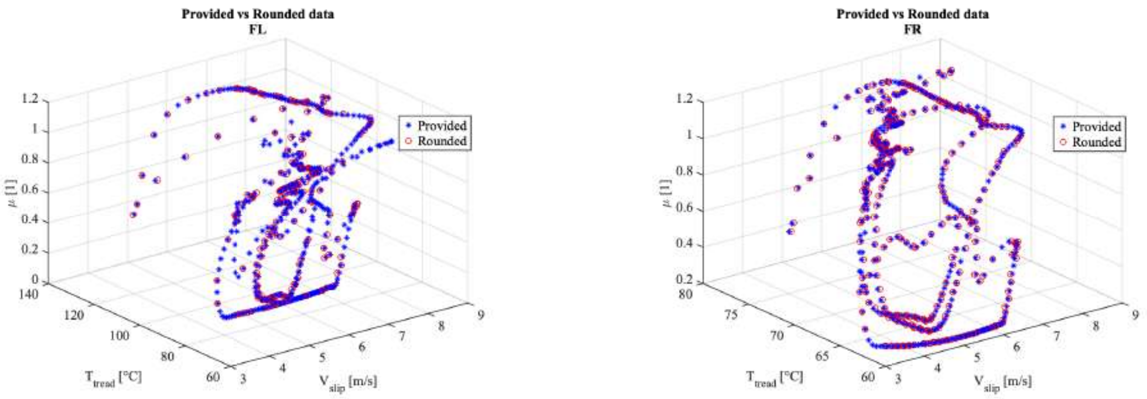
Figure 2.47: Friction coefficient rear

The first thing to be noticed is that the data set is very concentrated around certain values of slip velocity and temperature, with consequent areas in which information is very lacky. Moreover, for almost the same inputs values, very broad sets of $\mu$ values are observed.
Every step in the process will be intended to try to generalize, smooth out and improve the quality of the given poor dataset, to extract a more straightforward and easier to interpret one.

To face this problem, the following procedure has been adopted:

1. The input data has been rounded to the first decimal place
2. In the case of same rounded temperature and velocity, the maximum value of $\mu$ has been considered
3. The resulting data has been used to train an Artificial Neural Network algorithm
4. The trained ANN has been exploited to try to extract useful information also in the areas characterized by a lack of data

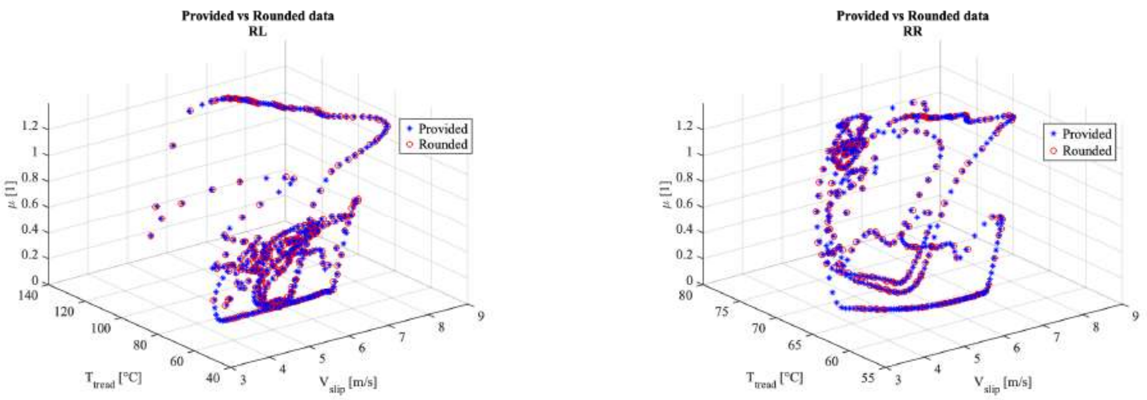
The rounding step is performed to try to smooth out a little the local oscillations in the $\mu$ values. The maximum value of $\mu$, for the same inputs, is considered to be representative of the rounded pair of inputs. The rounding to the first decimal place seemed to be a good compromise between smoothing the trend and avoiding to loose to much information. The results of the rounding procedure are shown in the following pictures, superimposed to the original set.




(a) Front Left Tyre

(b) Front Right Tyre

Figure 2.48: Rounding results front




(a) Rear Left Tyre

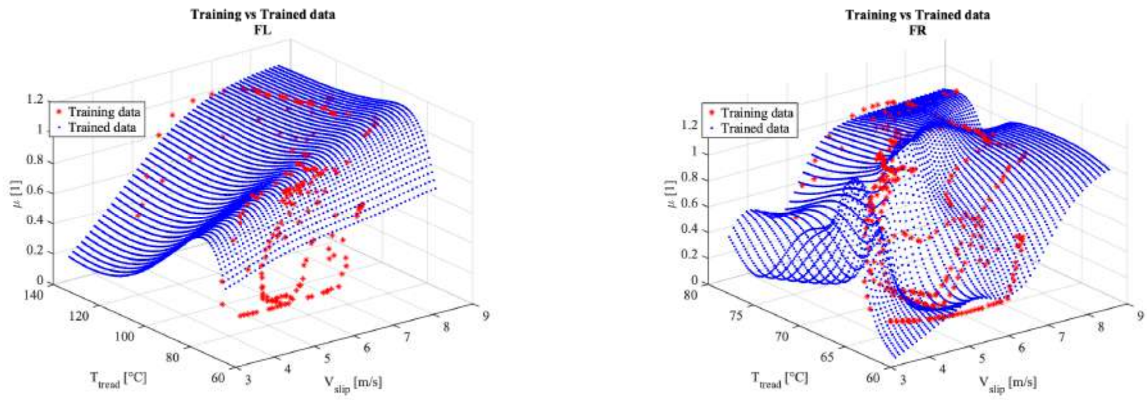
(b) Rear Right Tyre

Figure 2.49: Rounding results rear

Then, an Artificial Neural Network, with the following characteristics, has been trained:

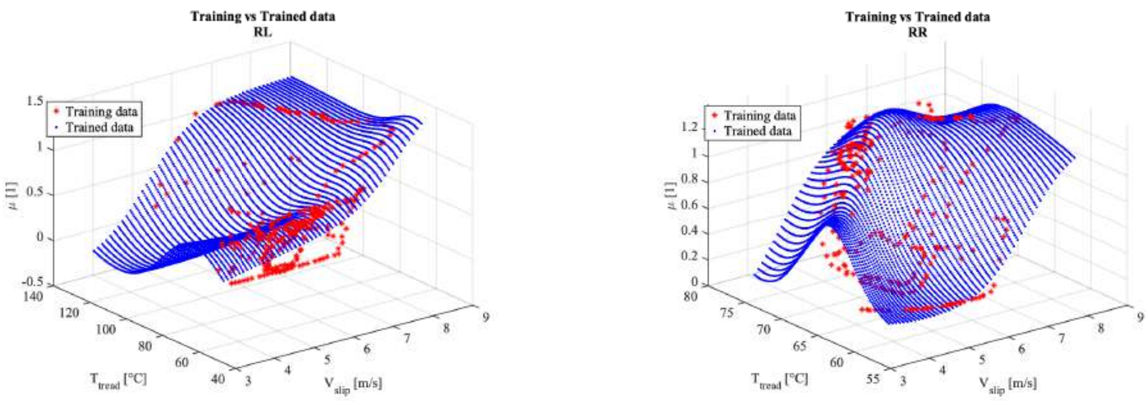
- 1 hidden layer with 10 neurons: best trade off between accuracy and avoiding too strong oscillations in the curves, to try to generalize as much as possible;
- Training algorithm: Bayesian Regularization, which generalizes better, while avoiding overfit;
- Layer transfer function: `tansig`;
- 80 % training set, 20 % validation set
- Maximum 6 validation fails

The results of the training procedure, repeated several times to ensure the results consistency, is here reported.




(a) Front Left Tyre (b) Front Right Tyre

Figure 2.50: Training results front

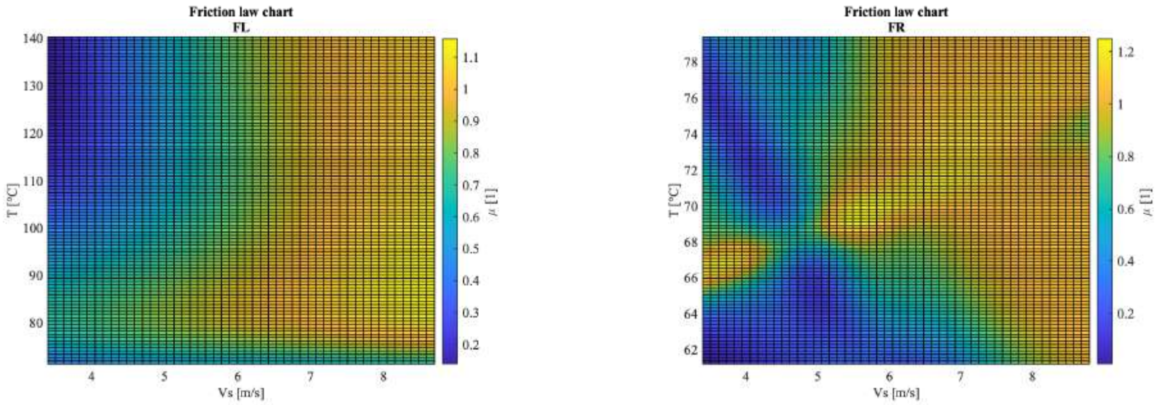



(a) Rear Left Tyre (b) Rear Right Tyre

Figure 2.51: Training results rear

FL seamed to be impossible to be approximated with better accuracy, resulting in a strongly generalized and smoothed result. On the contrary, FR appeared difficult to be generalized, with bad oscillations always observed. Both rear tyres resulted in a better trade off between accuracy and smoothing.

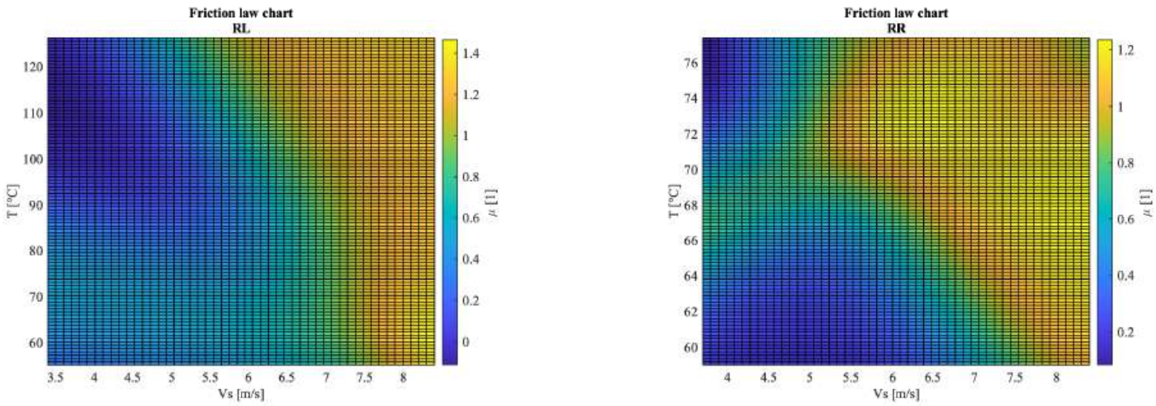
The resulting friction charts and friction law are reported in the following.




(a) Front Left Tyre

(b) Front Right Tyre

Figure 2.52: Friction charts front



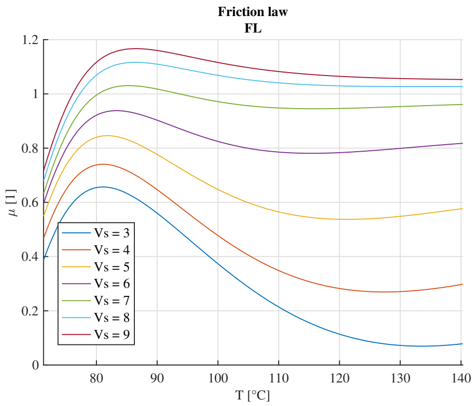

(a) Rear Left Tyre

(b) Rear Right Tyre

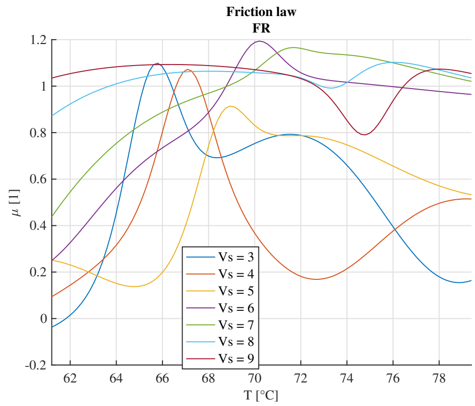
Figure 2.53: Friction charts rear

In all the charts some common elements can be observed. The very low friction values at low $V_s$ and large $T$ just come from the estimation performed by the NN, in a zone characterized by the absence of data, thus to be not considered as reliable. Higher friction values result for larger slip velocities, especially in the zone between $80\,°C$ and $100\,°C$. A smaller range of temperatures is observed in both right tyres. In general, the trends in many regions are shared between the four diagrams, with the front-right tyre clearly being the worst one.

In the next page, some slices for some fixed $V_s$ values are cut out from the friction chart to extract the friction laws.

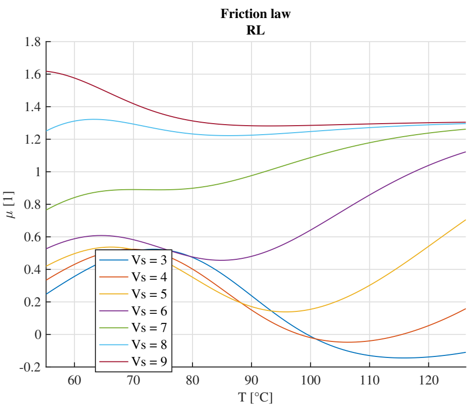



(a) Front Left Tyre

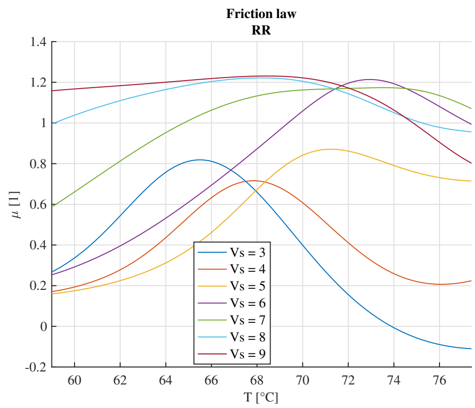



(b) Front Right Tyre

Figure 2.54: Friction laws front




(a) Rear Left Tyre



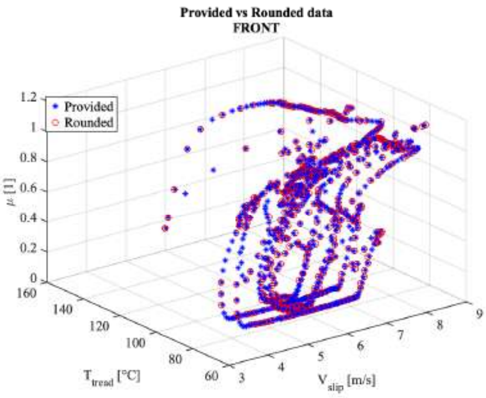

(b) Rear Right Tyre

Figure 2.55: Friction laws rear

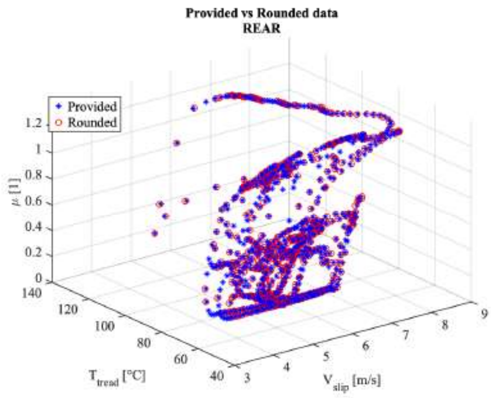
The zones at low temperature and large slip velocity were the ones characterized by the largest portion of the provided dataset. From the literature, this trend was more or less suggested: initial friction coefficient (at about 60 $°C$) slightly less than 1, larger for lower $V_s$, increasing with temperature; peak friction coefficient at around 100 $°C$, moved to higher temperatures with larger $V_s$; decreasing friction coefficient for larger temperatures, up to about 0.8, larger for larger $V_s$. The trends of all diagrams are definitely not respected for large temperatures, especially with lower slip velocities, which show extremely low $\mu$ values, as already stated. Curves only at large slip are at least reasonable for FR and RR tyres, with unexpected drops at larger temperatures (about 80 $°C$). Good trends are observed in FL tyre, even if maybe too much smoothed out (large mean error with respect to initial data). RL tyre shows some unexpected oscillations, some reasonable curve, some too large friction values (such as $V_s = 9\ m/s$) and some too low ones (even negative for lower slip velocities).

In general, these results can be considered as hints to try to roughly estimate the friction laws, in function of temperature and slip velocity, but definitely cannot be considered as quantitatively accurate, especially in some specific regions. Anyway, some reasonable trends has been observed in other zones, in which a rough consistency and correlation to what expected from the literature and from more precise experiments.

For completeness, in the following the results with merged dataset for the two front and the two rear wheels are reported: the idea is to try to increase the information at low temperatures with the right tyres data, which are useless alone for larger temperature values.

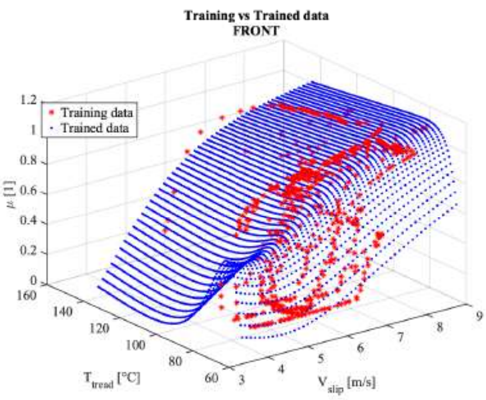



(a) Front Tyres

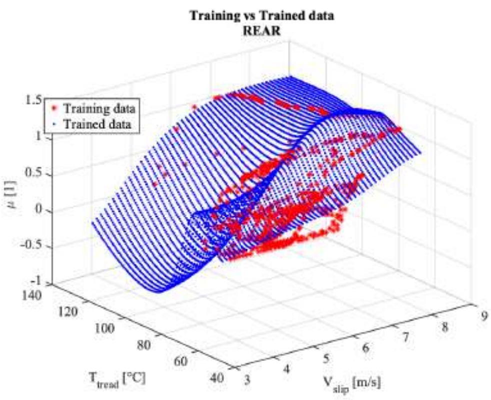



(b) Rear Tyres

Figure 2.56: Rounding results

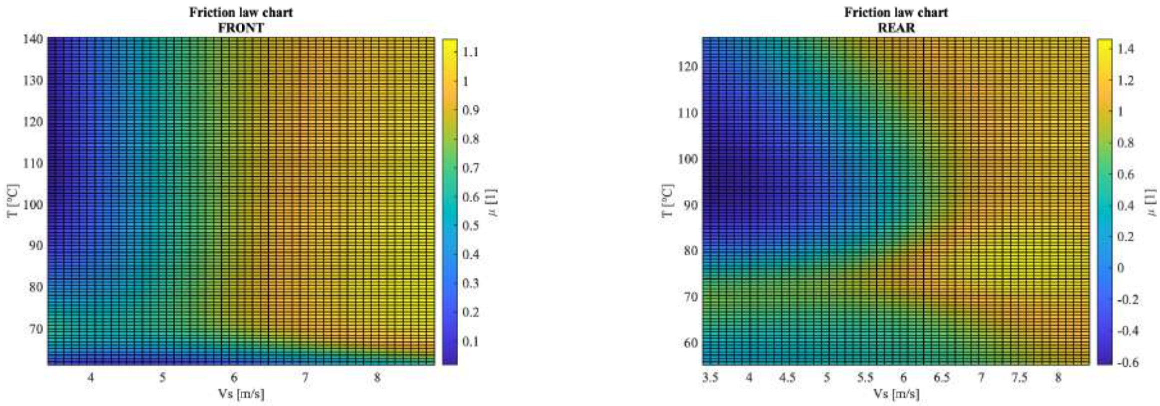



(a) Front Tyres



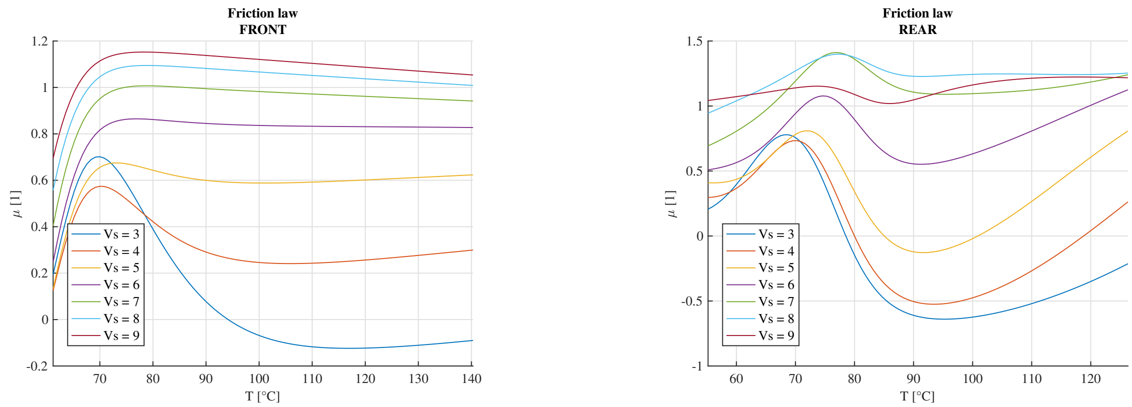

(b) Rear Tyres

Figure 2.57: Training results




(a) Front Tyres (b) Rear Tyres

Figure 2.58: Friction charts




(a) Front Tyres (b) Rear Tyres

Figure 2.59: Friction laws

The results are basically the combination of the previous plots, with some similarities between front and rear but with clearly a more reliable (at least qualitatively) information extracted from front tyres.

The procedure for the whole dataset considered together has been tested and not reported due to the fact the the resulting diagram were again a combination of the four tyres, with no improvements with respect the cases reported here; more fluctuating and not-expected trends were in fact observed in that case.

# 3 Lab.02 - Lap time

The aim of the laboratory was to analyze and study the effects of various setup parameters on the lap time of a *Tatuus FA010* running on different tracks.
The considered tracks are:

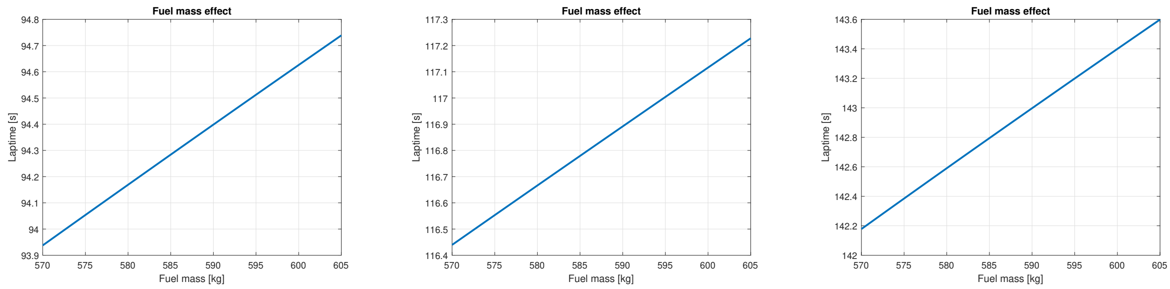
- **Donington:** A low speed track, with many corners and short straights.
- **Monza:** A high speed track, with many long straights.
- **Spa-Francorchamps:** A long track, with long straights and very high speed turns.

## 3.1 Fuel effect on lap time

Intuitively, as a car runs along a circuit, it consumes fuel, therefore as the laps increase, the car's weight decreases, therefore the car will be faster, improving its lap times as the fuel is consumed. To estimate this effect of fuel mass on the overall lap time, the car was run on the same track, each lap with a lighter fuel weight. The starting weight was the one provided in the `Main.m` file, equal to $m_{tot} = 605\,kg$, each iteration was run with the car $1\,kg$ lighter (arbitrary value just to show the trends), up to the minimum weight possible, meaning with an almost empty fuel tank. The total fuel weight can be computed as:

$$m_f = V_f \cdot \rho_f = 45\,l \cdot 0.789\,\frac{kg}{l} = 35.5\,kg \tag{3.1}$$

The results for the different tracks are reported in the following figures. All of the graphs show the same linear trend, meaning that a reduction in mass results in a reduction in lap time, as one could expect.




(a) Donington (b) Monza (c) Spa-Francorchamps

Figure 3.1: Fuel effect on laptime

The mass reduction influences both lateral and longitudinal dynamics: a lighter car is both able to turn faster and accelerate more quickly. (Tab. 3.1) shows the different lap time reduction percentages for the same $\Delta m_{fuel}$. Note that this analysis has been carried out considering the same mass reductions just to show the effects on the different layouts; clearly, a deeper analysis should also consider the different fuel consumptions per lap depending on the specific track in therms of both characteristics and length.

| | **Don** | **Mon** | **Spa** |
|---|---|---|---|
| $m_{diff}$ | 1% | 1% | 1% |
| $t_{diff}$ | 0.143 % | 0.114 % | 0.121 % |

Table 3.1: Lap time reduction

## 3.2 Isochrone maps

Isochrone maps are 3D surfaces, representing the lap time associated to different combinations of drag coefficient $C_d$ and lift coefficient $C_l$, with some superimposed lines highlighting different configurations which result in the same lap time. The feasible configurations can than be superimposed too, to extract the optimal one, which minimizes the lap time. The vehicle is considered with fully loaded fuel tank.

The following graphs were computed using as design variables all the possible aerodynamics coefficients, collected in two vectors: $C_d \in [0.6, 1]$ and $C_l \in [0.5, 2]$. Both vectors were populated using a 0.01 step. The objective function is the lap time, computed using `OpenLap`, computed for each possible combination of $C_d$ and $C_l$. After having created the isochrone map, the feasible setup combinations of $C_d$ and $C_l$ were plotted on the map to find the best feasible setup solution for each track.

| | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** |
|---|---|---|---|---|---|---|---|---|---|
| $C_l$ | 0.8401 | 0.8751 | 0.9101 | 0.9451 | 0.9801 | 1.0151 | 1.0605 | 1.1199 | 1.1793 |
| $C_d$ | 0.6594 | 0.6644 | 0.6694 | 0.6744 | 0.6794 | 0.6844 | 0.6918 | 0.7025 | 0.7132 |

| | **10** | **11** | **12** | **13** | **14** | **15** | **16** | **17** | **18** |
|---|---|---|---|---|---|---|---|---|---|
| $C_l$ | 1.2388 | 1.2982 | 1.3576 | 1.4170 | 1.4764 | 1.5359 | 1.5953 | 1.6547 | 1.7141 |
| $C_d$ | 0.7239 | 0.7346 | 0.7453 | 0.7560 | 0.7667 | 0.7774 | 0.7881 | 0.7988 | 0.8145 |

Table 3.2: Feasible configurations

## 3.3 Optimal aerodynamic configuration

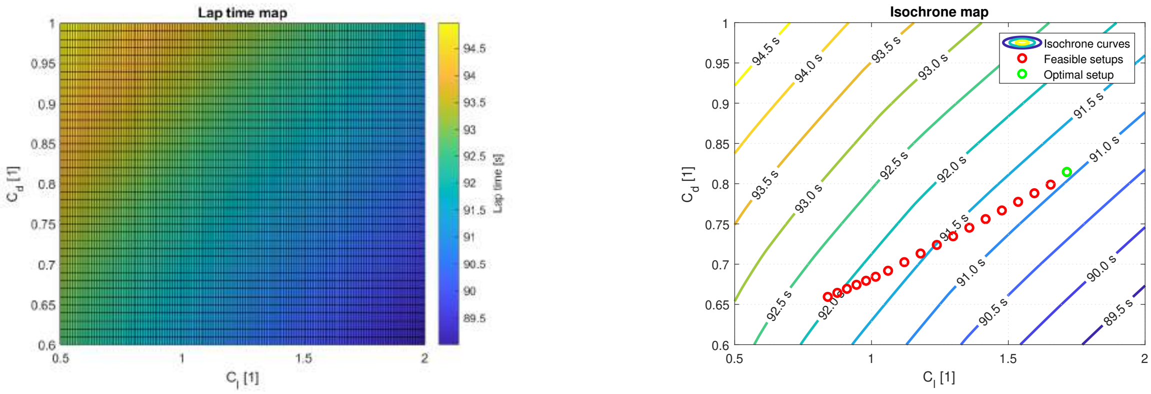
Considering the following isochrone maps it is possible to choose the optimal aerodynamic configuration that best suits the track layout. Note that the analysis is purely based on lap time, without any consideration on tyre management or weather condition, the combination of $C_d$ and $C_l$ chosen among the feasible ones is the one that minimizes the lap time.

This was solved by creating a lookup table with the computed data, fitting all the feasible configurations. The results are reported below.

- **Donington:** Being a track with many corners, the best configuration is the more loaded one, with the highest downforce, so configuration 18, resulting in a lap time of $t_{lap} = 91.08\,s$
- **Monza:** Due to the fast nature of the track with many straights the best configuration is the one with the lowest possible drag without compromising too much downforce, meaning configuration 6, resulting in a lap time of $t_{lap} = 110.96\,s$
- **Spa-Francorchamps:** Since the track has many high speed corners, downforce is important, but at the same time the long straights ask for low drag, therefore the optimal configuration is number 17, resulting in a lap time of $t_{lap} = 143.33\,s$

| | **Don** | **Mon** | **Spa** |
|---|---|---|---|
| $C_l$ | 1.7141 | 1.0151 | 1.6547 |
| $C_d$ | 0.8145 | 0.6844 | 0.7988 |
| $t_{lap}$ | $91.08\,s$ | $110.96\,s$ | $143.33\,s$ |

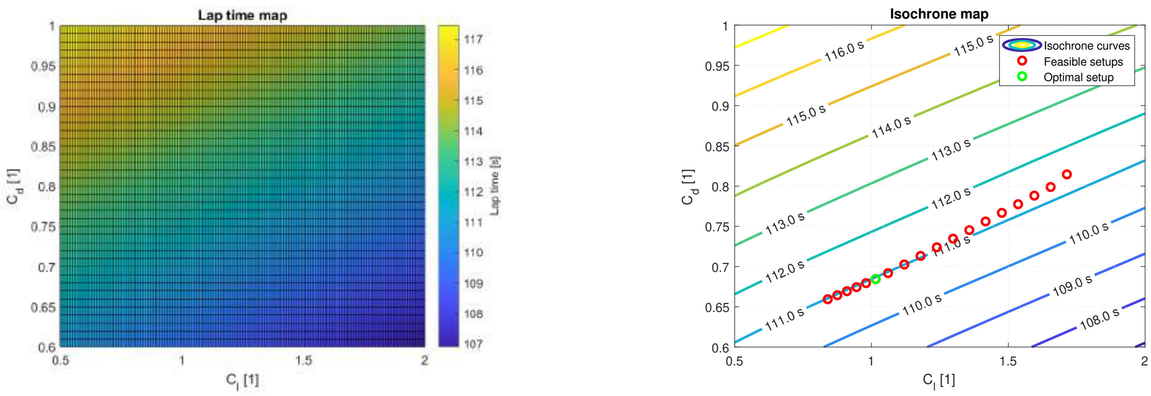
Table 3.3: Best configurations




(a) Time map

(b) Isochrone map

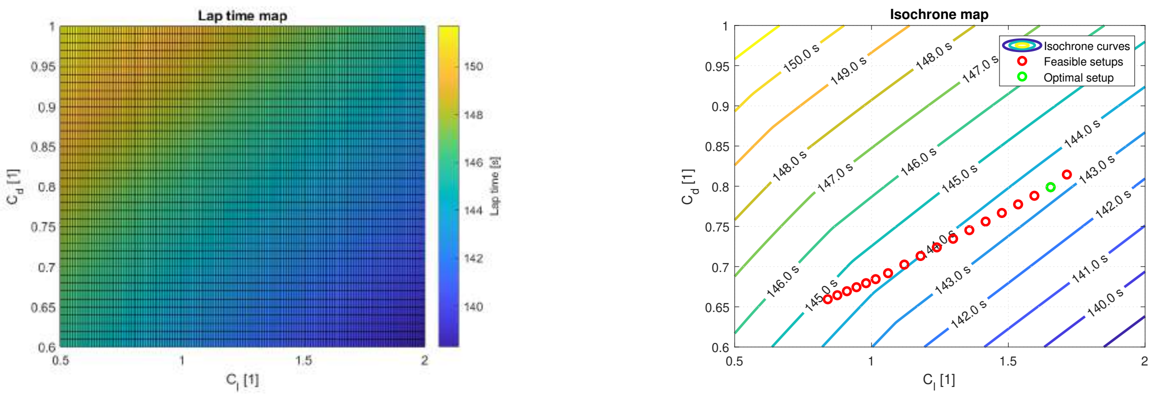
Figure 3.2: Donington track




(a) Time map

(b) Isochrone map

Figure 3.3: Monza track




(a) Time map

(b) Isochrone map

Figure 3.4: Spa-Francorchamps track

## 3.4 Other factors influenced by aerodynamics

Aerodynamics is one of the most important factors in racing cars as it can influence many other effects in different ways, thus is of the upmost importance to find an optimal setup able to produce a winning car: optimizing the lap time only, could not always result in the optimal choice, if a deeper and more complete analysis would be carried out. For instance, other factors to be taken into accounto could be:

- **Overtaking capability:** If a car is set up with a very high downforce, it will also have a larger drag, therefore in a straight it might not be able to reach a sufficient top speed to easily overtake. On the other hand, in presence of systems such as DRS, one should consider that being able to perform better with a high loads configuration, means improving the effect of the DRS activation, due to the larger drag which is produced when the system is not exploited.
- **Tyre management:** When a car is set up to minimize drag, it will also have low downforce, therefore the forces exchanged between the ground and the tyre will be lower. A tyre with a low vertical force tends to slide more, so heat up quicker and wears faster, meaning that lap times at the beginning of a stint will be very fast, but then they will drop after just a few laps, which is not ideal for a long race.
- **Driver feeling:** If a driver does not feel confident inside a car, it will not be able to drive it at its best, even if it is the car with the lowest lap time ever. A driver may not prefer a very low downforce car, opting for a more controllable one, especially in wet conditions.
- **Handling:** Aerodynamic forces acting on the car, transferred to the ground, influence also the suspension design and set up. Aero forces in F1 can reach up to three times the car's weight, so the suspensions will need very rigid springs to prevent the car from touching the ground, damaging the floor and the airflow under the car. So changing the downforce means that also a change in the suspension is needed, something that may also hinder handling or driver comfort (important if reduces the final performance) if not carefully taken into account.
- **Qualification vs Race:** During qualification one may want to optimize the car to reach the minimum possible lap time, since it is the only important thing. But, if the same setup must be used also for the race then this may create other problems, especially on tyre wear as described before. Therefore, if a *parc fermè* rule prevents teams to modify the setup between qualification and race, then the setup must be carried out thoroughly.

# 4 Lab.03 - Racing car braking system

The following exercise focused on brakes and braking system of sport cars. In particular, the vehicle used was the *Tatuus FA010*, the same as in (Sec. 3.1).
Starting from a point mass simulation in `Openlap` on the Monza track, it was possible to compute the forces and torques acting on the different axles using a single track model: these results were later used to define the overall braking system setup.

## 4.1 Disk sizing

The first request was to choose the correct brake disks diameter (both front and rear), out of a list of three available disks having $D = [250\,mm, 270\,mm, 290\,mm]$. The target of this choice was to have the brakes temperature profiles always confined in an optimal range between $200\,°C$ and $600\,°C$, over a race distance of at least 10 laps.
To compute the temperature profile, a thermal model, using Euler's method, was implemented in the function `BB_ThemalModel`, considering both convection and radiation to expel heat from the disks.

$$\begin{cases} T_{k+1} = T_k + \left.\frac{dT}{dt}\right|_k \cdot \Delta t \\ \left.\frac{dT}{dt}\right|_k = \frac{W_{net,k}}{mc_p} \\ W_{net,k} = W_{brk,k} - W_{conv,k} - W_{rad,k} \end{cases} \tag{4.1}$$

Where:

$$\begin{cases} W_{brk,k} = Torque_k \cdot \omega_k \\ W_{conv,k} = htc(v_{x,k}) \cdot A_{conv} \cdot (T_k - T_\infty) \\ W_{rad,k} = \sigma \cdot \varepsilon \cdot A_{rad} \cdot \left(T_k^4 - T_\infty^4\right) \end{cases} \tag{4.2}$$

Which translates into `Matlab` code as:

```
%% Euler's method
for i=1:length(time)-1
    htc_interp(i) = interp1(Vbpt,htc,V(i),'pchip','extrap');
    % =====================================================
    Wconv(i) = htc_interp(i)*Aconv*(T(i) - Tinf);
    Wrad(i) = sg*epsilon*Arad*(T(i)^4 - Tinf^4);
    Wnet(i) = Wbrk(i) - Wconv(i) - Wrad(i);
    dT_dt(i) = Wnet(i)/(M*cp);
    T(i+1) = T(i) + dT_dt(i)*dt;
    % =====================================================
end
```

Using this model implemented in the `Matlab` code, it was possible to define the best disk diamater for the analyzed car.

As one would expect, in a racing car the maximum braking torque required per lap is way larger at the front than at the rear due to the load transfer (in the case of the *Tatuus FA010* is basically twice as high (Fig. 4.1)) and it is crucial for the optimal braking distribution derivation. Is is therefore reasonable to select a disk with a larger diameter at front, to contemporarily allow for a larger toque with not too high temperatures (larger mass with same frictional thermal power) and smaller at rear brakes, to better heat them up. Clearly, for the same braking torque, a smaller disk will require a larger frictional force.

The disks diameters finally chosen are reported in (Tab. 4.1) because (after the initial transient) they both have the same temperature trend over a 10 laps distance, respecting also the request to remain inside the optimal temperature range.

| | **Disk F** | **Disk R** |
|---|---|---|
| $D$ | $270\,mm$ | $250\,mm$ |

Table 4.1: Optimal disk diameters

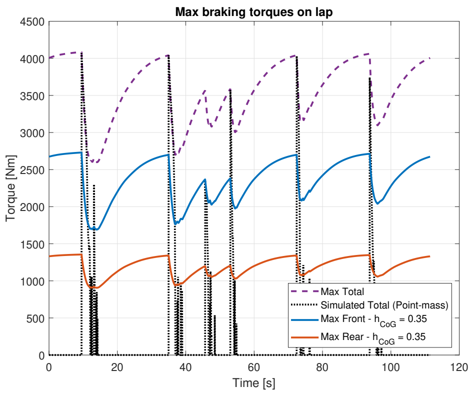



Figure 4.1: Braking torque per lap

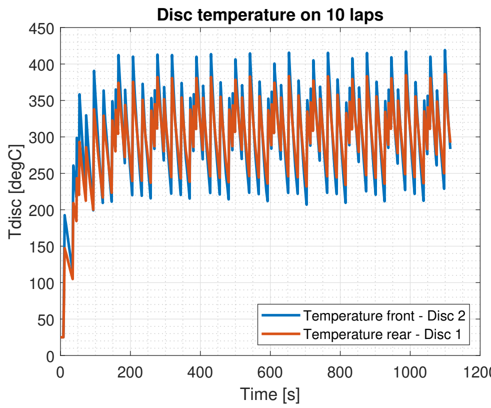



Figure 4.2: Temperature profiles

## 4.2 Brake system sizing

The objective now is to properly design the braking system in order to obtain the braking distribution (between front and rear) which corresponds to the best performance, which is the one closest to the optimal distribution but ensuring a stable maneuver at every experienced velocity. At the same time some constraints need to be respected: maximum pressure in braking system of 120 $bar$ (both front and rear), maximum braking force at the pedal of 785 $N$ (80 $kg$) due to the brake booster absence.

The possible variables and the related ranges/steps are:

- Disks diameter: fixed by previous step;
- Master cylinder diameters: from 17 $mm$ to 20 $mm$ with 1 $mm$ step for both front and rear;
- Number of pistons: 4 or 6 at the front axle, 2 or 4 at the rear one;
- Pistons diameter: 6 combinations for 6 and 4 pitons configurations, 3 combination for 2 pistons configuration;
- Balance bar: from 40 % to 60 % with 5 % step.

After some attempts, the following optimal configuration has been obtained.

| | D mast. cyl. | Pistons | Balance Bar | Pressure | Brake Balance |
|---|---|---|---|---|---|
| **Front** | 18 mm | 2x28 mm + 2x30 mm + 2x32 mm | 55 % | 63 bar | 66.86 % |
| **Rear** | 17 mm | 4x28 mm | 45 % | 58 bar | 33.14 % |

Table 4.2: Selected parameters

$F_{pedal} = 732\ N\ (75\ kg)$

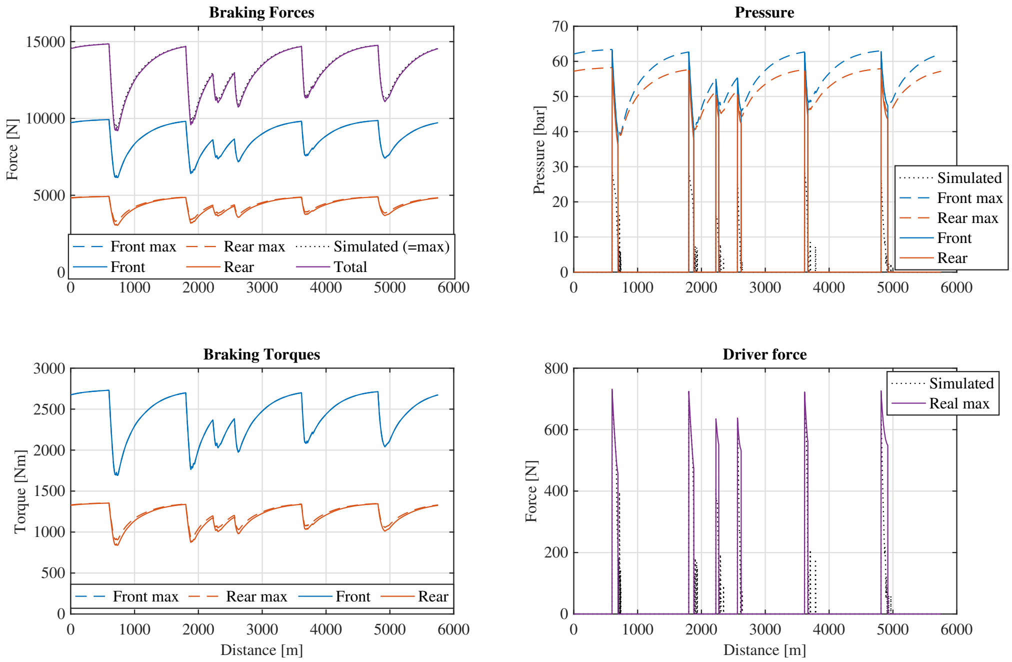



Figure 4.3: Forces, torques and pressures

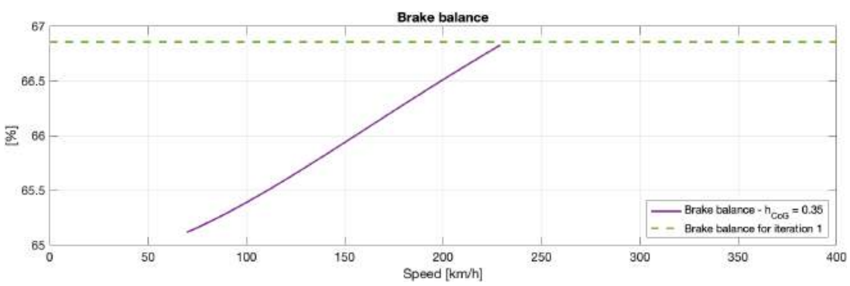



Figure 4.4: Brake Balance

As expected, the braking maneuver is always stable, meaning that the brake balance (which is fixed) is always grater (more towards the front axle) than the optimal one, which increases with speed instead. Consequently, the optimal balance is well approximated for higher speeds with respect to slower track portions, as confermed by the diagrams of (Fig. 4.3).

## 4.3 KERS implementation

For this last point, two different situations has been faced: a KERS (electric generator) braking force at rear axle only of 500 $Nm$ and 700 $Nm$ (constant torques) were introduced in the analysis. With the same reasoning, new setups has been found.

| <u>500 Nm</u> | D mast. cyl. | Pistons | Balance Bar | Pressure | Brake Balance |
|---|---|---|---|---|---|
| **Front** | 17 mm | 2x30 mm + 4x32 mm | 55 % | 58 bar | 80.87 % |
| **Rear** | 18 mm | 2x32 mm | 45 % | 42 bar | 19.13 % |

Table 4.3: Selected parameters

$F_{pedal} = 599\ N\ (61\ kg)$. As expected, the balance is strongly moved to the front.

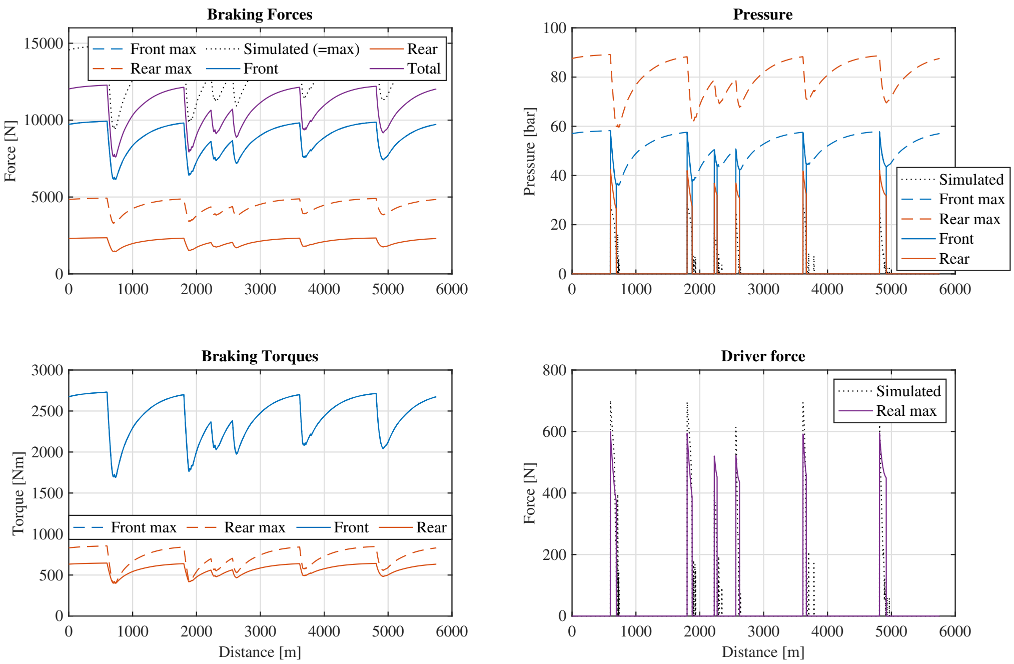



Figure 4.5: Forces, torques and pressures

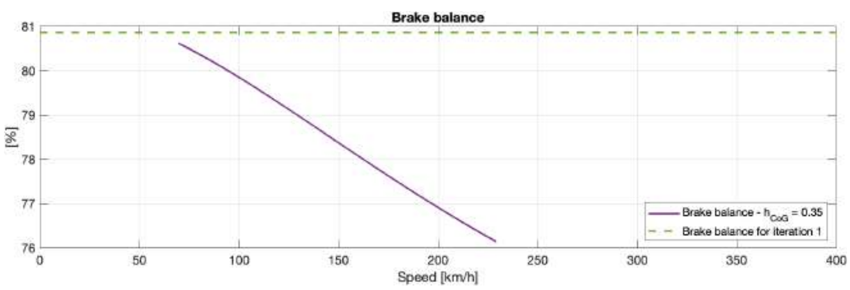



Figure 4.6: Brake Balance

In this case, the "max" values refer to "brakes + KERS", while the actual values to brakes only, to better highlight the difference with respect to the previous case (at the rear axle in particular).

For the 700 $Nm$ case instead:

| 700 Nm | D mast. cyl. | Pistons | Balance Bar | Pressure | Brake Balance |
|---|---|---|---|---|---|
| **Front** | 17 mm | 2x30 mm + 4x32 mm | 60 % | 58 bar | 89.32 % |
| **Rear** | 20 mm | 2x28 mm | 40 % | 28 bar | 10.68 % |

Table 4.4: Selected parameters

$F_{pedal} = 549\ N\ (56\ kg)$. In this case, the setup corresponds to the most front-oriented balance possible within the provided setups.

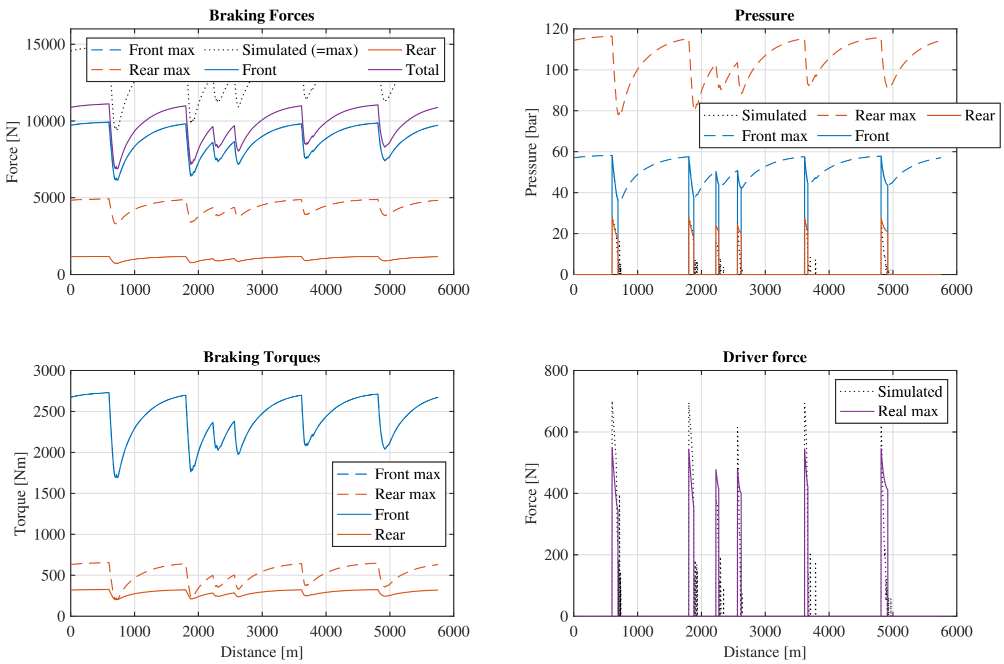



Figure 4.7: Forces, torques and pressures

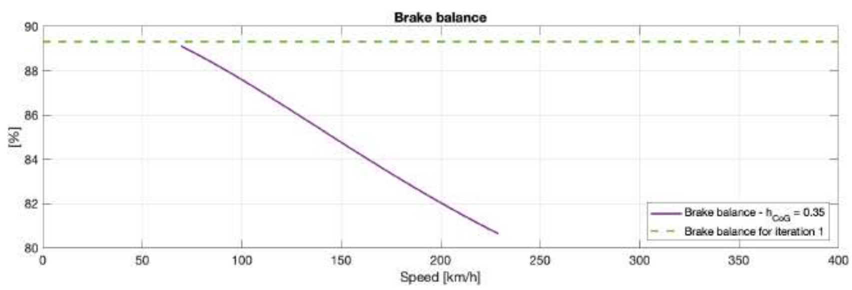



Figure 4.8: Brake Balance

# 5 Lab.04 - Ride height and lift and coast simulation

In this laboratory the aim was to analyze the effect of the static ride heights and spring stiffness on aerodynamics, and also the effect of the "lift and coast" technique, used to save fuel, on laptime. Both analyses were run using a modified version of OpenLAP, suited for each problem, using the *Tatuus FA010* formula car in Monza racetrack.

## 5.1 Ride height simulation

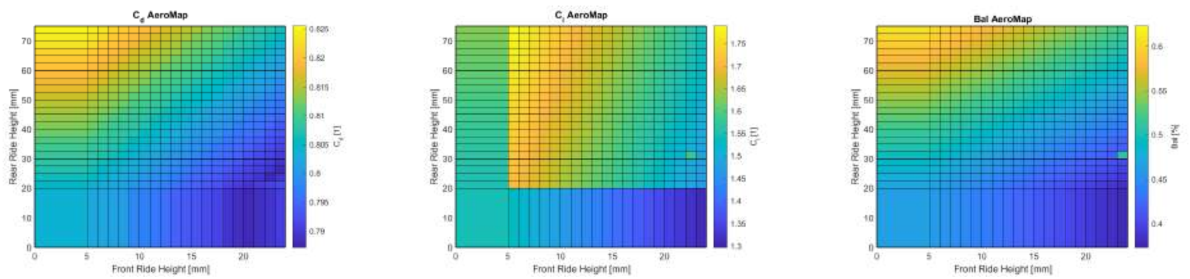
Aerodynamics is greatly influenced by the ride height of the vehicle. For example: the diffuser generates different downforce values depending on how far from the ground it is, the rear wing as well will produce different levels of downforce depending on how the flow moves around the car.
Therefore choosing the correct ride height is crucial to extract the best performance out of a car. The parameters on which engineers can act upon are the static ride height, which is the height of the car when static, and the suspension springs' stiffness for each axle. As the car reaches high longitudinal speed the downforce will push the car towards the ground, changing its ride height, therefore is important to choose the correct stiffness to maintain the desired car behavior and to prevent the car to touch the ground.

The spring stiffnesses that could be selected in our analysis, both for front and rear suspension are:

| $S_1$ $[N/mm]$ | $S_2$ $[N/mm]$ | $S_3$ $[N/mm]$ | $S_4$ $[N/mm]$ |
|---|---|---|---|
| 206 | 264 | 294 | 324 |

Table 5.1: Stiffnesses

The aeromaps showed in (Fig. 5.1) were provided together with the vehicle. They show the aerodynamic coefficients for drag and lift, and the aerodynamic balance in function of the ride height of the vehicle.




(a) $C_d$ (b) $C_l$ (c) Balance

Figure 5.1: Aerodynamic maps

### 5.1.1 Functions

The aerodynamic coefficients depend on the ride height, which in turn depends on the spring stiffness combined with the produced downforce, which depends on speed (and on the coefficients themselves): the aerodynamic coefficients are function of the speed. To grasp the evolution of the aerodynamic coefficients along a lap, functions depending on speed of $C_d$, $C_l$ and $Bal$ need to be computed. The adopted workflow to compute these functions is now described.

Starting from a chosen combination of static ride height front $SRH_f \in [5, 24]\,mm$ and rear $SRH_r \in [20, 75]\,mm$, and $0\,km/h$ of initial speed, a recursive algorithm was developed to compute $C_d(v)$, $C_l(v)$ and $Bal(v)$, increasing at each step the velocity, up to the maximum one achieved by the car on the selected circuit. Within the single step, the ride heights were exploited, thanks to the *Matlab* function `interp2`, to interpolate the coefficients from the available aeromaps. The computed coefficients were then exploited, together with the speed, to compute the downforce acting on each axle, needed to then compute, knowing the springs stiffnesses, the car squat towards the ground and consequently the new ride height. Note that at the beginning of each velocity step the step RHs are not know, being them dependend on the vertical load which, as already stated, depends on the lift coefficient which in turn depends on the RHs themselves. To face this issue, an initial guess for the coefficients is given through the previous step RHs and some iterations of the whole in-step computations are performed up to values stabilization. A check was finally introduced to ensure that the car didn't touch the ground at neither of the two axles. After having determined the values for $C_d(v)$, $C_l(v)$ and $Bal(v)$ for each integer value of $v$ between $0\,km/h$ and $230\,km/h$, their functions were saved as `interp1` to be then used in OpenLAP.

### 5.1.2 Preliminary considerations

To avoid performing too many simulations, some preliminary considerations need to be taken into account, which can be useful to correctly drive the analysis.
The first important remark regards the track type: Monza is a high speed circuit which requires a low-drag aerodynamic setup; the solution will probably be characterized by a not extreme rake angle, and a drop in $C_d$ may be desirable at the highest speeds (i.e. the lift and drag peaks should be reached before the maximum achievable speed). The second consideration comes from the following iso-chrone map.

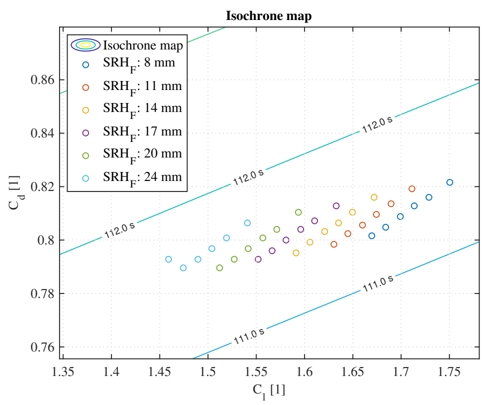



Figure 5.2: Iso-chrone map

The plotted dots are derived from the static ride heights (front $8-24\,mm$, rear $20-75\,mm$), thus they are not totally precise for our analysis but are helpful to understand some trends. In particular, it seems that the lap time is mostly influenced by the front height and less sensitive to the rear one (which spans a larger range): a low height for both axle seem to be generally advisable (within the solutions at same front RH, the worst are the ones with the highest rear RHs).

Some plots of the coefficients functions are now reported to analyze the general trends. Some extreme values combinations are exploited to highlights the different behaviors. In particular $SRH_f = [12,\ 24]\ mm$, $SRH_r = [25,\ 75]\ mm$, $s_f = [206,\ 324]\ N/mm$ and $s_r = [206,\ 324]\ N/mm$ are combined to generate the diagrams.

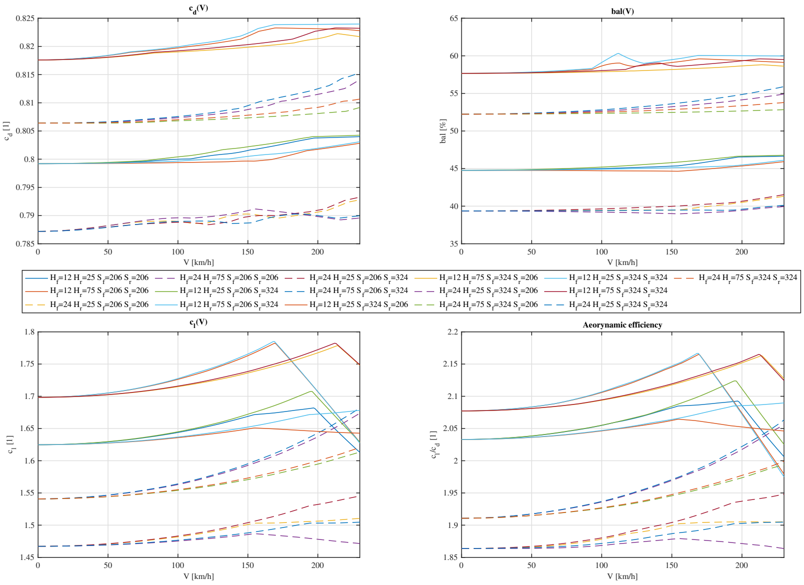



Figure 5.3: Coefficients vs Speed

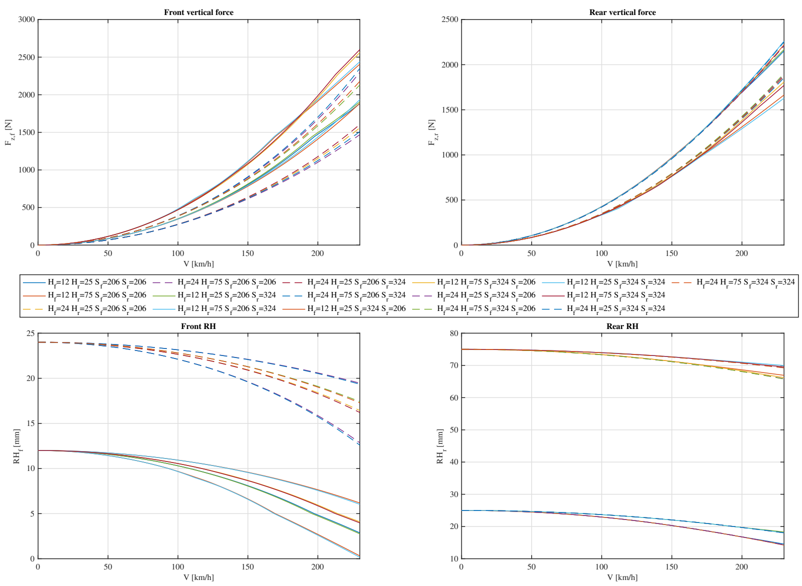



Figure 5.4: Forces vs Speed

Low $C_d$ is produced with small rear and large front heights; it is slightly affected by the stiffness (low drag with stiff front and soft rear) and strongly by the static heights (drag increases with this order: high front + low rear, low front + low rear, high front + high rear, low front + high rear).
The configurations that produce high drag also generate a large lift coefficient (and vice versa), with a

drop at high speed when the front height goes below 5 $mm$ or the rear one goes below 20 $mm$.
The aerodynamic efficiency follows the same trend as the lift, while the balance is moved towards the front as the drag increases.
An interesting observation is that the configurations with high front + high rear produce a larger $C_d$ with respect to low front + low rear, but also a lower $C_l$: in this case the increased drag is not payed back by an improvement of the downforce and the efficiency reduces. The aerodynamic vertical forces follow from the lift coefficients and balances.
Going back to the solution of the iso-chrone map, a both low heights solution is able to produce relatively low drag with a relatively high downforce, also characterized by a reduction of the lift at high speed and a ceiling of the drag coefficient; the balance is moved to the rear, important for the acceleration phase, the efficency is pretty high and the vertical forces are the highest for the rear axle.
It still seems to be a reasonable solution.

### 5.1.3 Simulations

The different functions were then saved and given to the `OpenLAP` algorithm to compute the final lap-times. Remembering that the cases which would have resulted in touching the ground have been discarded, a first rough series of simulations is performed to investigate the general behaviors; in particular, the feasible combinations between $SRH_f = [10,\ 24]\ mm$, $SRH_r = [25,\ 50,\ 75]\ mm$ and all the stiffensses are tested and the best combination (in terms of lap times) is reported in (Tab. 5.2).

| $S_f\ [N/mm]$ | $S_r\ [N/mm]$ | $SRH_f\ [mm]$ | $SRH_r\ [mm]$ | **Lap Time [s]** |
|---|---|---|---|---|
| 206 | 264 | 10 | 25 | 111.303 |

Table 5.2: First result

Again, a low set-up for both front and rear axles is suggested.
Keeping this in mind, the low-low region is further investigated, resulting in the following lap times, shown in (Fig. 5.5) and (Tab. 5.3). The heights were limited to: $SRH_f = [6,\ 8]\ mm$, $SRH_r = [20,\ 22.5]\ mm$.

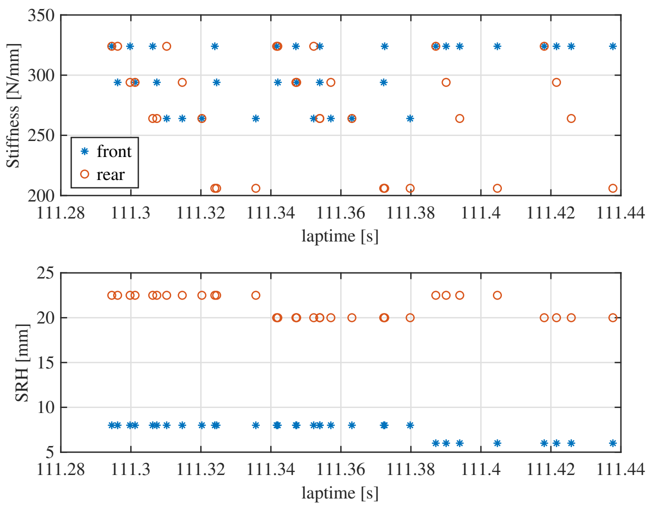



Figure 5.5: Final results

| $S_f$ $[N/mm]$ | $S_r$ $[N/mm]$ | $SRH_f$ $[mm]$ | $SRH_r$ $[mm]$ | **Lap Time [s]** |
|---|---|---|---|---|
| 324 | 324 | 8 | 22.5 | 111.295 |
| 294 | 324 | 8 | 22.5 | 111.296 |
| 324 | 294 | 8 | 22.5 | 111.300 |
| 294 | 294 | 8 | 22.5 | 111.301 |
| 324 | 264 | 8 | 22.5 | 111.306 |
| 294 | 264 | 8 | 22.5 | 111.307 |
| 264 | 324 | 8 | 22.5 | 111.310 |
| 264 | 294 | 8 | 22.5 | 111.315 |
| 264 | 264 | 8 | 22.5 | 111.320 |
| 324 | 206 | 8 | 22.5 | 111.324 |
| 294 | 206 | 8 | 22.5 | 111.324 |
| 264 | 206 | 8 | 22.5 | 111.336 |
| 324 | 324 | 8 | 20 | 111.342 |
| 294 | 324 | 8 | 20 | 111.342 |
| 324 | 294 | 8 | 20 | 111.347 |
| 294 | 294 | 8 | 20 | 111.347 |
| 264 | 324 | 8 | 20 | 111.352 |
| 294 | 264 | 8 | 20 | 111.354 |
| 324 | 264 | 8 | 20 | 111.354 |
| 264 | 294 | 8 | 20 | 111.357 |
| 264 | 264 | 8 | 20 | 111.363 |
| 294 | 206 | 8 | 20 | 111.372 |
| 324 | 206 | 8 | 20 | 111.373 |
| 264 | 206 | 8 | 20 | 111.380 |
| 324 | 324 | 6 | 22.5 | 111.387 |
| 324 | 294 | 6 | 22.5 | 111.390 |
| 324 | 264 | 6 | 22.5 | 111.394 |
| 324 | 206 | 6 | 22.5 | 111.405 |
| 324 | 324 | 6 | 20 | 111.418 |
| 324 | 294 | 6 | 20 | 111.422 |
| 324 | 264 | 6 | 20 | 111.426 |
| 324 | 206 | 6 | 20 | 111.438 |

Table 5.3: Final results

Finally, a low configuration with stiff springs seems to be the best one. Soft springs are not suitable for these static heights because they result in the car hitting the ground. Too low *SRH* are not the best solutions, as also going too close to the ground means loosing performance. The final solution functions are plotted below. As explained before, the most influencing parameters are the ride heights, particularly the front one, then the springs' stiffnesses have minor influences, but still the stiffer springs proved to be the optimal choice.

Between all the configurations tested, the best one is the one showing a good compromise between a relatively low drag configuration, but that is able to produce a very high lift respect to the others, therefore one with a high efficiency.

The best solution with $SRH_f = 8\,mm$, $SRH_r = 22.5\,mm$, $S_f = 324\,N/mm$ and $S_r = 324\,N/mm$ is presented.

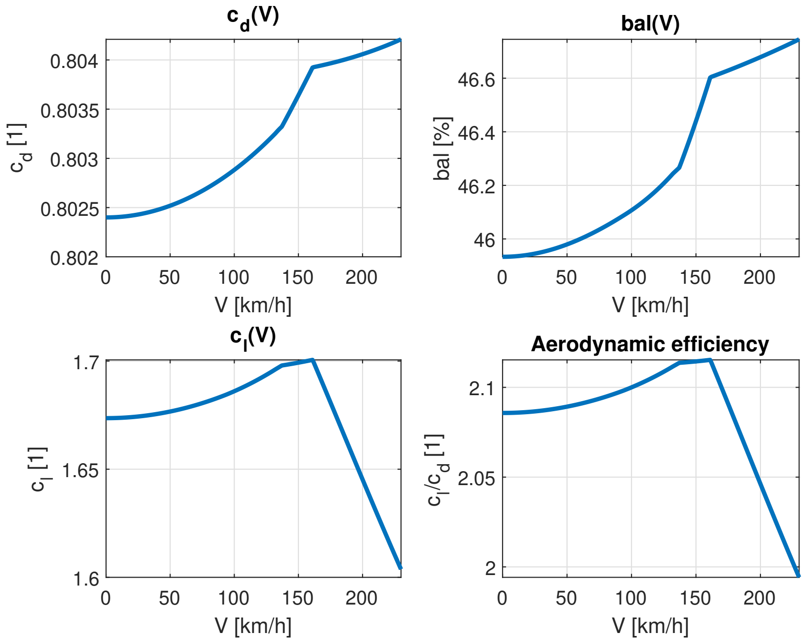



Figure 5.6: Coefficients vs Speed - Final

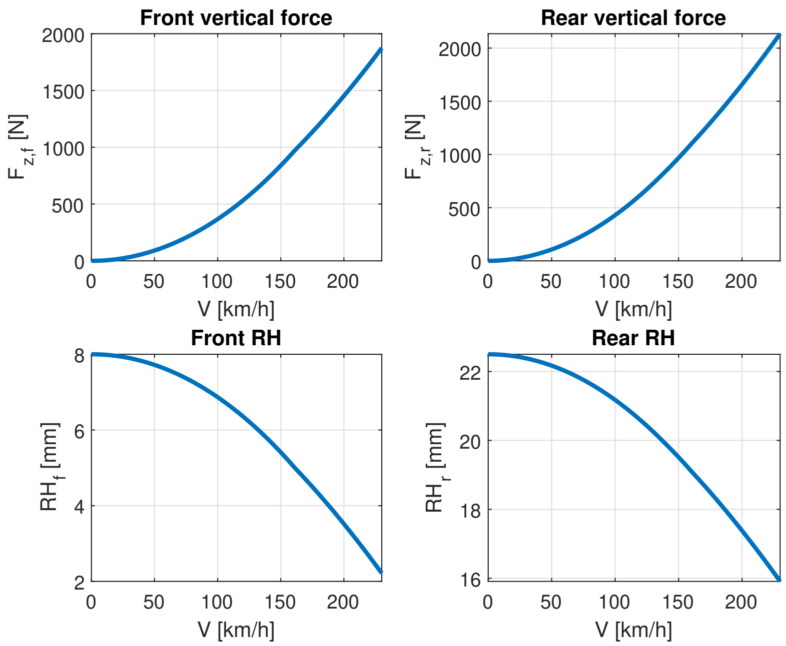
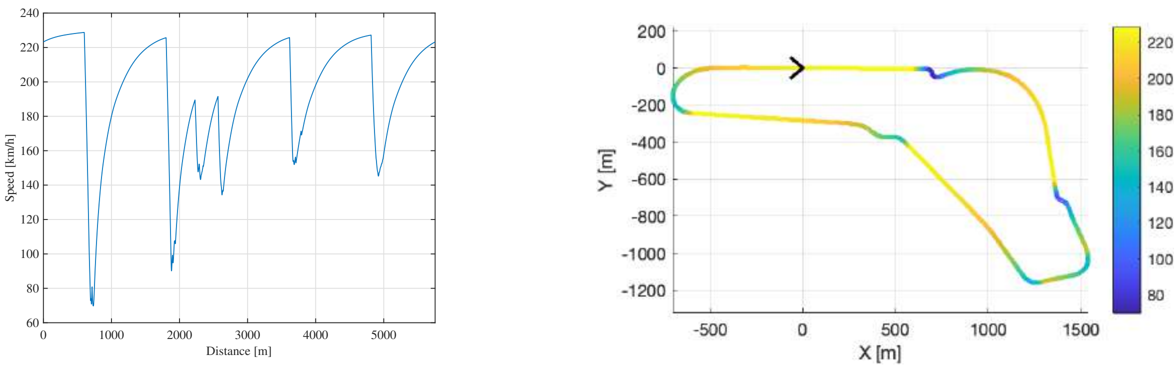



Figure 5.7: Forces vs Speed - Final

As expected, the peak lift is achieved before the maximum speed, at about 150 $km/h$ , the drag coefficient becomes almost constant at the same speed and the balance is moved towards the rear axle. Note that with this configuration, the turns *"Prima Variante"* and *"Seconda Variante"* are travelled at about 100 $km/h$, thus at high lift coefficients, while all the other turns are travelled at about 150 $km/h$, around the peak value. All the other regions are instead straights travelled at higher speeds, in which the drag coefficient ceiling is exploited and without any interest in the lift drop. The following representation shows the explained speed trends.




(a) (b)

Figure 5.8: Velocity profiles

## 5.2 Lift and coast simulations

The so called "lift and coast" is a technique aimed at saving fuel during a race or keeping under control too high temperatures or too large tyre wear, affecting as less as possible the laptime. It consists in releasing the accelerator pedal some meters before the actual best braking point, allowing the car to coast down until the braking point. In this way some fuel is not burned, at the expense of a slight laptime increase. It is therefore crucial to identify where this technique can be efficiently used to minimize the drawbacks. To study this phenomena two corners of the Monza racetrack were analyzed, namely:

- **Turn 1:** *"Prima variante"* a low speed corner after a very long, high speed, straight
- **Turn 6:** *"Lesmo 1"* a medium speed corner after a short, medium speed straight

### 5.2.1 Simulations

The simulation was carried out by creating a combination of braking strategies that the driver would implement to save fuel. The strategy consisted in lifting the accelerator $[0:25:100]\,m$ before the actual braking point in *"Prima variante"* and $[0:20:80]\,m$ in *"Lesmo 1"*. All possible combinations of lifting points in (Tab. 5.4) were studied to extract the more convenient one.

| **Strategy** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** | **12** | **13** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Turn 1** $[m]$ | 0 | 0 | 0 | 0 | 0 | 25 | 25 | 25 | 25 | 25 | 50 | 50 | 50 |
| **Turn 6** $[m]$ | 0 | 20 | 40 | 60 | 80 | 0 | 20 | 40 | 60 | 80 | 0 | 20 | 40 |

| **Strategy** | **14** | **15** | **16** | **17** | **18** | **19** | **20** | **21** | **22** | **23** | **24** | **25** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Turn 1** $[m]$ | 50 | 50 | 75 | 75 | 75 | 75 | 75 | 100 | 100 | 100 | 100 | 100 |
| **Turn 6** $[m]$ | 60 | 80 | 0 | 20 | 40 | 60 | 80 | 0 | 20 | 40 | 60 | 80 |

Table 5.4: Braking strategies

### 5.2.2 Instantaneous quantities

In (Fig. 5.9) are collected the most relevant quantities useful to study lift and coast strategies, the black line shows the ideal lap, without any lifting.

Already by analysing these graphs is possible to notice the effect of lifting before a corner. The top graph represents the speed profile along the circuit, which shows a drop before turn 1 and 6, as the throttle is released. Since at turn 1 the car is already near its maximum speed and it is accelerating slowly, releasing the accelerator before braking is reducing less the speed than in turn 6. Here releasing earlier means also cutting a part where the acceleration is high, so the impact on speed respect to the ideal lap is greater.
In the central graph the braking energy is presented. Following the reasoning made on the speed, the braking energy needed to slow down the car for turn 1 when lifting is similar to the ideal case, since there is not a great speed reduction, while in turn 6 the energy required is much lower due to the lower entry speed.
The latter graph shows the instantaneous fuel consumption, whose integration is the fuel consumed over one lap. It shows that, with the same meters of lift and coast, lifting in turn 6 is more effective than in turn 1 (if the sole objective is fuel saving, without any consideration for the increase in lap time), as the instantaneous consumption is higher before the turn due to the high acceleration.

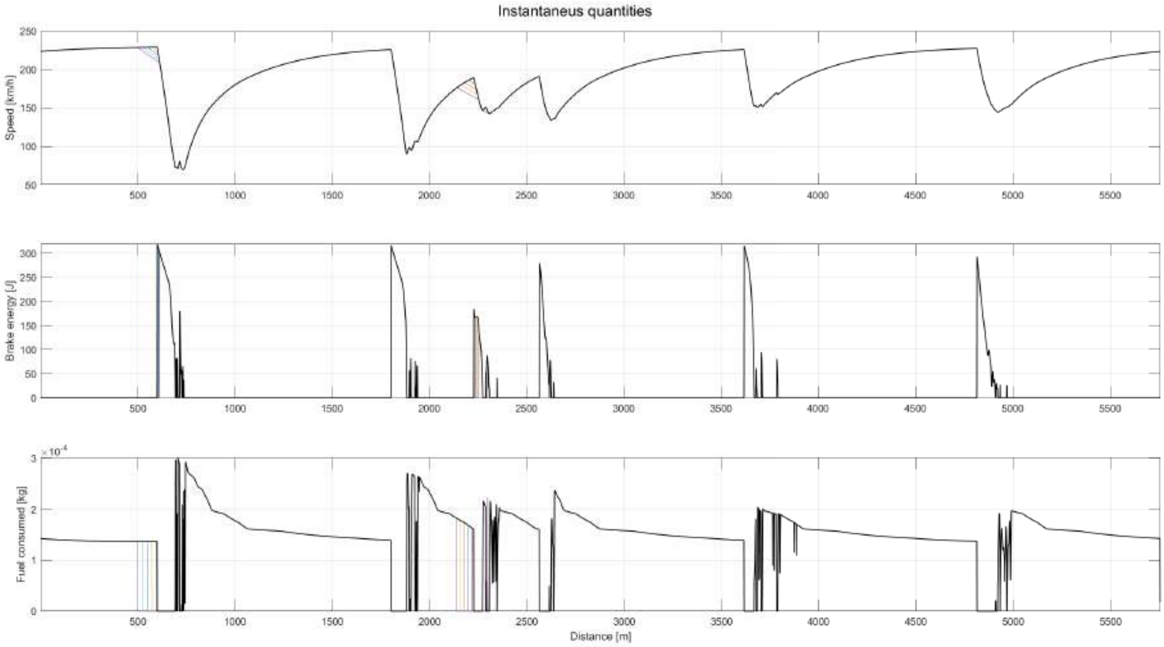



Figure 5.9: Instantaneous quantities

### 5.2.3 Total quantities

To compare the different strategies the total quantities of laptime, fuel and braking energy were plotted together, to see which strategy proved to be the best one. The results on the $y$ axis are plotted as a difference respect to the ideal lap, to show how the different strategy would change the performance respect to not lifting at all. The braking strategy numbering in (Fig. 5.10) corresponds to the one in (Tab. 5.4).

As a comparison method it was chosen to select strategies that led to the same quantity of fuel saved, namely all the strategies with around $0.017\,kg$ of fuel saved per lap. For the same quantity of fuel saved, the strategy resulting in the smallest laptime increase is the number 17: the driver would have to lift the accelerator $75\,m$ before turn 1 and $20\,m$ before turn 6. Since the maximum liftoff distances imposed where $100\,m$ for turn 1 and $80\,m$ for turn 6, it means that to save some fuel, without losing too much laptime the driver shout lift more in turn 1, therefore is more convenient to lift and coast in turn 1 rather than in turn 6.

| **Strategy** | **Fuel** $[kg]$ | **Laptime** $[s]$ | **Brake energy** $[J]$ |
|---|---|---|---|
| **5** | 0.0176 | 0.186 | $1.131 \cdot 10^4$ |
| **9** | 0.0174 | 0.115 | $1.128 \cdot 10^4$ |
| **13** | 0.0173 | 0.077 | $1.181 \cdot 10^4$ |
| **17** | 0.0173 | 0.071 | $1.162 \cdot 10^4$ |
| **21** | 0.0174 | 0.092 | $1.138 \cdot 10^4$ |

Table 5.5: Results

The convenience of lifting in turn 1 instead of turn 6 is also coherent with the theory: when carrying out a sensitivity analysis on laptime, the difference in laptime between a two laps is given by (Eq. 5.1).
The higher the velocity $v$ at which lifting is carried out, the smaller the laptime difference $\Delta t$ will be, as the velocity difference $\Delta v$ due to lifting will be divided by a greater number.

$$\Delta t \approx -\int_0^L \left(\frac{\Delta v}{v^2}\right) ds \tag{5.1}$$

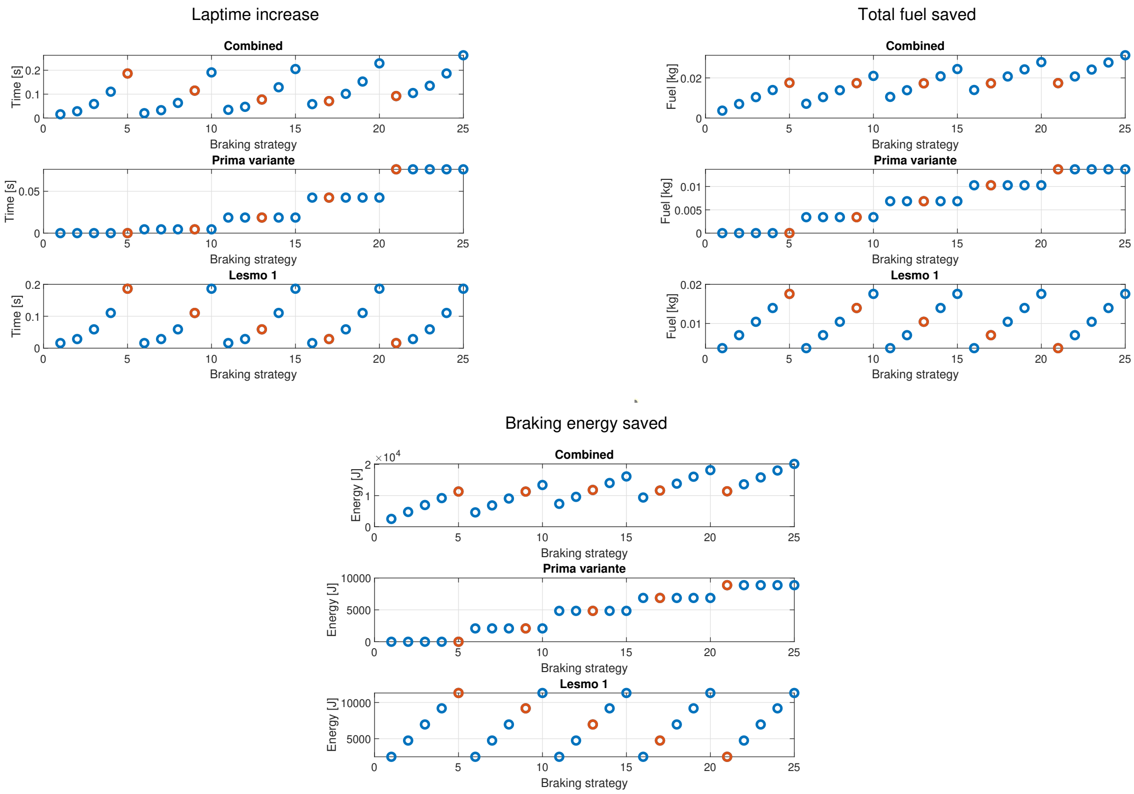



Figure 5.10: Total quantities

Considerations on the different quantities:

- **Laptime:** Confirming once again what described before: even if the lifting distance is greater in turn 1 the time lost is much smaller. Lifting $100\,m$ in turn 1 results in an increase of $0.076\,s$, while lifting $80\,m$ in turn 6 of $0.186\,s$
- **Fuel:** Lifting in turn 6 allows to save more fuel as the driver is effectively cutting down on the throttle in a place where the acceleration is high (and so the fuel consumption) saving around $0.019\,kg$ of fuel, whereas in turn 1 there is almost no acceleration and fuel consumption is relatively lower, saving only around $0.015\,kg$.
- **Brake energy:** Once again since in turn 6 the acceleration is cut before the turn, the entry speed is much lower than normal, therefore less braking is required, saving $11306\,J$, which much more braking energy than in turn 1, where this speed difference is less evident, where you save just $8869\,J$.

### 5.2.4 Lift and coast in long runs

When deciding the fuel strategy for a long race with a fixed distance (namely 40 laps), it is important to decide whether to carry more fuel at the start but racing without concerns on fuel management, or start with less fuel and then manage it along the race by doing lift and coast, with the advantage of carrying less mass during the whole race.
Just relying on what observed in this analysis, without any consideration on possible controllable and uncontrollable conditions that may happen during a real race, and remembering what commented in (Sec. 3.1), the general trend suggests that the laptime increases linearly with the mass (as a heavier vehicle will be harder to accelerate and to handle), and quadratically (or similar) while lifting (Fig. 5.10). This means that starting with 1 $kg$ of fuel more on board causes a linear increase of the laptime, while starting with 1 $kg$ less but saving it through lift and coast causes a linear decrease (less mass on board) added to a more than linear increase. For instance, on this track, a $\Delta m = 1\ kg$ resulted in about $\Delta t = 0.04\ s$ in (Sec. 3.1), while for the best strategy here, a fuel saving through lift and coast of 1 $kg$ corresponds to about $\Delta t = 4\ s$.

A deeper sensitivity analysis could be carried out, comparing simulations in which the vehicle mass decreases as the laps pass by, to understand which strategy is better. But, by just looking at the trends, is probably more convenient to start with enough fuel to complete the race without managing.

Moreover, starting with a full fuel load allows the driver to be more flexible and push when needed, allowing for more aggressive strategies if needed and less concerns to the driver himself. On the contrary, a driver forced to lift and coast for the entire race needs to be less aggressive and more conservative, with possible problems especially with overtakes, but can be favored in case of a safety car, where he can save a lot of fuel without laptime problems.

A benefit of starting with less fuel can also be tyre consumption, as a heavier car will slide more and wear the tyres faster, but once again this benefit is more evident only at the first laps of a race and decreases while the race approaches the end and the strategies converge, when the overall difference in mass is not such to create an important advantage.

### 5.2.5 Model's assumptions

Many simplifications are adopted while running the `OpenLAP` script.

The first strong assumption, already introduced in the previous laboratories, is the point mass model, which neglect all the considerations that should be carried especially when talking about mass: a different fuel consumption corresponds to different behaviours of the vehicle, in terms of both longitudinal and lateral dynamics; moreover, as the tank empties, the car centre of gravity moves, with obvious consequences on the car response. This means that a more accurate analysis may also consider different car dynamics corresponding to different strategies.

A second important assumption regards the different driving style that need to be adopted in the case of lift and coast. Lifting means loosing speed and loosing speed means having less kinetic energy to be dissipated, thus a different braking point. Lift and coast is a technique which could generate unwanted stress in the driver, which has to pay additional attention in a delicate phase such as braking, to start the lifting phase in the correct point and then find a new (delayed) optimal braking point: errors in evaluating the initial position for the procedure and the modified best braking point are factors that could generate even more laptime loss.

Another simplification could be the fact that, during the entire simulated single lap, the car mass is considered to be constant. This is in general not true, due to fuel and tyre consumption, and becomes relevant especially when analysing the lift and coast case: even if the $\Delta m$ is low, a precise simulation should consider the increasing of mass with respect to the ideal drive when releasing the throttle. Furthermore, still thinking about the mass evolution, a consideration should be made on the fact that there are two main reasons why a driver is forced to do the lifting: either he previously consumed too much fuel thus after lifting he'll be lighter and his performance will be better (effect that partially counteracts the increasing in laptime), or the car shows other types of problems (such as temperatures), thus the mass is the same as the ideal case, but, after a series of laps of lift and coast, will be larger than the ideal one and the performance will be even worse.

# 6 Lab.05 - Genetic Algorithm

The Genetic Algorithms are important tools for solving optimization problems of complex systems, such as a vehicle during a race.

They are based on the Darwin's *Theroy of Evolution* and start with an initial population of individuals, where each individual is composed by genes that correspond to a value of each optimization variable, genes reproduce themselves by mixing up their genes (crossover), to obtain a new set of individuals. Other steps can be introduce to promote sparsity in the solutions (such as mutation) or convergence (such as elitism).

The main characteristic is that it relies on random improvement of the population, in which the quality of the single individual is evaluated through a fitness function. The intrinsic working principle of the GA remove the necessity to know and understand the system itself, which is an advantage because it simplifies very complex problems and a disadvantage because is reduce the capability of understanding the achieved solution. The system is thus treated as a black-box and the only elements of our interest are the input variables and the output fitness function. Moreover, the important advantage of the GA technics is the strong reduction in the number of simulations to be performed, being the optimization development mainly driven by the best individuals in the set.

Aim of this laboratory is therefore to properly select the Design Variables (DVs) of the 8 DOF model and the weights of the Fitness Function (FF), which is defined as a weighted sum of different normalized Key Performance Indicators (KPIs), in order to obtain a fast, reliable and controllable vehicle (depending on the required behaviour).

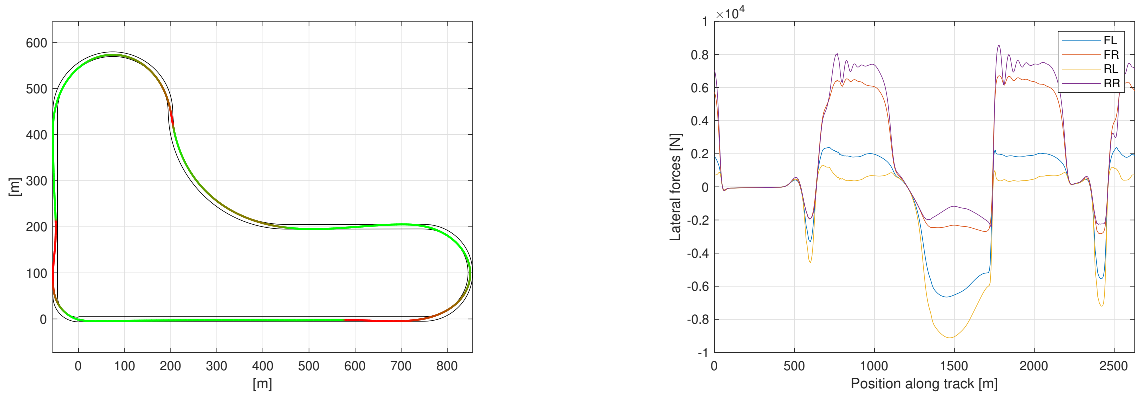
## 6.1 Reference solution

Being the lap time the most important parameter to be minimized for a race, a first simulation is run with only that as Optimization Function (OF). The solution will be analysed and corrected if necessary with the introduction of different KPIs and will be considered as the reference case in terms of best lap time. All the cases ere run with the following parameters: four variables have been chosen as DVs, namely the distance between the front axle and the COG ($L_f$), the roll stiffness distribution ($RSD$), the aerodynamic lift distribution ($ALD$) and the braking moment distribution ($BMD$); the chosen track is Track 1 (Fig. 6.1a); 20 individuals per generation and 40 generations max were adopted. The obtained solution is reported below.

| **Lf [m]** | **RSD [1]** | **ALD [1]** | **BMD [1]** | **Lap Time [s]** |
|---|---|---|---|---|
| 1.49 | 0.39 | 0.59 | 0.41 | 59.570 |

Table 6.1: Solution

Being the wheelbase $L = 2.74$ , $L_f$ corresponds to the 0.54 % of the total length, thus the COG is closer to the rear axle, promoting an oversteering behaviour. The same for the $RSD$ which causes the 61 % of the lateral load transfer to occur at the rear axle, again favouring oversteer. Also the aerodynamic balance is moved towards the front axle, enhancing the turning performance especially at high speed and the lateral slip of the rear tyres. The $BMD$ is moved towards the rear, again promoting oversteer when braking in turn (remember that no longitudinal load transfer is considered in the vehicle model).




(a) Track (b) Lateral forces

Figure 6.1

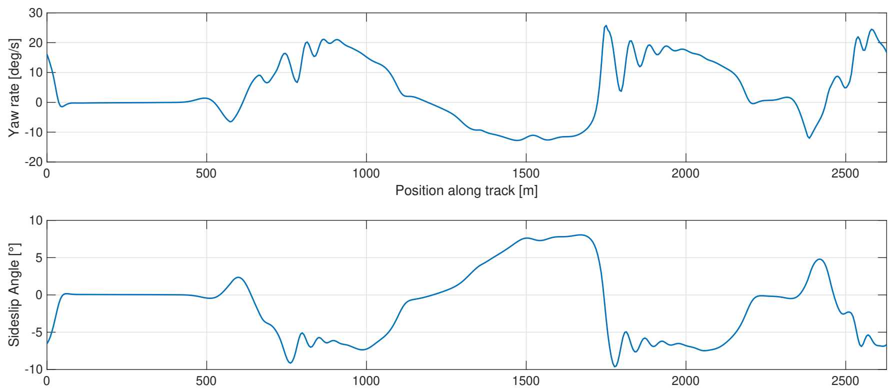



Figure 6.2: Car variables

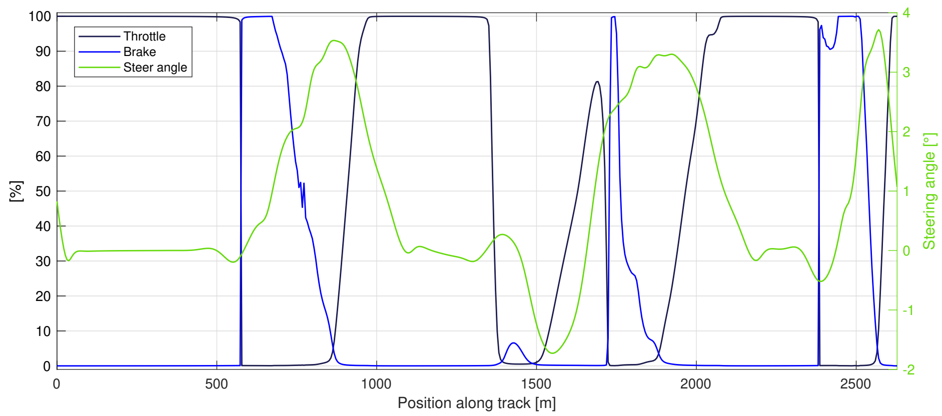



Figure 6.3: Driver demands

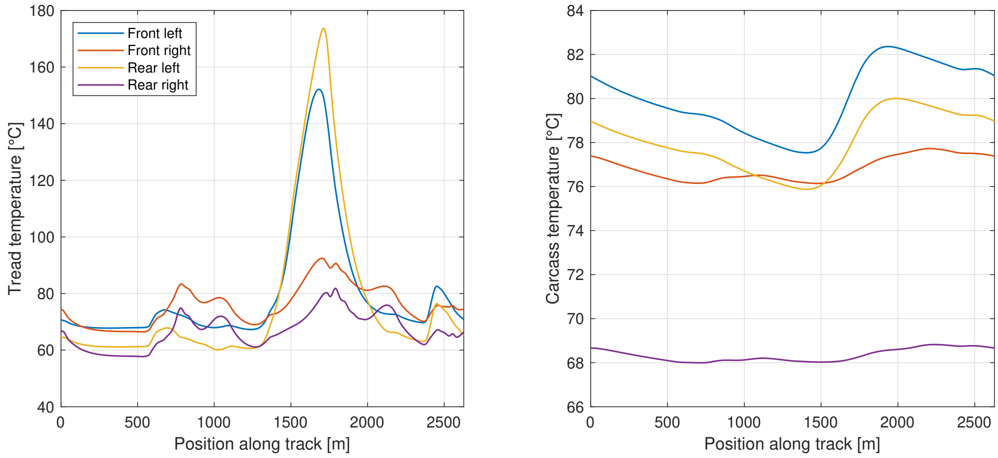



Figure 6.4: Tyres temperatures

Large lateral forces are produced especially in turn 2 (high speed right turn) and in particular in both the left (outer) tyres (Fig. 6.1b), which also show a high peak in the tread temperature, at almost 180 °$C$ and 150 °$C$. Driver demands (Fig. 6.3) appear to be reasonable, apart from some small portions of non-smooth trend of the braking pedal percentage. The peaks in the sideslip angle (Fig. 6.2) of almost $\pm 10°$ are not extremely detrimental, but, assuming a driver that doesn't like a too nervous car, one could try to reduce it in the most critical parts by reformulating the problem.

Let's now to see what happens if a KPI is added to control the maximum absolute value of the sideslip angle $\beta$.

## 6.2 Laptime 0.75, sideslip angle 0.25

A test is now performed trying to make the car more controllable: the `max(abs(`$\beta$`))` KPI is introduced into the FF, with a weight of 0.25. The weight on the laptime is consequently reduced to 0.75.
The resulting parameters are:

| Lf [m] | RSD [1] | ALD [1] | BMD [1] | Lap Time [s] |
|---|---|---|---|---|
| 1.51 | 0.37 | 0.52 | 0.40 | 59.595 |

Table 6.2: Solution

It can be observed here that some different directions are developed: all the variables are slightly moved towards the rear axle, being the *ALD* change the only one that promotes understeer. The introduced KPI here refers only to the peak absolute value of $\beta$ and the laptime still possess a great importance in the analysis: the reduction of the max $\beta$ value could be compensated by its increase in other sectors, with a final time loss of 25 $ms$, which may be acceptable.

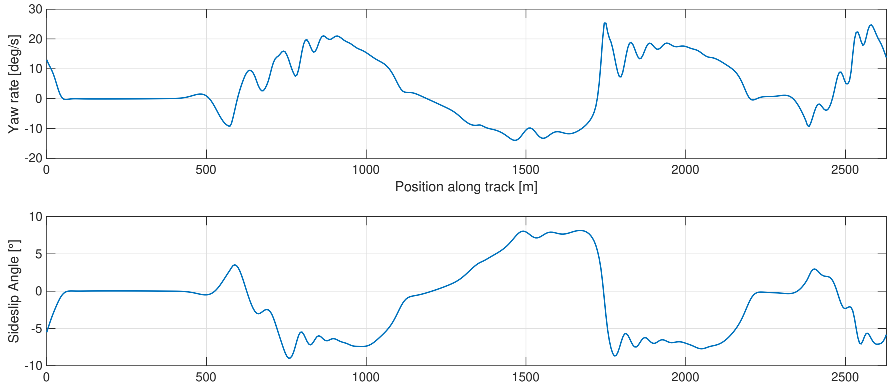
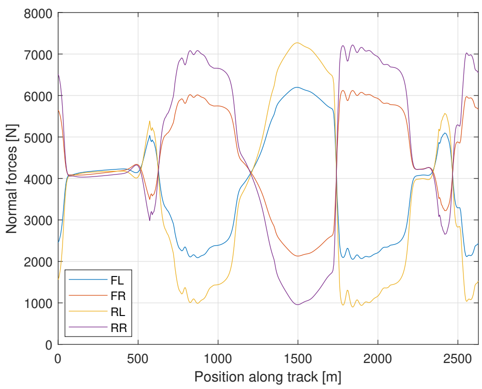


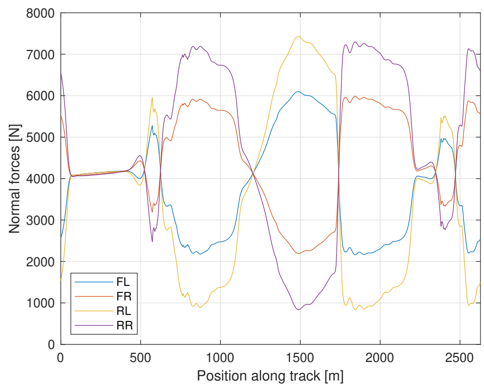

Figure 6.5: Car variables

The first comments regard the $\beta$ diagram, which successfully shows a reduction of the peak value from about $10°$ of the previous case to about $8.5°$ of the current one. Also note that an increasing can be observed in other points, such as in turn 1, just after 500 $m$, passing from about $2.5°$ to about $4.5°$. Finally, closer values of the peaks between each others are produced.

Let's now focus on the difference in normal forces (Fig. 6.6).




(a) Normal forces - $w_t = 1$

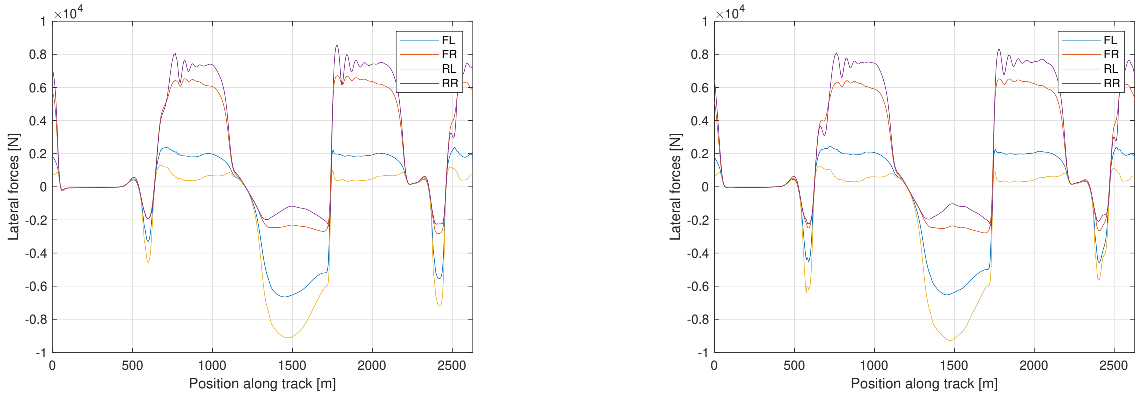



(b) Normal forces - $w_t = 0.75$, $w_\beta = 0.25$

Figure 6.6

In the second case, all the normal forces on the rear axle are larger with respect to the front one, result compatible with all the changes in $L_f$, $RSD$ and $ALD$.

Focusing on lateral forces instead:




(a) Lateral forces - $w_t = 1$

(b) Lateral forces - $w_t = 0.75$, $w_\beta = 0.25$

Figure 6.7

The main differences are highlighted while entering turn 1, where much larger forces are developed (with increasing of $\beta$) and while entering turn 4 (just before 2500 $m$), where instead smaller forces (and smaller $\beta$) are observed. Inside the turns, smaller peak forces are produced at the rear axle and larger ones at the front axle, especially in turn 3, at about 1750 $m$, where the peak of $\beta$ has been reduced the most.

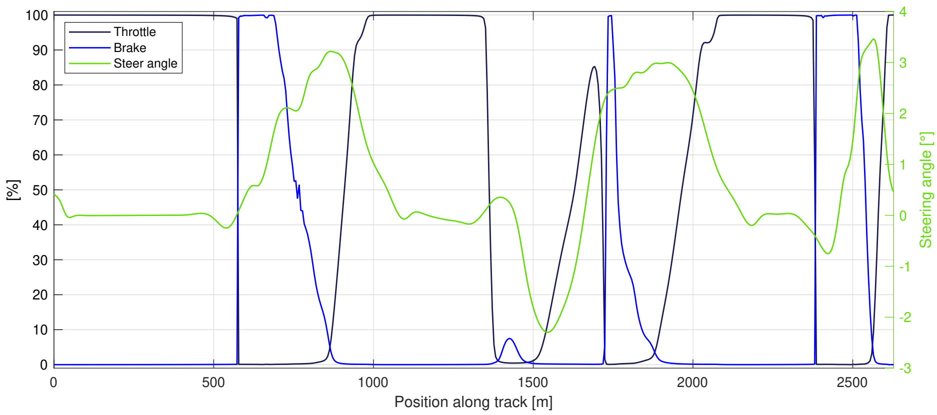



Figure 6.8: Driver demands

The driver inputs show a smoother trend for the braking action and a smaller steering angle in turn 1, 3 and 4, larger in turn 2 and entering turn 4, showing a stabler car.

Not evident differences are experienced in the temperatures. The engineer, together with the driver, should decide how much they are willing to lose in terms of laptime to obtain an advantage in stability, depending on driver's preferences and driving style.
Let's now investigate the effects of introducing a KPI on the maximum dissipated power.

## 6.3 Laptime 0.75, dissipated power 0.25

The $P_{diss}$ KPI is introduced into the FF, with a weight of 0.25. The weight on the laptime is consequently reduced to 0.75.
The resulting parameters are:

| Lf [m] | RSD [1] | ALD [1] | BMD [1] | Lap Time [s] |
|---|---|---|---|---|
| 1.52 | 0.39 | 0.59 | 0.43 | 59.658 |

Table 6.3: Solution

Compared to the reference solution, here the *COG* changes by 30 *cm* and the *BMD* by 2 %, both towards the rear axle. A more evident loss in laptime is produced, with a reduction of 88 *ms*.

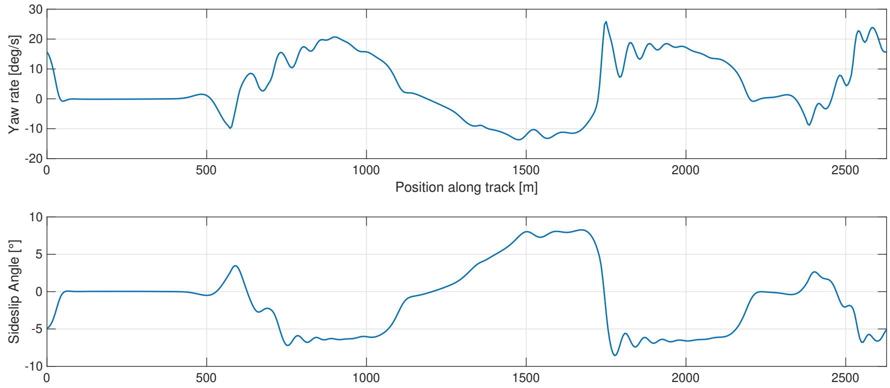



Figure 6.9: Car variables

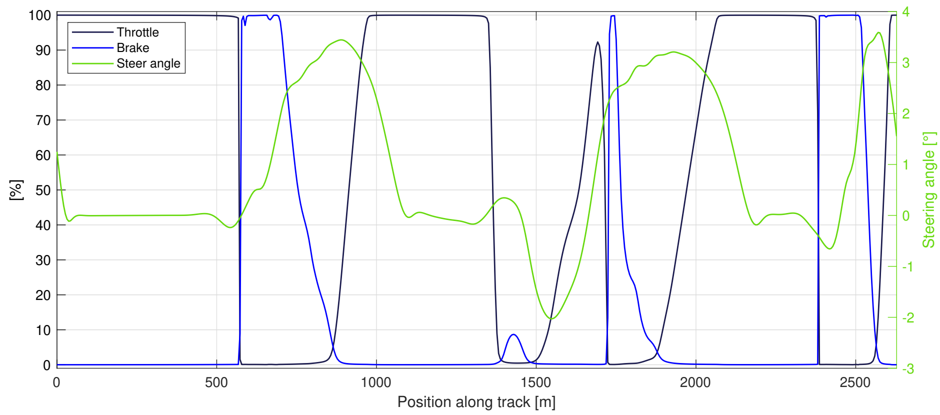
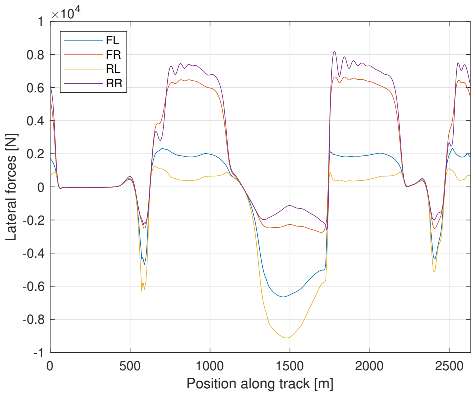
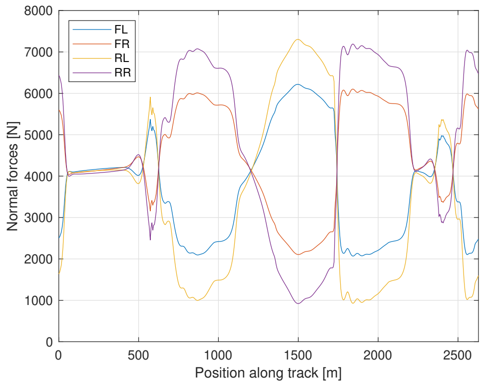



Figure 6.10: Driver demands




(a) Lateral forces




(b) Normal forces

Figure 6.11

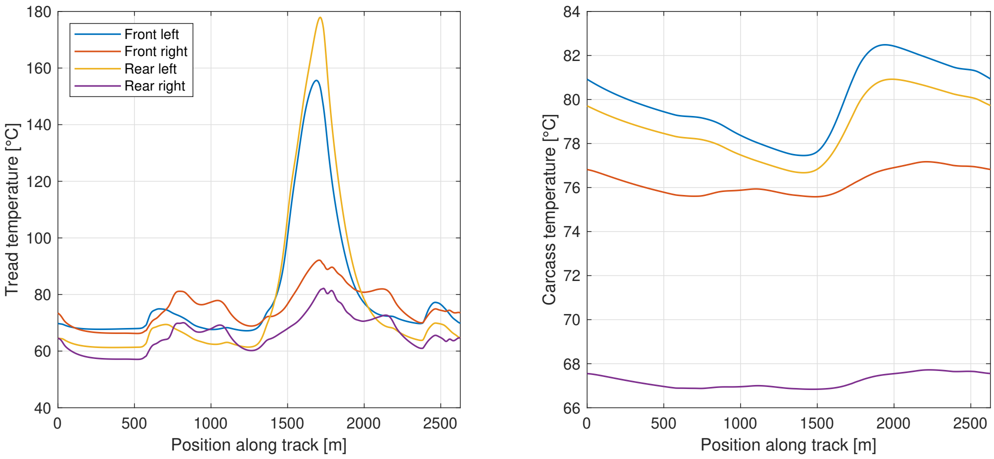



Figure 6.12: Tyres temperatures

A strong reduction of the $\beta$ values is observed, especially in turn 1, 3 and entering 4. The KPI on $P_{diss}$ seems to be more effective in reducing the $\beta$ values globally and not only at the peaks, with the drawback of a larger laptime loss. The steering angle is quite similar, with just a larger peak in turn 2. A smoother trend of the brake input is produced. Smaller lateral forces and similar normal ones are shown.
The tyre temperatures behaves in different ways depending on the considered wheel and on the peaks of interest, both for tread and carcass; in the main peak, outer tyres have larger temperatures, which is unadvisable.

The final simulations are performed with all the three KPIs together.

## 6.4 Laptime 0.8, sideslip angle 0.1, dissipated power 0.1

With this configuration, the following results have been obtained.

| Lf [m] | RSD [1] | ALD [1] | BMD [1] | Lap Time [s] |
|---|---|---|---|---|
| 1.49 | 0.39 | 0.53 | 0.42 | 59.599 |

Table 6.4: Solution

Little differences are observed here with respect to the reference case, mainly on the aero-balance which moves backwards by 6 %, with a small change (1 % frontwards) on the *BMD*. The laptime loss is 29 *ms*, similar to case 1.2.

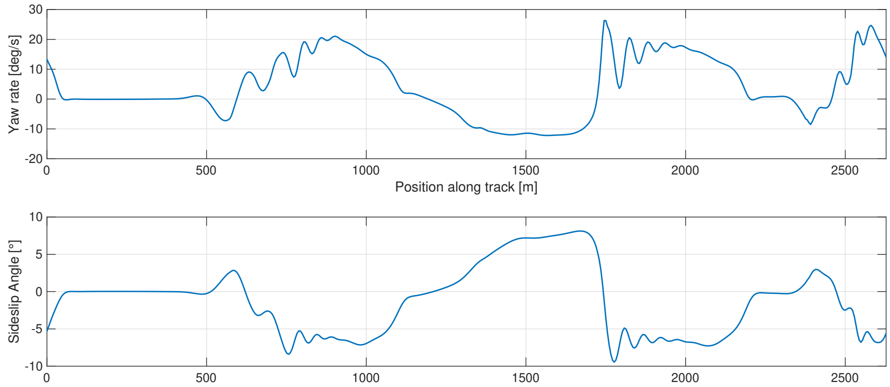



Figure 6.13: Car variables

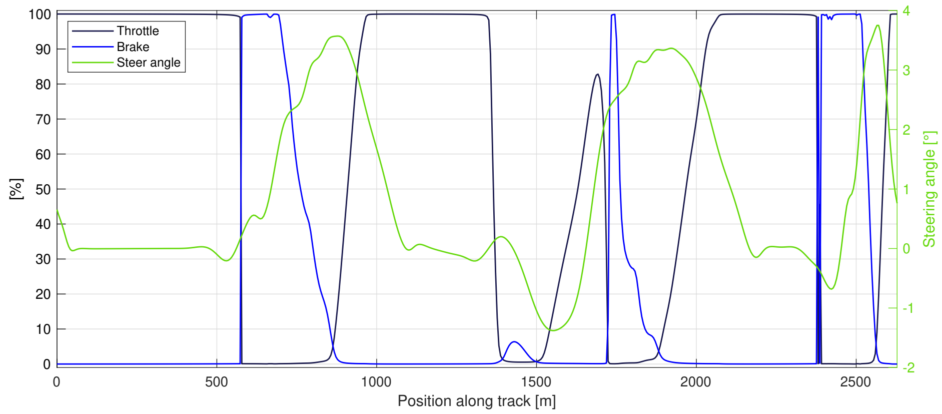
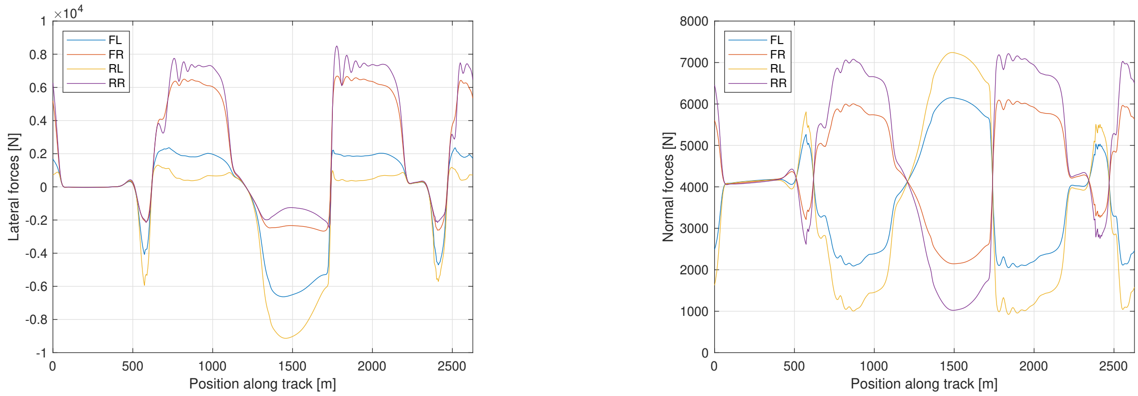



Figure 6.14: Driver demands




(a) Lateral forces

(b) Normal forces

Figure 6.15

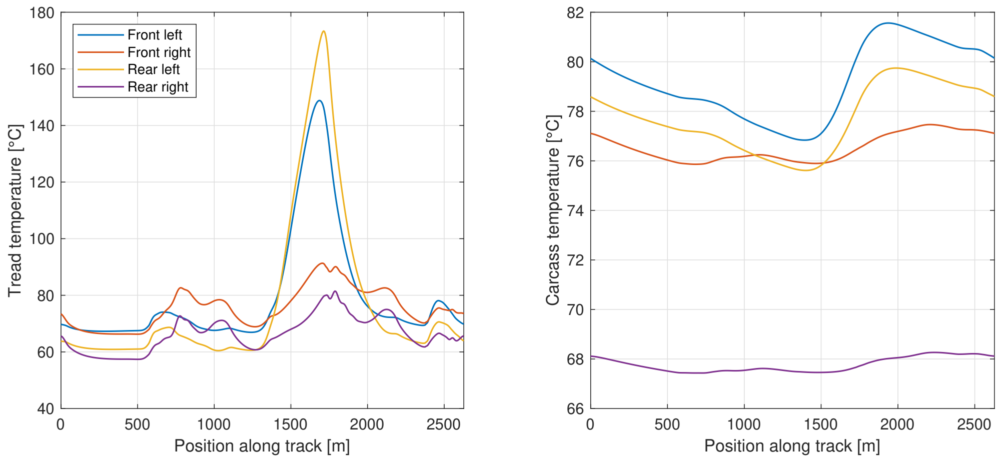
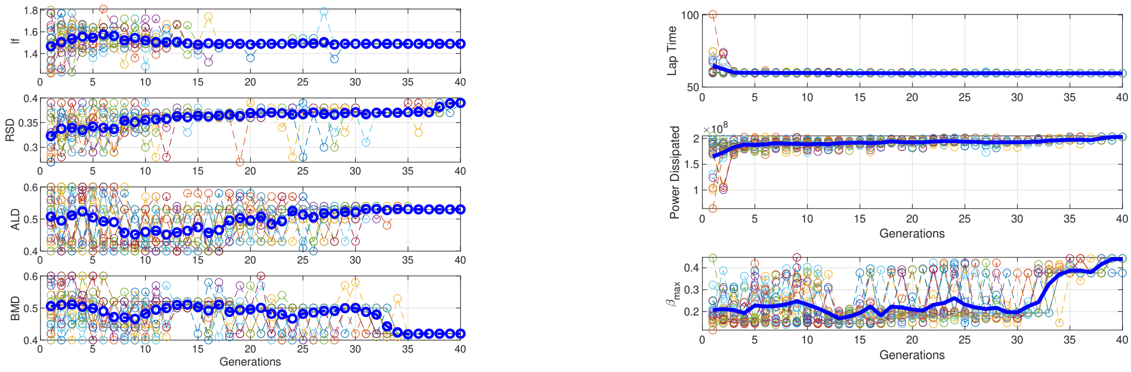



Figure 6.16: Tyres temperatures

The results are generally halfway between case 1.2 and 1.3. Compared to the reference case: slightly smaller reduction of $\beta_{max}$ which peaks at about $-9°$ with also a component of reduction also in the other zones and peaks; the driver inputs are slightly smoother in the brake pedal and with a generally slightly lower steering angle; normal forces are not affected by strong changes, apart from some reduction in front tyres and increasing in rear ones in the higher speeds portions (aero-balance); lateral forces are similar to case 1.2 (25 % on $\beta$). Temperatures show a general reduction, which is positive for our case; still the peaks can be detrimental for the tyre durability during a full race. This final result appears to be a good trade-off.

Stronger advantages in terms of controllability, temperatures, wear and so on may be reached with larger weights on the KPIs and possibly with the introduction of different KPIs in the FF, always at the expenses of the achieved laptime. The optimal solution is to be chosen also considering the final goal: for instance, a qualifying set-up optimization requires less attention in tyre decay and wear, but can be anyway affected but a too large temperature peak which would bring the tread out of the correct window with a consequent instantaneous grip reduction; on the countrary, for an entire race, durability, controllability and wear may assume prior importance with respect to the pure laptime itself, if this would cause a lower total race time and/or a better driver feeling.

As a reference, the convergence plots of this last case are reported in the following




(a) Variables convergence

(b) KPIs convergence

Figure 6.17

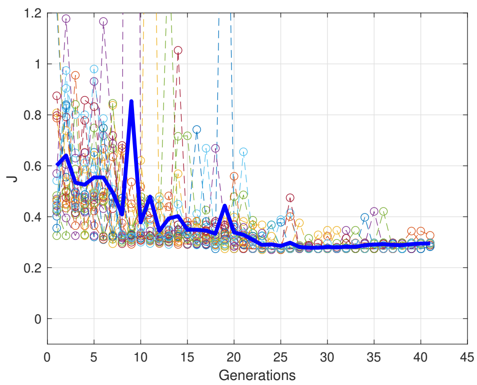



Figure 6.18: Fitness function convergence

## 6.5 Different track

As a test, the same setup is tried on a different track, namely an oval, to see if the resulting behaviour is still acceptable or if a new optimization would be required.

In general the results are not satisfactory. The plots showing the major issues to take into account are plotted here below.

As it can be seen, both yaw rate and side slip angle show many oscillations at each turn, with also pretty large peak value of $\beta$ (almost 10 °). At the same time, another concern regards the temperatures, which in this case are possibly too low instead of too high. A possible GA optimization may be run with larger attention to stability and temperature window.

The final laptime obtained from this simulation is 47.034 $s$

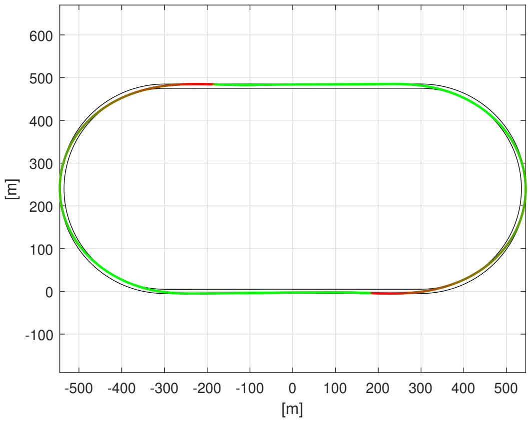



Figure 6.19: Track

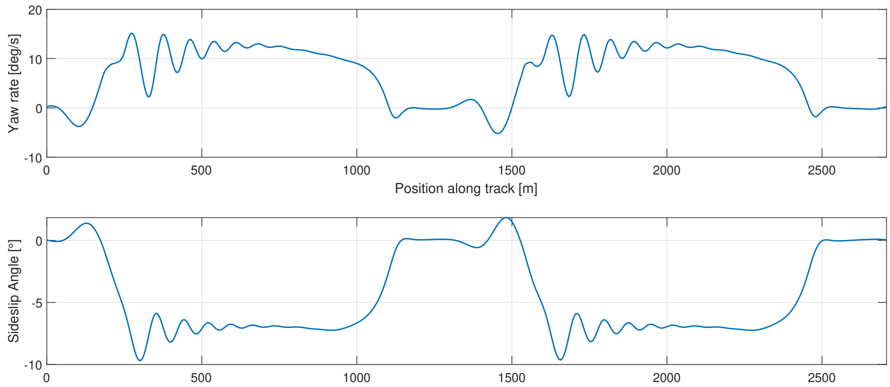



Figure 6.20: Car variables

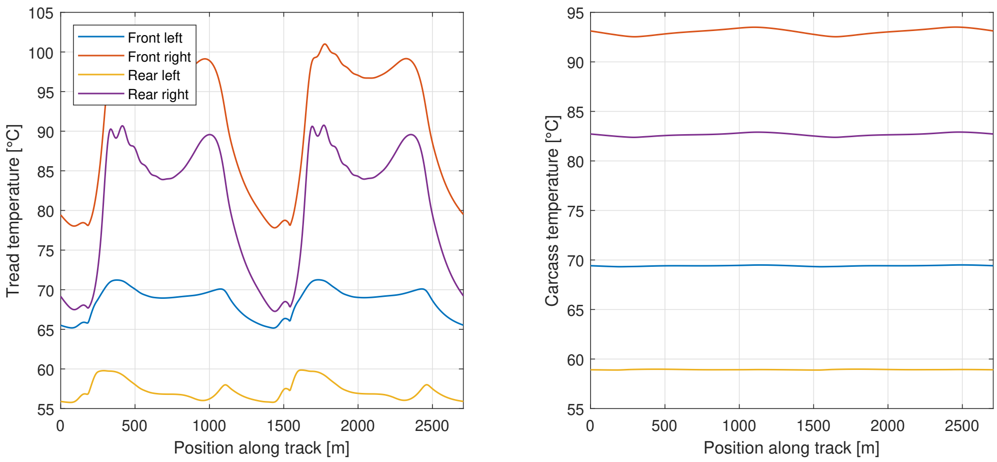



Figure 6.21: Tyres temperatures

# 7 Lab.06 - Acceleration optimization

## 7.1 Introduction

Aim of this project is the acceleration optimization for a car equipped with four independent electric motors, from 0 to 70 $m/s$. In particular, the torque split between front and rear is to be selected instant by instant, to achieve the best acceleration. The engines had limitations, which must be respected:

1. Maximum torque at each wheel: 800 $Nm$
2. Maximum power of each electric motor; 200 $kW$

The following data, regarding the vehicle, are given:

| **Mass** $[kg]$ | **Weight Dist.** $[\%]$ | **Wheelbase** $[m]$ | **COG Height** $[m]$ | **Wheel Rad.** $[m]$ |
|---|---|---|---|---|
| 800 | 45 | 3.7 | 0.35 | 0.28 |
| **Drag Area** $[m^2]$ | **Lift Area** $[m^2]$ | **Aero Bal.** $[\%]$ | **Static fr. coef.** $[1]$ | **Tyre Load Sens.** $[1/N]$ |
| 1 | 3 | 42 | 1.5 | -0.0003 |

Table 7.1: Vehicle data

## 7.2 Procedure

To solve the problem, the equilibria of the following external loads acting on the car must be computed.

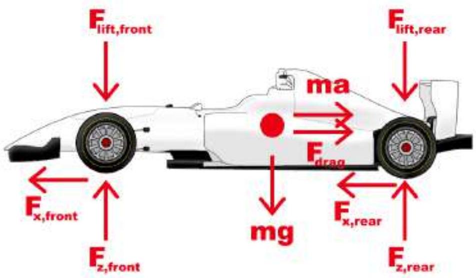



Figure 7.1: External loads

For acceleration optimization purposes, the $F_{x,front}$ and $F_{x,rear}$ need to be considered as the maximum available, thus, using the Coulomb friction representation:

$$\begin{cases} F_{x,front} = \mu_{front} F_{z,front} \\ F_{x,rear} = \mu_{rear} F_{z,rear} \end{cases}$$

The only exception will be when the maximum torque or power of the electric motors is reached; in that case, torque or power will be the limiting factors, therefore the longitudinal forces will be calculated starting from those quantities.

To solve the problem, the following system needs to be solved for each discrete velocity value, from 0 to 70 $m/s$. In particular, the chosen velocity step is 0.1 $m/s$.

$$\begin{cases} F_{z,front}+F_{z,rear}-F_{lift,front}-F_{lift,rear}-mg=0 \\ F_{x,front}+F_{x,rear}-ma-F_{drag}=0 \\ (F_{lift,front}-F_{z,front})l_f+(F_{z,rear}-F_{lift,rear})l_r-(F_{x,front}+F_{x,rear})h_{COG}=0 \\ F_{lift,front}=\frac{1}{2}\rho C_l S v^2 BAL_{aero} \\ F_{lift,rear}=\frac{1}{2}\rho C_l S v^2(1-BAL_{aero}) \\ F_{drag}=\frac{1}{2}\rho C_d S v^2 \\ F_{x,front}=\mu_{front}F_{z,front} \\ F_{x,rear}=\mu_{rear}F_{z,rear} \end{cases}$$

Where $\rho = 1.225\ kg/m^3$, and:

$$\begin{cases} l_f=l(1-WD) \\ l_r=l-l_f \end{cases} \qquad \begin{cases} F_{z,front,static}=mg\cdot WD \\ F_{z,rear,static}=mg-F_{z,front,static} \end{cases}$$

The friction coefficient is computed using the load sensitivity as:

$$\begin{cases} \mu_{front}=\mu_{front,static}+SENS\cdot\frac{(F_{z,front}-F_{z,front,static})}{2} \\ \mu_{rear}=\mu_{rear,static}+SENS\cdot\frac{(F_{z,rear}-F_{z,rear,static})}{2} \end{cases}$$

Note that the longitudinal force depends on the vertical load $F_z$ and on the friction coefficient $\mu$, which in turn depends on the vertical load itself, thus the last two equations of the system are of the second order with respect to the vertical load, which makes them not linear.
The solving procedure must be iterative and the first iteration for each velocity value uses the previous velocity step $\mu$ values to estimate the results, that are then iteratively corrected up to a variation between each result of two consecutive iterations smaller than $10^{-5}$ (stable values).

To implement the system of equations inside `Matlab`, it was written in matrix form as:

$$\begin{bmatrix} 1 & 1 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & -1 & -m \\ -l_f & l_r & -h_{COG} & -h_{COG} & l_f & -l_r & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ -\mu_f & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\mu_r & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} F_{z,front} \\ F_{z,rear} \\ F_{x,front} \\ F_{x,rear} \\ F_{l,front} \\ F_{l,rear} \\ F_d \\ a \end{bmatrix} = \begin{bmatrix} mg \\ 0 \\ 0 \\ \frac{1}{2}\rho C_l S v^2 BAL_{aero} \\ \frac{1}{2}\rho C_l S v^2(1-BAL_{aero}) \\ \frac{1}{2}\rho C_d S v^2 \\ 0 \\ 0 \end{bmatrix}$$

Where the matrix was called $A$, the vector of unknowns $x$ and the known term $b$.
The system was then solved as:

$$x = A^{-1}b$$

At each velocity step a check was introduced for the maximum power and torque that can be produced by the motors. If the limit is reached, the longitudinal force are then dictated by that limit and all the other results are derived accordingly; again, the solution is then iterated up to the results stabilization.
When the longitudinal forces are fixed, the following system is solved (similarly to the previous one).

$$\begin{bmatrix} 1 & 1 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & -1 & -m \\ -l_f & l_r & -h_{COG} & -h_{COG} & l_f & -l_r & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} F_{z,front} \\ F_{z,rear} \\ F_{x,front} \\ F_{x,rear} \\ F_{l,front} \\ F_{l,rear} \\ F_d \\ a \end{bmatrix} = \begin{bmatrix} mg \\ 0 \\ 0 \\ \frac{1}{2}\rho C_l S v^2 BAL_{aero} \\ \frac{1}{2}\rho C_l S v^2 (1 - BAL_{aero}) \\ \frac{1}{2}\rho C_d S v^2 \\ F_{x,front,max} \\ F_{x,rear,max} \end{bmatrix}$$

Finally, the required time to reach 70 $m/s$ is obtained from the acceleration at each velocity step.

$$t_{step} = \frac{v_{step}}{a_{step}}$$

The total time is the sum of all the steps.

## 7.3 Results

The resulting torque split, defined as $T_f/T_{tot}$, which optimizes the acceleration is:

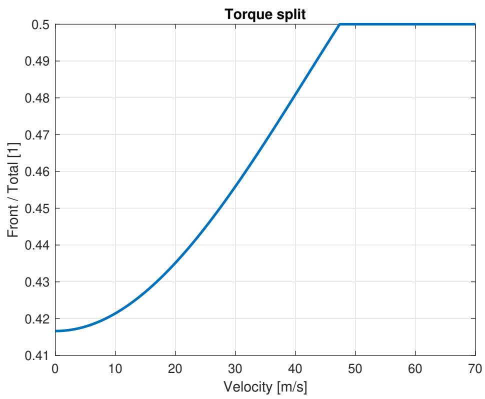
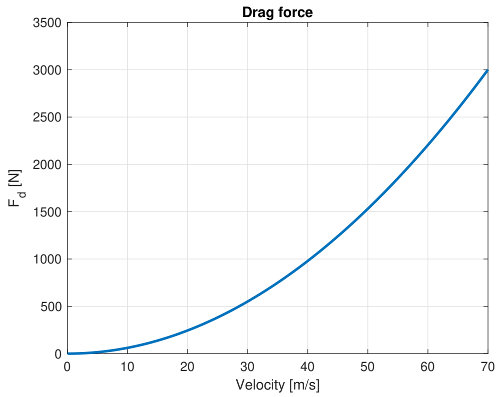



Figure 7.2: Torque split

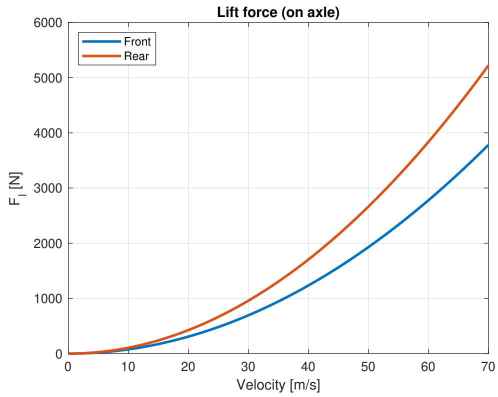
With this configuration the total time required to accelerate from 0 to 70$m/s$ is:

$$t_{tot} = 5.7765\,s$$

As it was expected, the aerodynamic forces increase quadratically with the speed.



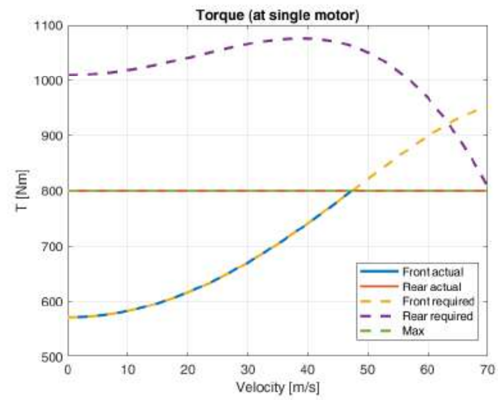

(a) Drag



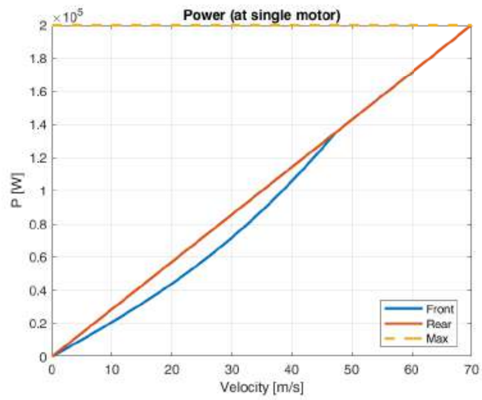

(b) Downforce

Figure 7.3: Aerodynamic forces

The torque and power curves are presented as the quantities relative to each single motor: since the manoeuvrer is in a straight line, the quantities on the left and right of the same axle are equal. In the plots also lay the respective maximum limits and the dashed line in (Fig. 7.4a) representing the ideal required torque, which is then limited to $800\,Nm$



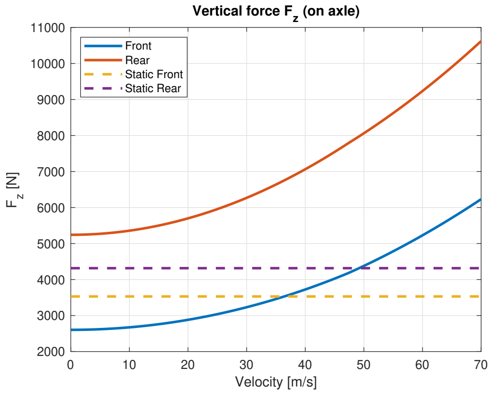

(a) Torque



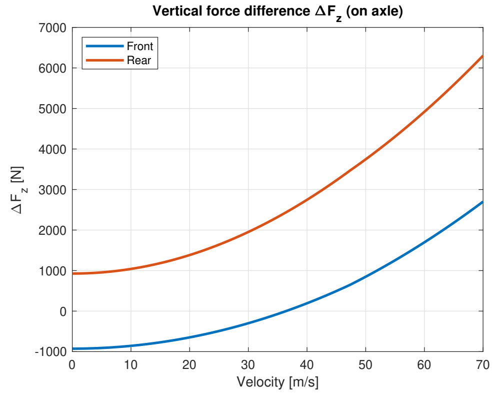

(b) Power

Figure 7.4: Engine demands

As required, the friction coefficient values depend on the vertical load variation, and the maximum developable longitudinal force changes according to both. The three quantities are reported below. Finally, the resulting acceleration and the cumulative time are plotted, too.

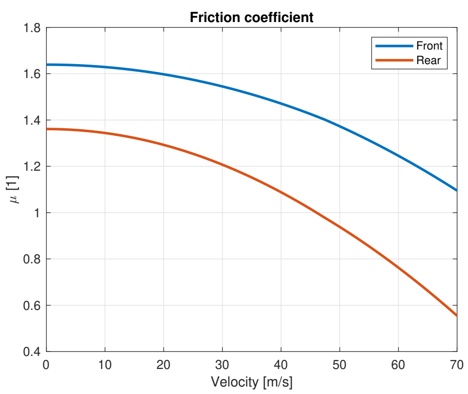



(a) Total

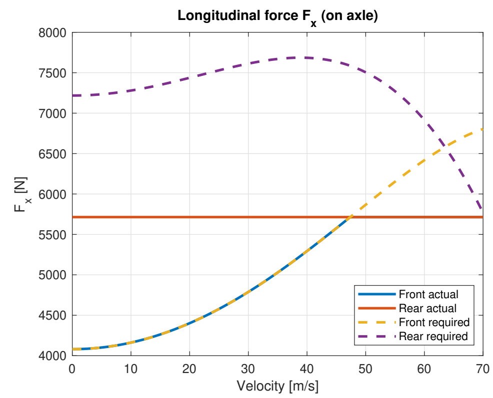



(b) Variation

Figure 7.5: Vertical loads

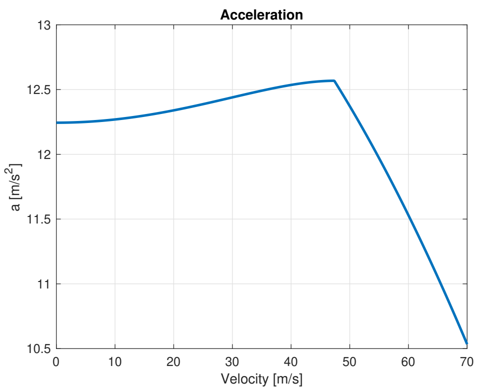



(a) Friction coefficients

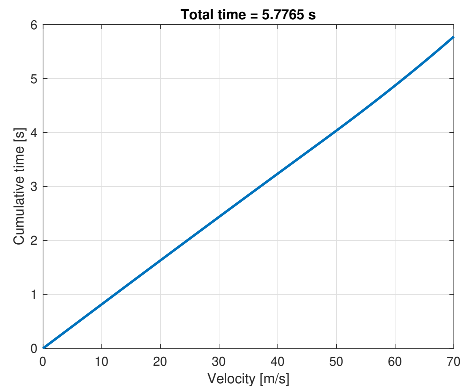



(b) Longitudinal forces

Figure 7.6: Results




(a) Acceleration




(b) Cumulative time

Figure 7.7: Results

# 8 Lab.07 - VI-grade

## 8.1 Introduction

VI-grade offers a complete software suite aimed for vehicle simulations, in which the user can modify any vehicle parameter he wants to observe effects and results, also in terms of on-track performance.

Aim of this laboratory is to modify some simulation parameters and observe the consequent effects.

## 8.2 Gear ratios

The first setup modification regards the gear ratios, which are made longer with respect to the baseline configuration. The main reason for this choice is the fact that the rpm limiter is reached in the long straights, causing a strong performance loss. Obviously, longer ratios are also connected to slower accelerations, especially at low speed.

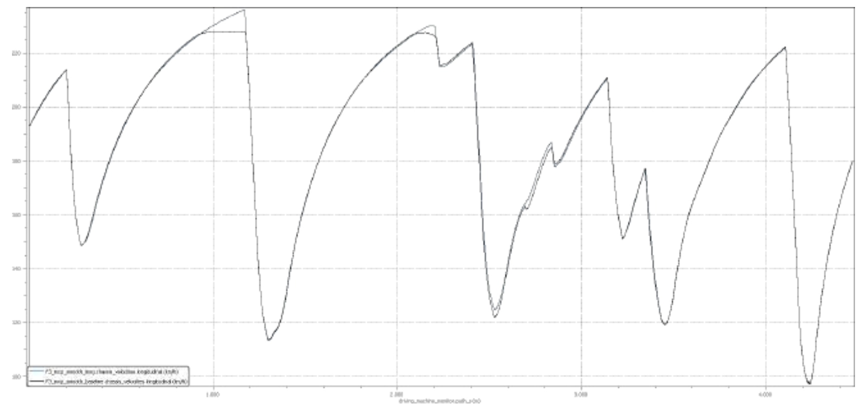


Figure 8.1: Speed comparison

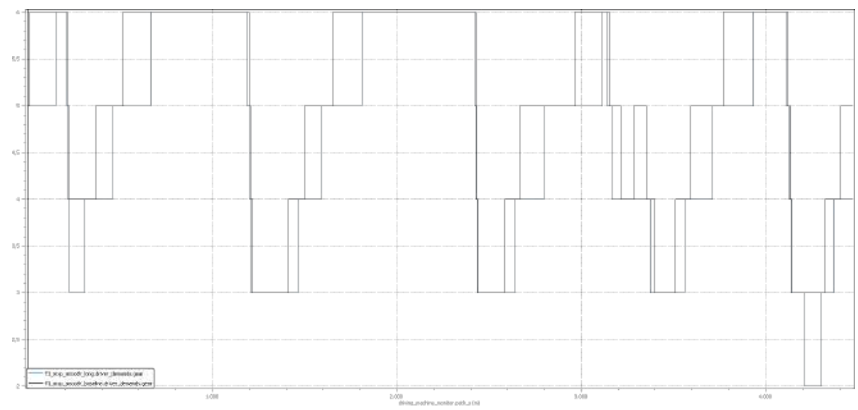


Figure 8.2: Gears comparison

As expected, the main differences can be observed in the speed profile: the long configuration (blue line) removes the problem of the rpm limiter and shows strongly larger speeds in the straights, while loosing

mainly in the acceleration phases. Moreover, a speed loss is experienced after braking for high speed turns, such after 2000 $m$ and before 3000 $m$, possibly because of different position in the engine map. Also the gears usage is obviously different, with the longer gears generally shifted later and also using the second gear, never selected in baseline configuration.
The longer gear configuration helps improving the laptime.

$$laptime_{baseline} = 90.211\ s$$

$$laptime_{long} = 90.051\ s$$

## 8.3 Aerodynamic configuration

The second setup modification is related to the aerodynamic loads: a higher-downforce configuration, with a totally new aero-map, is tried, to try gaining advantage in high-speed turns, at the cost of slower accelerations and maximum speed due to the increased drag.

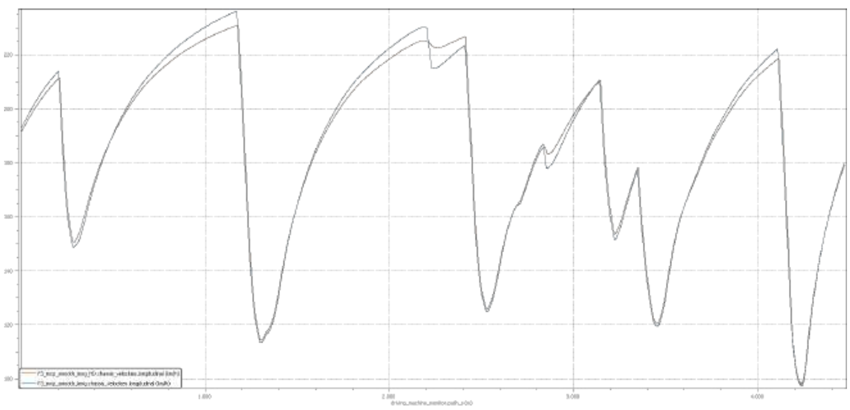


Figure 8.3

The maximum speed is reduced in every straight, with strong advantages in high speed turns and slight advantage in lower speed turns. During braking phases, the longitudinal deceleration is stronger with the

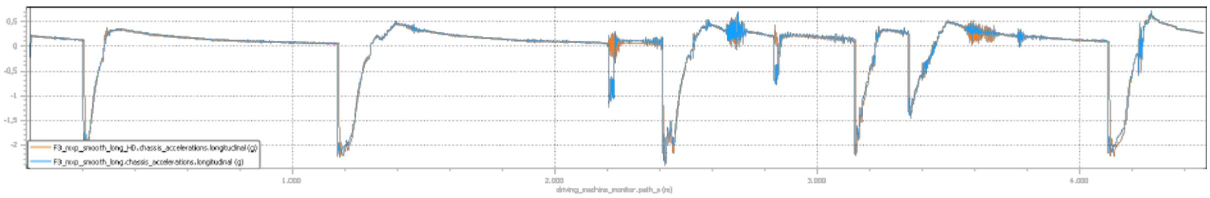


Figure 8.4

HD configuration, while in turn exits it the acceleration is slightly lower.

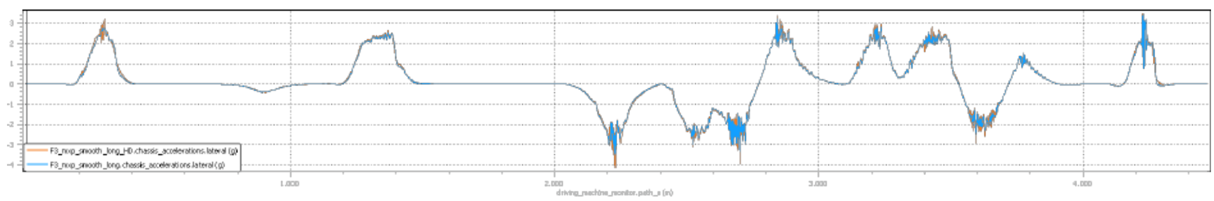


Figure 8.5

The lateral acceleration is generally larger in the HD configuration, especially when the speed is higher, due to the increased vertical loads which allow for more grip.

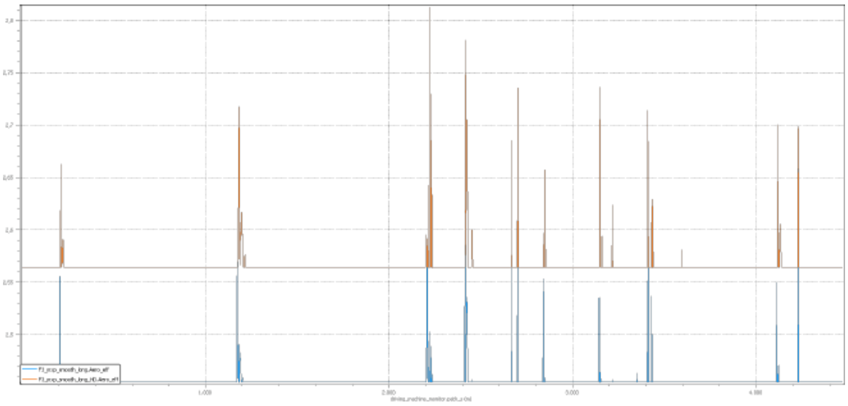


Figure 8.6

The aerodynamic efficiency ($C_l/C_d$) is constantly larger (about 0.1 more than baseline configuration) with the new map. Some spikes are observed at the beginning of the braking phases and when passing over kerbs.

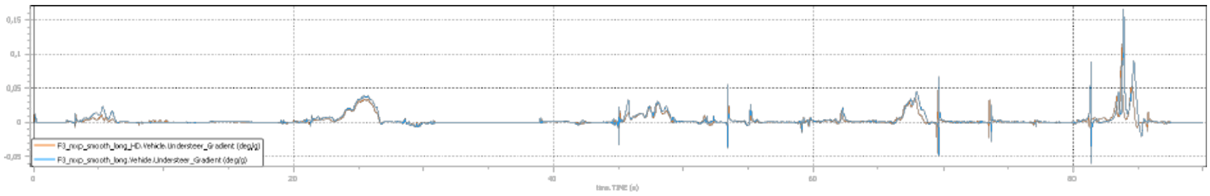


Figure 8.7

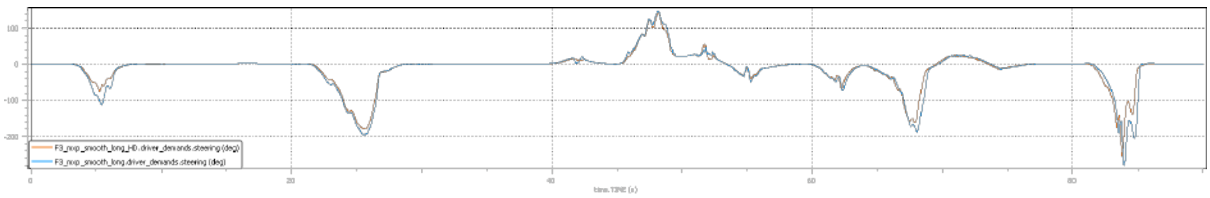


Figure 8.8

Thanks to the increased vertical loads and grip, the understeering behaviour is generally strongly reduced in the HD configuration, with the effect of requiring much less steering wheel angle for the same turn.

$$laptime_{long} = 90.051\ s$$

$$laptime_{long,HD} = 89.861\ s$$

## 8.4 Anti-roll bar

The third test is a so-called "investigation", that is a full factorial DOE (Design Of Experiments) to have a first idea on how the anti-roll bar stiffness (front and rear) affects the car behaviour. In particular, two scaling factors has been chosen for the front stiffness and two for the rear one, namely 0.7, and 1.3, where 1.0 corresponds to the reference value. All the combinations have been thus simulated.

The final laptimes are reported below.

| | | | | |
|---|---|---|---|---|
| v0001 | 0.700000 | 0.700000 | ⚠ | 90.4710 |
| v0002 | 0.700000 | 1.30000 | ⚠ | 90.1110 |
| v0003 | 1.30000 | 0.700000 | ⚠ | 90.4510 |
| v0004 | 1.30000 | 1.30000 | ⚠ | 90.9210 |

Figure 8.9

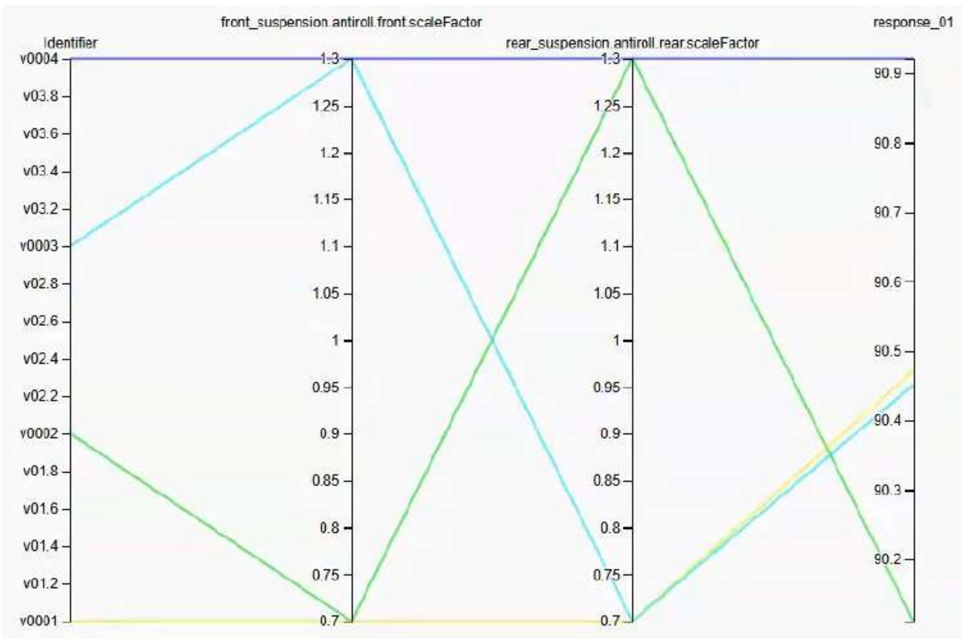



Figure 8.10

The best laptime is reached with a less understeering configuration, with 0.7 as front scaling and 1.3 as rear one. The worst solution is the one reached with rigid axles both at front and rear.

# 9 Sem.01 - Race Track Engineering

## 9.1 Introduction

The first seminar, held by Marco Fainello, focused on the importance of the team in motorsport.
Team management, race preparation and race strategy play a pivotal role to achieve success in the always fast and evolving world of motorsport.

Motorsport is a very complex and competitive sport, in which every tam component works to achieve the maximum possible performance in the available time, trying to maximize the results with only one final goal in mind: win. For this reason, motorsport is first and foremost a team sport. A team is made up of a set of individuals, each with its own strengths, weaknesses, fears, ambitions, that must collaborate together to thrive and achieve a common goal: winning competitions.

## 9.2 Competition

Competition is an activity were many different teams race against each other, under strict rules, in order to win something. A positive, fair and respectful competition is always required.
Competitors always lose more than win, that is why the key is learning how to improve and learn from defeats, which requires a strong method to analyze mistakes and possible areas of development.

Competition is not entirely about learning from mistakes but also luck and opportunities: staying always ready, waiting for the right moment, recognizing a possibility and fully exploiting it in the quickest way possible when it comes, are all keys for the win.

## 9.3 Goals

As said, the final goal of motorsport is always win. This can be sub-classified into two macro-goals: performance and reliability. These two goals can be divided in many different and smaller objectives, each one possibly achieved by one of the groups.
Keys to tackle different goals are:

- Clear definition and management of the explicitly shared goals, setting a strategy and dividing in smaller goals trying to define the weight and contribution of each sub-goal to the final one
- Clear definition of a goal achievement timeline, defining the proper priorities
- Division and allocation of human and tools resources
- Definition and measure of success and failure

Always remember that motorsport is compromise, at the end you don't need the best possible solution, but the best you can obtain in the time that you have: sometimes it is worth giving less importance to some goals to prioritize others (maybe due to time lack, issues or changes in rules).

A compromise needs to be achieved also in the timeline definition step: it is not always true that, if possible, it is always better to start earlier, a trade-off exists: the later you start, the more tools and knowledge on methods and rules you have; the earlier you start, the safer you are, with more capability of recovering in case of issues.

## 9.4 Assets

Assets are the basis to create a winning team, defining what the whole group and set of resourced, which is achieved by:

- Knowing your tools to define what you are capable to design and produce in-house and what you need to buy
- Knowing your team, personal likings, cultures, interests and ambitions
- Knowing your technical knowledge and know-how, to carefully define what and how needs to be designed and/or improved

## 9.5 Constrains and Boundaries

In a real world scenario there are many constraints and boundaries that can limit your development, improvement, design and much more. These can be due to strict timetables, limited money, regulations from FIA or from the team itself and many more.

Everyone in the team needs to perfectly know the regulations. Some rules are more explicit, some less: it is often a matter of interpretation, you never are totally sure that a choice will be considered as legal, but you can estimate the possibility that it will be, especially when working withing the so-called "gray areas".

Two approaches exist in following the timetable:

1. Designing the best possible in the allocated time for a specific task, then, when the time is over, freeze everything without the possibility of modifying anything: safer but possibly not optimal for performance maximization
2. Accepting the risk of something going wrong and the possibility to need to fix something last minute, maybe when production already started, so that you can correct and recover: not everyone can do this, you need strong resources to work like that

## 9.6 Team structure

A typical team is structured in very small groups with few interactions between them, with precise tasks to be accomplished, coordinated by a team manager which needs to perfectly know his team and its vision, ensuring the correct distribution of people in the groups, that every group properly works for the team vision, that cadences are respected, always trying to extract useful ideas from possible (positive) conflicts: communication and trust are the keys.

## 9.7 Conclusion

In the end motorsport can be summarized as a team sport, which is founded on preparing to grab an opportunity that might present itself casually, and exploit it to win. To do this is important to find a compromise between all the different variables, to be as close as possible to the best performance car at the best moment , trying to stick as close as possible to known principles, avoiding to over-complicate things.

# 10 Sem.02 - Goodyear

## 10.1 Introduction

In this seminar, a focus on tyre behavior was carried out, especially going more in detail on the differences between road and race tyres, in terms of construction, behavior, requirements and characteristics. The main differences between road and race vehicles regard power, mass, acceleration (both longitudinal and lateral), speed and lap time.

## 10.2 Main differences between road and race tyres

**Stresses and degradation:** The combined (longitudinal + lateral) sliding power per tyre can be computed as $P_{comb,i} = F_{comb}V_{comb}$, which integrated over the lap gives the energy applied to the tyre itself, a useful parameter for instance to compare stresses on different turns and see which is the most demanding one.
In general, the stress level is much larger in racing tyres, which leads to a very different design, focusing on grip and handling instead of durability, wear, noise and consumption.
Racing tyres have and expected life of about $100 - 600$ $km$ ($50000$ $km$ or more for road), have no grooves if for dry asphalt and work at higher temperatures, between $80°C$ and $120°C$.
The level of degradation is estimated (on average between the teams) through some simulations (e.g. 30 laps with soft tyres), then each team makes its estimate considering the specific case (vehicle, driver, conditions). After every event, telemetry data are shared with the tyre manufacturer and exploited to find correlation in performance, temperatures, degradation and so on to improve the models accuracy.

**Stiffness:** In racing tyres the stiffness needs to be maximized: low tread thickness ($3 - 4$ $mm$), slick tyres and reinforced sidewalls (through aramid and steel elements) allow for steering precision and a fast response to driver inputs, while in road tyre the main task of the sidewalls structures is to protect from sidewalks impacts.

**Camber and inflation pressure:** When setting a car for a race, engineers always try to maximize the (negative) camber angle and to minimize the inflation pressure, both the two techniques allow to maximize the grip, especially while cornering.
At the same time, a threshold for both exists: a too strong camber angle causes contact patch reduction and a drop in the grip, a too low pressure causes issues in terms of fatigue loads with a consequent risk of explosion: for these reasons, often a tyre expert supports the team in setting the parameters.

**Shape design:** Much less rounded angles are used while designing the cross-sectional shape of a racing tyre, which leads to a less elliptical and more squared contact path. A square footprint is indeed preferred to obtain a better handling, dry traction and wear uniformity. In standard vehicles, rounded footprint are exploited for a better wet traction (due to a better water film rupture) and lower noise (due to the more progressive impact of the contact patch ribs). This also affects the peak in lateral force, which for square contact patches is reacher at lower values of side slip angles.
The tyre shape also influences the stresses, which for a more squared one, result in smaller belt and carcass tension near the contact patch, and larger stresses near the beads, effects that in general lead to better

handling a contact patch in corners. At the same time this shape improves the shoulder flexibility: the tyre needs to be as stiff as possible but some soft points are needed to ensure a certain level of deflection, to avoid wear problems.
At the end, a topology optimization process is always required, in which the objective function is the pressure deviation in the footprint (to be minimized, even more importantly while cornering).

**Compound:** The tyre stress with respect to strain can be represented by a complex elastic modulus, in which the real part represents the storage modulus (elastic effects) and the imaginary part the loss modulus (viscous effects). Differently from road tyres, in which a trade-off between performance and consumptions is reached, in race situations the loss modulus is always maximized, to obtain a larger hysteresis cycle which corresponds to larger grip. For this reason racing compounds are typically softer than road ones.
In road tyres, both Carbon Black and Silica are added to the compound, while in racing ones only CB is added to slick tyres (Silica is used only for wet conditions): CB gives lower storage drop with temperature increase, which is aimed to obtain a constant and predictable behavior of the tyre during the race.

Two effects regard tyre softening:

- Payne (strain induced softening): the more the strain, the more the softening.
- Mullins (load induced softening): the more the number of runs the more the softening; this is the effect that bring to tyre degradation.

## 10.3 Data analysis

As already stated, after every event, telemetry data are exploited to find correlation for instance in performance, temperatures (of tyre and cavity air), inflation pressure, grip and degradation.
All the quantities always show an evolution during the initial laps and stabilize at a certain point: the car must always be set up considering the stable value and consequently won't be in optimal condition while running the first kilometers with fresh tyres.

**Grip:** Different parameters regarding the grip (in terms of obtained accelerations) can be exploited and analyzed over a entire stint, such as the overall grip, the braking grip, the acceleration grip, the cornering grip and the aerodynamic grip. To decide which tyre is better in a certain situation, spider charts can be used.

**Temperature:** Temperature is one of the most important parameters to be observed, in terms of evolution during the laps in all the structures of the tyre and on the cavity air. The combination of temperature and lateral acceleration data can be exploited to define the optimal temperature window for the tyre, which maximizes grip and lap time. Moreover, temperature analysis can be used to check if the camber is set properly: a correct angle should lead to a temperature increase of about $20°C$ during a turn, to avoid leaving the optimal interval with a consequent drop in lateral acceleration.
This can be exploited also for a better car set up: if for instance $\Delta T_{rear} > \Delta T_{front}$, $F_{y,rear} < F_{y,front}$, leading to a more oversteering behavior at the end of the turn.

**Pressure:** As for the temperature, pressure increases during laps and stabilizes after a certain drove distance. An important parameter is the IP ratio, which gives an idea of how much inflation pressure you have at front tyres with respect to the overall one.

## 11 Sem.03 - Racing cars braking systems | Brembo

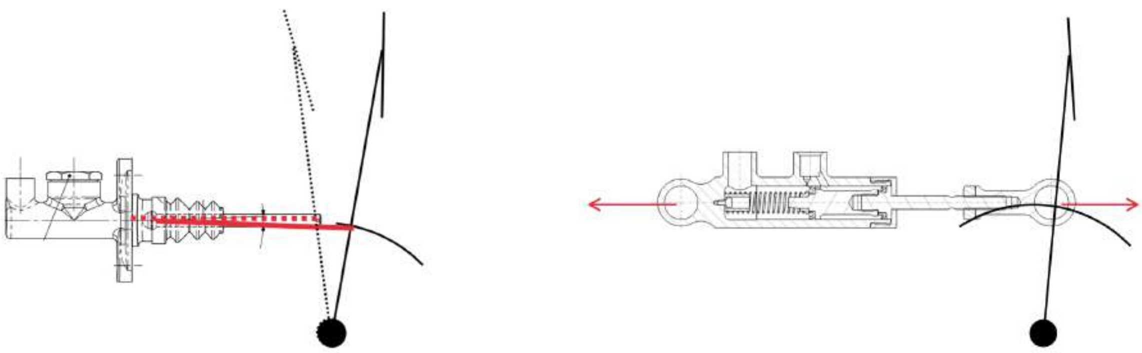
### 11.1 Introduction

Despite the similarities which link, on a conceptual level, racing braking systems to the standard vehicles ones, some strong differences have instead to be accounted for in terms of loads, operational conditions, systems, materials, solutions and so on. For this reason, it is important to study in details the architecture of the former, which presents strong differences also due to the lack of driving aids, such as ABS or brake booster, which are often banned by regulations.

### 11.2 Components and working principles

**Master cylinder:** this is the component which physically translate the force applied to the pedal into hydraulic pressure. A distinction can be introduced between two types of master cylinder (Fig. 11.1):

- Wall mounted: cheaper, in which the cylinder is fixed in the space, meaning that the push-rod, rotating with pedal stroke, will end up being not aligned with the cylinder itself.
- Rod type: more expensive, in which the cylinder is a rod, meaning that it can transfer only axial forces (i.e. it rotates around a hinge), allowing for a perfect alignment between push-rod and cylinder in every working condition. In this case the transmitted force is totally useful force. This solution is widely adopted in high level racing.



(a) Wall mounted type

(b) Rod type

Figure 11.1: Master cylinder configurations

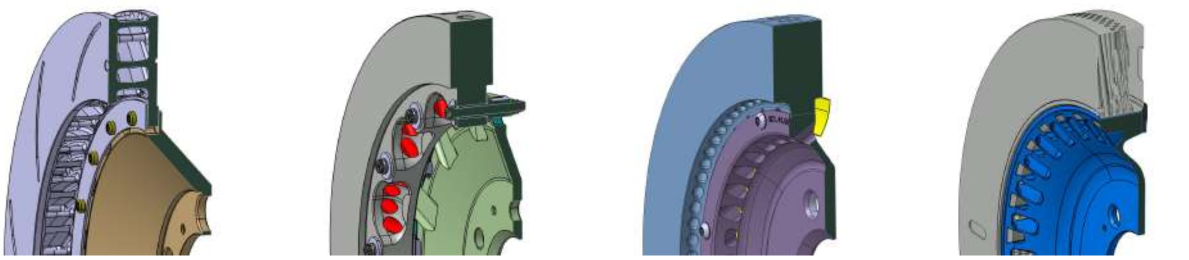
**Caliper and disk:** Differently for production vehicles, which often exploit floating calipers, in racing the fixed caliper solution is always the choice: the pistons, which act on both sides of the caliper, are the only component moving while braking, making the pads to be pressed against the disks.
Fixed calipers are mono-blocks: they are typically machined from billets, or alternatively pre-forged and then machined; all the oil channels need to be drilled and all cavities to be obtained by material removal. A particular attention is required when drilling the channels: air bubbles must be avoided inside the oil circuit (to avoid sponge-effect and strong air expansion when subjected to high temperature, typical of racing braking systems). To achieve that, the holes that generate the channels need to be positioned on the highest part of the chambers to avoid air to get stuck without the possibility to reach the bleed nipple. With this configuration, while filling the caliper with oil, the air is naturally forced to escape the system.

**Pistons:** The main design variables in this case are the total number of pistons per caliper and their diameter, which can be chosen to be equal or different between the pistons of the same pad.
The reason for choosing a variable diameter is the fact that an as uniform as possible contact pressure between pad and disk is aimed, to increase performance and obtain a consequent uniform wear on the pad itself (non-uniform consumption would be detrimental for performance, especially in long races such as endurance). As for many other applications nowadays, preliminary designs can be made starting from simple equations, but FEM simulations are then exploited to get the best possible solution.

**Pad:** An interesting analysis has been done on a Brembo's patent regarding braking pads, which instead of being designed with a rectangular cross section, are characterized by a parallelogram shape: this allows to exploit the reaction forces applied by the caliper to the pas to generate an additional component in the force applied to the disk, which increases the breaking action: an additional braking force is thus introduced "for free", just by cleverly exploiting the constraints geometry.

**Disks:** One of the most crucial parts, responsible of effectively dissipate the thermal energy produced while braking by friction, is the braking disk. A good disk is able to dissipate a very large amount of heat power, in a very fast way, both using convection with air, conduction with other solid elements and radiation. Different types of disk can be produced:

- Cast iron disks: differently from production cars, they are not made of a single part, but two, namely the disk itself and a bell, which connects the disk to the hub. The bell is usually made from aluminum, to promote lightweight. Note that cast iron and aluminum have very different thermal expansion coefficients, which must be carefully taken into account while designing the coupling between the two components.
- Carbon-carbon disks: made from carbon fibers inserted in a carbon matrix (in place of resin, which cannot sustain the temperatures reached in these applications, up to 1000 $°C$), this type of disk is widely adopted in all top racing categories. More expensive bells are used here, made from titanium. Also, strict tolerances need to be ensured to reach the required performances and to avoid unwanted vibrations which are detrimental for a proper braking action. Cheaper solutions use coupling only at one side of the disk; more expensive solutions have splined couplings on both, to better sustain the loads by avoiding them to be transmitted from one side of the disk to the other, causing strong shear forces in structurally weak regions (due to the presence of ventilation holes). Also note that a double-sided coupling reduces the cooling performance due to air flow area reduction: specific holes are thus created inside the connection teeth, to increase the cooling flow up to the one of the one-sided solution, but with a better force distribution.



(a) Cast Iron (b) T-Bobbing (c) One-sided spline (d) Two-sided spline

Figure 11.2: Possible disk configurations

## 11.3 Brake pressure and balance

All drivers, when braking, produce triangular brake pressure diagrams: the pressure suddenly builds-up at the beginning, then the driver starts to smoothly release the pedal while the car is slowing down. This behavior is due to the aerodynamic loads, which make the wheels' vertical loads to decrease with the square of the vehicle speed, which gradually reduces the maximum admissible longitudinal force that avoids wheel locking. Note that the optimal brake balance is instead not affected by this aerodynamic loads reduction.

Brake balance (BB) is not a trivial concept in racing, for several reasons. It needs to be correctly set up to reach the maximum possible deceleration (when both tyres lock together).
Practically, from a single-track model, the BB can be firstly estimated; from this result, the number of pistons, their dimension and disks radii can be chosen so that they fit the specific case. From here, further adjustments are required while driving, due to the fact that the friction coefficient between tyres and ground $\mu$ is not constant during a race: for instance it varies between dry and wet condition and between new and worn tyres. A specific device is thus needed to face this issue: this device is the so called balance bar. This bar is connected to the brake pedal on one side and to two separated master cylinders (one for front and one for rear) on the other side. By rotating, it can press the two push-rods independently and with a different force, which is inversely proportional to the two lever arms (i.e. the distance between the pedal connection and the master cylinder one). By modifying this length turning a knob, the BB can be changed by the driver during the race.

Figure 11.3: Balance bar

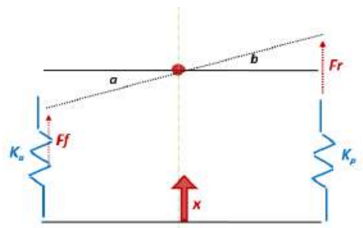



Figure 11.4: Balance bar model

Another complication for the optimal BB comes from the hybrid/fully electric motorization, in which the KERS unit is exploited to convert the car kinetic energy into electrical energy while braking to recharge the batteries. Usually, it acts on one axle only (typically rear).
The main difficulty comes from the fact that the contribution of the electric motor is not constant and not always present: FIA regulations on maximum energy recover per lap, full battery, too large temperature in battery are examples of causes which make KERS not exploitable.
Moreover, supposing constant power of the electro-magnetic braking action (maximum allowed, to minimize lap time and maximize efficiency), the torque increases while the speed decreases. This means that with the same optimal total force on the axle, torque from KERS increases while braking, so the mechanical action needs to be reduced. At the same time aerodynamic vertical forces reduce, and so the total optimal force needs to be reduced accordingly. These elements generate three different phases during the braking action (Fig. 11.5): a first part at high speed in which a large mechanical and low KERS forces are required; a second opposite phase characterized by large KERS and low mechanical action and a final part at slow speed in which the KERS component alone is able to produce the (lower) required braking force.

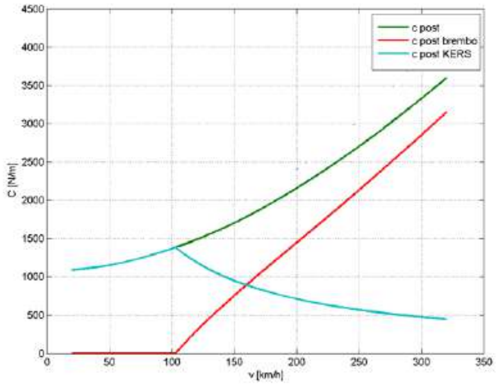



Figure 11.5: Braking torque

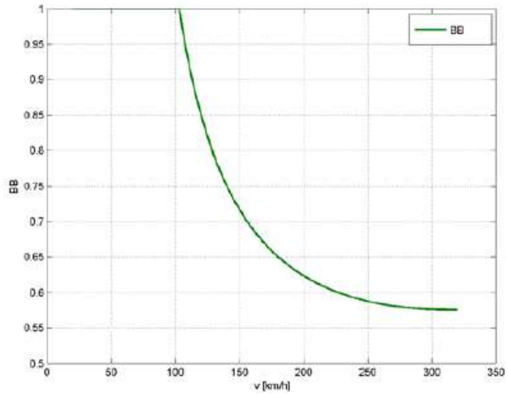



Figure 11.6: Brake Balance required

Thinking to the hydraulic system alone, this generate the need for a largely variable BB while braking (Fig. 11.6), which progressively increases (moving towards front by definition) up to 100 %, while slowing down. With the balance bar it is not possible to continuously act on the BB and to move the whole force to a single axle, reasons for which a change in the regulation was introduced by the FIA with the hybrid/electric era: brake-by-wire systems replaced fully hydraulic ones.
Anyway, mainly for safety reasons, a valve in the system allows to switch between brake-by-wire - in which the oil coming from the master cylinder enters the so called Compliance Chamber where the input pressure is transformed into braking force by a control unit - and traditional hydraulic system, in which the oil directly connects the master cylinder to the pistons.

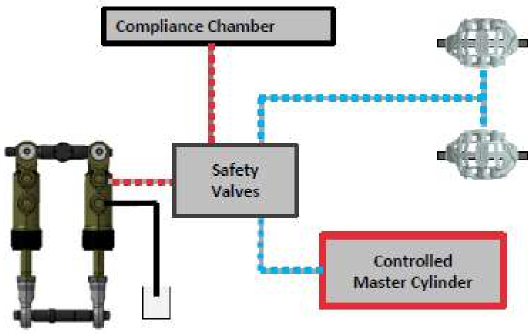



Figure 11.7: Safety valve setup

# 12 Sem.04 - Vehicle and Data analysis | Ferrari Ges

A race performance engineering team mainly focuses on vehicle performance, in which the lap-time is the main aspect to be improved within the various ones. Such team usually works on many subsystems at once: in order to define the optimal setup compromise it needs to consider both each subsystem singularly and the vehicle as a whole and how different subsystem interact with each other. Therefore, members of such teams are not specialist of one single subsystem, but are engineers able to make all the subsystems work together harmoniously.

As an example, let's think to the suspension design, which is extremely linked to aerodynamic: suspensions need to achieve desired kinematic targets, such as allowing the vehicle to squat to increase the downforce generated in the floor, without touching the ground; at the same time suspension arms are actually aerodynamic components and their shape and positioning can affect the flow around the car. Other examples of subsystems to be tuned can be active control associated to electronic differential, brake balance, regenerative systems, power unit, gearbox and so on.

## 12.1 Objectives

The track-side team, responsible for the actual running of the vehicle, has to achieve mainly two objectives, by exploiting the initial vehicle package provided during the design phase, and needs to optimize all the components so that the car can behave according to how it was designed and intended to work:

- Quickest car on a single lap
- Quickest car over a race distance

Achieving this, means having a vehicle that shows a consistent behavior along a race, with the lowest possible lap-time and behavior variation.

## 12.2 Setup optimization

In a such competitive world, the orders of magnitude are important. Often, the aim is to reduce the lap-time, for instance, of $0.1\,s$ over an $85\,s$ lap: even such an apparently minimal reduction can potentially result in improving the track position and help turning the tides of championships. A $0.1\,s$ reduction can be worth months of development and millions of euros so it is crucial to know which direction to take.

When on the track, lots of parameters can be tuned, but time during a race weekend is limited. Three practice sessions may not be enough or may even be shorter than expected, thus it is crucial to get on track with a clear knowledge of what has to be actually tested and what can actually be changed. This process is promoted if a preparation is performed through virtual simulations, both off-line and with driver, to identify the most important areas to work on.

## 12.3 Parameters

The parameters that can be modified to set up car are:

- **Front/Rear wing angle:** The optimal setup is a balance between the time gained in a corner thanks to higher angles that generate higher downforce and the time lost on a straight due to the increased drag. Therefore a trade-off depending on each track and its corners has to be found: usually to speed up the process a lot of lap-time simulations are run to evaluate the best possible configuration.

- **Ride height:** Aerodynamic forces are extremely dependent on the height from the ground, especially with the current set of rules where the ground effect is producing the majority of the downforce: the closer a car can get to the ground, the better the ground effect and the larger the downforce will be. The rear wing is mainly insensitive to the height from the ground, while the front wing shows lot more dependency on height, since the lower part of the wing creates suction again through ground effect. Even for cars with less powerful aero-devices, such as GTs, a lower ride height is beneficial, as it lowers the center of gravity of the vehicle.

  Finally, the aim is to run as low as possible, avoiding hitting the ground due to the extreme vertical forces produced. This would be detrimental for three main reasons: the ground effect would not work properly anymore; a vertical reaction force would be generated on the plank thus reducing the vertical load and the grip on the tyres; the straight line speed would reduce due to the floor scraping, which could also lead to excessive plank wear.

- **Camber angle:** The aim is to have it negative, to produce a wider contact patch area in corners, resulting in more grip. On the other hand, an excessive negative camber angle can be detrimental for straight line speed and for tyre uneven wear.

- **Weight distribution:** It affects handling and also traction because it allows to put more vertical force on an axle. It changes during the race mainly due to fuel consumption.

- **Heave stiffness:** It is highly dependent on aerodynamics and ride height. The aim is to be as low as possible to have aerodynamics working at its best, keeping in mind the issues if touching the ground. Therefore, the usual solution is to have stiff springs overall, especially at the rear, to limit the squat. If the track is bumpy, it might be useful to soften a bit the springs, which would require to rise the static ride height to avoid touching the ground, affecting aerodynamics.

- **Roll stiffness:** It is one of the most powerful tools to manage the lateral behavior of a vehicle. The overall load transfer depends mainly on the height of the center of gravity (and how it moves while cornering), on the lateral forces and accelerations and on the vehicle's track-width. In track-side conditions it is not possible to easily control the overall load transfer, but it possible to manage the partition of load going in one axle with respect to the other acting on the anti-roll bars (*ARB*). Stiffening the *ARB* at one axle means having more load transfer on that axle with consequent reduced axle characteristic, meaning that stiffening the front *ARB* will generate more understeering while stiffening the rear one will reduce it.

  *ARB* are U-shaped and actuated (twisted) by a rocker moved by the push/pull rod suspension arm, which is directly connected to the upright: the length and positioning of the linkage mechanism will change how the *ARB* will twist kinematically and therefore its stiffness. So the *ARB* stiffness is easily controlled by changing the ratio between the wheel travel and how much the *ARB* is actually twisted.

*ARB* are also dependent on the vehicle speed: as speed increases the aerodynamic forces will squat the car down, effectively changing the suspension's kinematics and the ratio at which the *ARB* is twisted. Depending on the kinematics of the rockers, the *ARB* can both result softer or stiffer as speed increases. Finally, the same *ARB* can act softer or stiffer depending on how fast the wheel vertical motion is happening.

- **Dampers:** They are set without considering comfort, up to the point where the car becomes extremely difficult to be driven and the driver cannot drive it to the maximum of its abilities. Dampers in racing cars are used to control the vertical displacement, since springs are usually very stiff, producing high forces, dampers play an important role in controlling them. Given the fixed vertical stiffness (for aerodynamic purposes), dampers are the only modifiable parameter to limit the tyre vertical load oscillations. Usually, the vertical motion of the car is limited as much as possible, to keep the center of gravity always at the same height (as low as possible): the dampers must work to minimize the oscillations.
  If a suspension oscillates the road holding increases and the grip reduces due to tyre hysteresis and non-nonlinearities: the grip improvement that you get from a tyre when $F_z$ reaches a peak is smaller than the grip loss when $F_z$ reaches a valley ($F_y$ increases less than linearly with $F_z$). Moreover, this behavior is often unpredictable and the grip can be different in every turn, therefore the car becomes more difficult to drive.

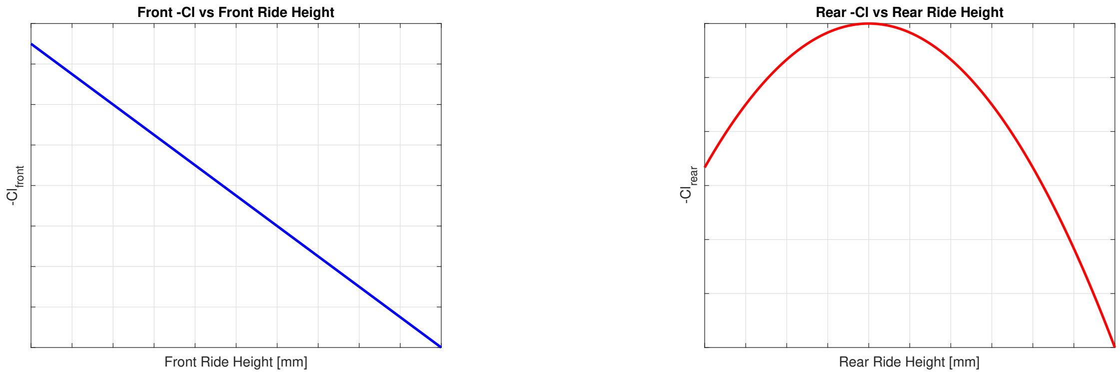
- **Cooling:** High cooling need means more drag and less downforce: to create cooling ducts you have to effectively "cut" the bodywork, which means less surfaces where aerodynamic forces can act upon. Moreover, the radiators create high drag due to their internal construction.

- **Differential and brake balance:** In modern-era cars, when braking, the driver only controls directly the pressure of the front brakes, while in the rear the pressure is controlled by a brake-by-wire system. This system is required due to the presence of the engine braking and energy recovery systems, together with the usual mechanical brakes. The optimal setup is therefore achieved through the correct balance between all the systems at the rear axle and the front/rear forces ratio.

- **Tyres:** Always not produced in-house but purchased from dedicated companies: a team has limited influence on their construction and must use them as they are. Due to their construction it is important to heat up and use them in the right temperature window: considering that thermal blanket use is restricted both in temperature and time, a good heat-up process on-track is of the upmost importance for performance.
  Within some predefined margins, the teams need to select the best tyre pressure: low tyre pressure means bigger contact patch area and increased grip, but with enhanced structural risk in the tyre itself, which means that a trade-off must be found as always.

## 12.4 Aerodynamics

When studying aerodynamics is also important to analyze how aeroforces and ride heights influence each other. For this purpose aero-maps are extracted in wind tunnels to study how much a single component produces in terms of downforce and drag. In particular, is possible to divide the contribution of $C_l$ at the front and at the rear axles. As visible in the following figure $C_l^R$ shows the presence of an optimal ride height: if the floor is too low the Venturi effect will not work effectively, while at too high ride heights it will fade away. Instead, $C_l^F$ decreases as the car rises since the front wing lower plate works purely by ground effect.




(a) Front

(b) Rear

Figure 12.1: $C_l$ dependence on ride height

Another important analysis is $C_l$ in function of the angle of the attack of the front wing $\alpha$.

- $C_l^F$ increases less than linearly as $\alpha$ increases
- $C_l^R$ decreases linearly as $\alpha$ increases since there will be less airflow going towards the rear
- $C_l^{tot}$ initially increases less than linearly, then reaches a plateau before dropping drastically due to wing stall: it is designed to work not exactly on the maximum $C_l$ to avoid stalling and to have a large set of conditions to exploit before getting to the maximum possible $C_l$

$C_l$ and $C_d$ depend also on yaw, the effect of yaw is measured in wind tunnels by rotating the model against the airflow, studying how much aero changes due to side winds. When side winds are present, turbulence due to tyres may be redirected by the wind towards the rear wing or under the floor. This high pressure turbulent flow will impact the efficiency of all the affected aerodynamic devices. The car which aerodynamically behaves better, is the one less affected by this effect.

Plotting $C_l$ against $C_d$ polar plots are created. Through this curve it is possible to extract the optimal aerodynamic configuration by doing laptime simulation for each different track. To find the optimal setup, also lift and drag sensitivities are studied, defined as the gained/lost time while modifying lift/drag divided by the number of points of lift/drag variation.

$$C_l^{sens} = \frac{\Delta t}{\Delta C_l}\left[\frac{ms}{Points}\right] \qquad C_d^{sens} = \frac{\Delta t}{\Delta C_d}\left[\frac{ms}{Points}\right]$$

By dividing $C_l^{sens}$ by $C_d^{sens}$ is possible to plot the iso-chrono maps: the iso-chrono line which is tangent to the polar line is the best aerodynamic configuration.

## 12.5 Tuning

To tune the ride height of the vehicle, a starting point is (Fig. 12.1b). Looking especially at the graph of the rear axle, is possible to define $C_l$ in function of the speed. The rear ride height of the vehicle depends on the speed with a trend shown in (Fig. 12.2), which is the typical trend of a complex racing suspension with non-linear springs.
Combining the two graphs is possible to find $C_l$ at a given speed: the aim is to tune the vehicle to have the peak of $C_l$ at a speed smaller than the maximum speed of the vehicle. The car has to be set up to develop the highest $C_l$ possible at the speed at which the corner is actually tackled: each circuit will have a speed range in which all the corners will be run and, by changing $RRH$, this range has to be centered around $C_l^{max}$. If, for example, we have a circuit with many corners traveled on average at $150\,km/h$ the car should be set up to develop $C_l^{max}$ at that speed.

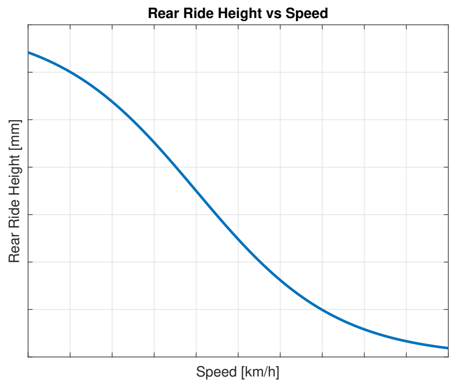



Figure 12.2: $RRH$ dependence on speed

Changing the static $RRH$ will effectively shift the curve of (Fig. 12.2), if it is reduced, then $C_l^{max}$ will be reached at lower speed. Reducing the $RRH$ also often results in stiffening the suspension stiffness, which produces a further beneficial effect: the increased downforce will allow to travel the corners faster (reducing the speed variations) and the stiffer spring will reduce the variation of $RRH$; as a consequence of this two factors, the car will work in a narrower region around $C_l^{max}$.

## 12.6 Simulations

### 12.6.1 Quasi-static lap simulations

Simulations are made on a non-linear solver using a multi-body model with about 50 degrees of freedom, through the which it is possible to derive the $g-g$ diagrams and the performance envelope of the vehicle. The simulations are performed computing the longitudinal accelerations, while the speed is found by integration. To find the optimal setup, an optimization problem is studied, which solves all the sections of the track to minimize the laptime, while ensuring that car dynamics and limits of the model are respected (considered as constraints). An extra applicable constraint is the driver, with all its limitations.

### 12.6.2 Driving simulator

Off-line sims are very accurate but cannot precisely simulate the interaction between the vehicle and the human. For this reason, simulations with the driver have always great importance. The aim is to make the simulation as real as possible, allowing the driver to feel itself as in the a real vehicle.

## 12.7 Telemetry

### 12.7.1 Data acquisition

A wide set of data is collected by plenty of sensors and some other quantities are derived from the measurements. For example, usually a load cell is positioned on the push rod, from which it is possible to extract the vertical forces acting on each wheel.
Usually the data are collected to feed some multi-body car model to estimate the quantities which cannot be measured directly, such as the forces on the contact patch.

### 12.7.2 Ride measurements

Vertical load measurements by themselves are hard to compare, due to high noise. Instead, to compare two different setups is possible to analyze the data using spectral analysis, studying resonances and RMS values.

## 12.8 Corner stability

In racing cars it is important to correlate what drivers feel in the car to data, especially about under/over-steering. In first approximation, this can be done through the handling diagrams, which however work well only with steady state maneuvers, which never happens in reality. Additional metrics are thus used to correlate such data:

- $N_\beta = \frac{dM_z}{d\beta}$ is a measure of the understeering coefficient, works better than handling diagram and $K_{us}$ as it does not have the problem of having multiple values for the same steering angle $\delta$ as it happens in very complex handling diagrams.
- $k_{yaw}$ is the yaw stiffness, it describes how much rotation the car gets for a given yaw moment. As long as $k_{yaw}$ is positive, the side slip angle will counteract the yaw moment, leading the car back towards the equilibrium position; if $k_{yaw}$ is negative, the car will be pushed away from it. Still, this is a function of front/rear cornering stiffness.
- $c_{yaw}$ is the yaw damping, it works to dissipate oscillations of the yaw motion. As long as it is positive, the car works as a dissipative system, making the oscillations to fade away and the car itself to go back to the equilibrium position. When negative, some energy is introduced in the system and instabilities can be produced.

All these quantities to be computed require the cornering stiffness for each tyre: the noisy data regarding $F_y(\alpha)$ is given to fitting models to produce a cleaner curve, which is then exploited to compute the derivative and find the cornering stiffness.

# 13 Sem.05 - Sustainability in Motorsport

## 13.1 Introduction

The lecture held by Marelli Motorsport focused on how sustainability has today become a central part of the industrial strategy and a fundamental aspect for a company's success.

Sustainability is a global priority and it is defined as "meeting the needs of current generation without compromising the ability of future generation". This is a very broad definition which touches many topics and very different aspects.

Sustainability is founded on three pillars:

1. Social impact, regarding population and workers;
2. Economic impact, based on creating jobs and profit for communities;
3. Environmental impact, that is to say pollution and similar.

These three pillars must be balanced to build an effective sustainable model. Referred to the pillars, 17 global goals has been defined by the United Nations, each of them regarding one or more of the three.

Some useful concepts are:

- Biocapacity (BC): capacity of the environment to supply humans of what they require.
- Ecological Footprint (EF): how much humans require from the environment in terms of resources for their activity.
- Non-sustainability: when the EF is larger than the BC.
- Earths: measurement unit equal to 1 when $EF = BC$, grater than 1 when $EF > BC$; it describes how many earth we would need to be sustainable with the current EF.
- Earth Overshoot Day: the day of the year after the which the ecological deficit starts, that is when we are producing non-renewable resources.
- Carbon neutrality: the produced emissions are compensated a-posteriori in some way (e.g. by paying someone to plant threes); not the best way, doesn't eliminate directly the pollution.
- Net-zero emission: the emissions are compensated directly inside the production and supply chain; better way, the final outcome of the industrial process itself is zero-emissions.
- Green-washing: practice of producing environmentally responsible image without any real and positively impactful action.

Some global consequences of non-sustainable behaviours are:

- Climate change: scientific community agrees on human impact on it. It causes an increase of temperatures especially at the poles.
- Ozone layer depletion: fortunately solving thanks to the new rules. It's a barrier for radiation, thus again related to temperatures.
- Consumption on non-renewable resources.
- Biodiversity loss

While local consequences are:

- Deforestation: due to the need of more space to build
- Acidification: increase of $CO_2$ underground
- Eutrophication: increase of growth of algees

A strong impact on climate change comes from Greenhouse Gases (GHGs), which are those substances that when released in the atmosphere make the earth surface temperature increase, by preventing the sun radiations to leave the earth after having been reflected by its surface.
The main contribution to GHGs is $CO_2$, which, being the most present in the atmosphere, is set as a reference for the normalization in the Global Warming Potential. $GWP_{CO_2} = 1$, while for instance $GWP_{CH_4} = 11$ meaning that the effect of 1 $kg$ of $CH_4$ is the same as 11 $kg$ of $CO_2$: the impact of $CO_2$ is mainly driver by the amount released in the air.

To better understand companies' environmental impact, the lecture introduces the concept of Scope 1, 2, and 3 emissions (GHG Protocol):

1. Scope 1: direct emissions. They are generated directly by the company activities;
2. Scope 2: indirect emission from energy sources. When a company buys electricity, a part of the emissions caused by its production is a company's responsibility.
3. Scope 3: indirect emissions. They are the most challenging to control and they include all the indirect emissions produced by the supply chain, product use by customers and logistics.

As an example, in Marelli Motorsport, scope 3 emissions represent a very large share of the total environmental footprint. This means that it's crucial not only to work internally to reduce emissions, but also to actively engage suppliers and customers into improving the overall environmental performance of the entire chain (aim of the Life Cycle Assessment).

The, the ESG standard has been introduced: it is a criteria used to assess how attentive a company is to Environmental, Social and Governance (risk management, ethical business, integrity and so on) issues.
The ESG is similar to the three aforementioned pillars, but with a more practical and less ideological approach. It was explained that respecting these standards is no longer just a matter of ethics, but also a strategic business advantage: companies that show strong sustainability practices are nowadays better positioned on the market.

## 13.2 Sustainability in motorsport

Always in motorsport, particular attention is given to the environmental impact: The FIA Environmental Accreditation is the only motorsport-related certification.

One particular aspect involves motorsport. Despite it is often perceived negatively in terms of environmental impact, it is the main source of innovation in the automotive field and it serves as a fundamental laboratory and test bench to develop sustainable technologies that are later applied to everyday vehicles. In motorsport, cutting-edge solutions such as ultra-lightweight materials, more efficient systems, high-performance electric motors, aerodynamics and energy recovery technologies have been developed, tested and then transported to the automotive industry. Note that an industrialization process always follows the innovation introduced in motorsport, in which the parts are not design to last long, as it is required in everyday applications.

Another interesting consideration, against the typical misperception, was made considering the F1 world as an example: an F1 car burned fuel corresponds to less than 1 % of the total impact of the F1 world; instead, the most comes from logistics (49 %) and travelling (29 %).
Also note that often technical requirements are so strict that it's difficult to improve sustainability, which is extremely complex to balance with performance. In other situations instead, performance and substantiality match same targets, such as in the case of lightweight components development.

The FIA Environmental strategy tries to involve and guide the whole chain in their vision for sustainability. Some ideas to reduce $CO_2$ emissions are (in order of priority):

1. Innovation: the concept of inventing new and highly efficient technologies.
2. Social impact: motorsport has a large influence on people's perception; a strong marketing campaign is important to encourage people to act for a sustainable lifestyle.
3. Best practices: the strategies to (for instance) reduce consumptions and wastes.
4. Quantitative reduction of emissions: obtained by practical actions and by adopting efficient technologies.

### 13.2.1 Case studies

Moving to the practical examples, Marelli presented us some case studies. The whole process does not just consists of creating greener products, but also of rethinking the entire way of working, from adopting more sustainable materials to choosing suppliers who meet strict sustainability standards, and reorganizing the production processes to consume fewer resources and reduce pollution.

**Carbon Fibre Chassis** Life Cycle:

1. Material sourcing: energy-intensive production;
2. Manufacturing: high energy consumption and waste;
3. Usage: lightweight, good also for emissions;
4. End of life: difficult to recycle because it is a composite material and also because when recycled it does not recovery the same high mechanical properties as the original product.

Examples of actions to improve sustainability:

1. New materials: for some parts (where less resistance is required), natural fibres can be used instead of traditional ones.
2. Use different technologies to optimise the process (such as cuttings to reduce wastes).
3. Design the product for easier disassembly and recycling.

**VCU** The case is made from Aluminium, sometimes Magnesium.

Life Cycle:

1. Material sourcing: mining of precious materials;
2. Manufacturing: complex processes, high energy consumption;
3. Usage: optimized race strategies to reduce fuel consumption;
4. End of life: recycle of electronic parts and rare earths is critical (composite materials).

Examples of actions to improve sustainability:

1. Reduce material removal processes when possible, e.g. by using additive manufacturing, stamping or moulding, to reduce scraps.
2. Try to work on retrieving precious materials at end of life of boards.

**Case study: MGU** Life Cycle

1. Material sourcing: permanent magnets are one of the most impactful parts;
2. Manufacturing: high precision processes, high energy consumption;
3. Usage: energy recovery, very high power density and efficiency;
4. End of life: the parts are difficult to separate and recycle, windings and magnets in particular.

Examples of actions to improve sustainability:

1. Try to work on recycling and materials retrieving.
2. Try to work on recycling and materials retrieving.
3. Design for longevity

In conclusion, the Marelli Motorsport's lecture made it clear how the industrial sustainability is not an easy task, but it is now an unavoidable path to be followed. It's not just about protecting the environment but also building stronger companies, capable of innovating, adapting to changes, being competitive and work in an ethical way, so to have the possibility to succeed in the long term.

# 14 Sem.06 - Data Anlaysis in Motorsport

## 14.1 Introduction

The seminar held on the $12^{th}$ of May 2025 by Marelli Motorsport dealt with the topic of "Data Analysis in Motorsport".

Data analysis has a great importance in Motorsport for many fields, such as performance, modelling and development of the car, entertainment, driver training and so on.

The data recorded during a session are used for many different reasons by many different actors, like engineers, drivers, $R$ & $D$, suppliers, sanctioning body, race director, timing, media and sponsor. One of the branches in which data analysis is particularly exploited is safety: as an example, biometric sensors record the bio-activity of the drivers while racing, and, in case of accident, the related data are instantaneously transmitted to the medical car via Bluetooth, in this way medics can get a real time information on the severity of the situation, which can be crucial for a correct treatment, and potentially could save a life.

Two common examples of data acquiring/coptrol systems on race cars are:

1. **High Speed Camera (HSC):** a very light and small camera to record whatever happens during a session, and in case of a crash it is useful for the medical crew to see the movements of the driver's body (Fig. 14.1).
2. **Surveillance Data Recorder (SDC):** a black box, positioned under the driver seat and capable of surviving up to $1000\,g$ accelerations and high temperatures, to save and protect every acquired data, to later study them to improve safety and security in case of accidents.
3. **Engine Control Unit (ECU):** initially used only to control basic engine functions (such as ignition), later evolved to incorporate many functions. Nowadays it is called Vehicle Control Unit (VCU). Some of the VCU's tasks are: injection, ignition, RPM limiter, active suspensions, pressure sensors and steering wheel controls.

Figure 14.1: HSC-120 camera

Initially, in early years, data were all analogic and it was mainly a driver task to feel the car behaviour and transmit his feeling to the engineers when returning to the pits, who had to translate them into interpretable data. In more recent times, new technologies, such as digital acquisition before, and networks then, changed the whole approach: nowadays everything is instantaneously sent to the factory and back to the track thanks to internet connection, so the engineers could receive live numbers and were able to analyse them more easily, unburdening the driver's job.

## 14.2 Telemetry history

One first remark to be done is the difference between telemetry and data acquisition: the former consists of transferring the data from the car to the box and reading them in real time; the latter comes from the logger, inside the car, from which, after the session, data are exported and then post-processed, thus not in real time.

Telemetry is extremely important: in F1 it is even mandatory; a driver is not allowed to race if its telemetry doesn't work, for safety reasons.
Its working principle is: some sensors measure the related quantities, the data logger collects and formats data, which are then transferred via radio frequency and finally received by the team and displayed for real time analysis.
The on-track hardware to receive/transmit data, is composed by antennas (omni or multi directional), which receive the signals and transmit them to the access points through cables, to be than elaborated by switches and filters. Of great importance at the beginning of a race weekend is always to verify the track layout, check the signal coverage and the updating frequency, to avoid issues deriving from (for instance) interference, bad weather and broken fibres.

The main data types are three:

1. Telemetry
2. Voice (team radios)
3. Video (on board cameras)

Which can be affect by problems like packet loss, latency and bandwidth saturation.

## 14.3 Data analysis

To perform a complete and effective data analysis, Marelli developed a specific software, namely *Wintax*, which is able to comfortably elaborate, confront and visualize the data, through plenty of features for the post-processing phase.

In order of importance, the teams' priorities must always be:

1. Safety
2. Reliability
3. Performance

In the following, some examples of the normally studied telemetry are reported.

**Data in turns** The typical driver inputs order depend on the turn type:

- Low-speed corners: braking + steering up to apex and then throttle to induce rotation;
- Medium-speed corners: braking up to turn entry, pure steering up to turn exit and then throttle;
- High-speed corners: braking up to turn entry and then steering + throttle.

Some different approaches can be followed, depending on the driver level:

- Non-expert: two phases corner analysis (braking and acceleration);
- Expert: three phases corner analysis (braking, pure cornering and acceleration);
- Extremely expert: five phases corner analysis (pure maximum braking, braking + steering, pure steering, steering + throttle, pure maximum acceleration).

The reason for this differences lays in the more precise feedback that the expert drivers are able to give to the team.

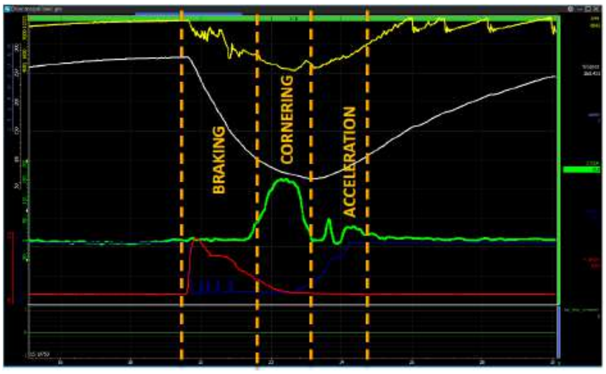



Figure 14.2: Three phases corner analysis

**Trajectory analysis** Trajectory analysis and comparisons are performed through the position data, thanks to the GPS technology used to map the car in the circuit.

**Driver performance comparison** Some of them are: laptime comparison (over the whole track length), through the typical dataset composed by speed, RPM, gear, steering angle, brake pressure, throttle and whatever necessary for the analysis. The analysis can be performed sector by sector, corner by corner (in and out), meter by meter, to understand where exactly the driver is loosing or gaining time compared to another one.

**Under/oversteering** The car behaviour can be analysed in general and/or in terms of particular issues.

**Suspensions** Suspension elongations can be acquired, as well as damping coefficients in all the ways separately (such as compression/extension both in high/low frequency). Data analysis is important also for useful and cost effective diagnosis: for instance, in the damping vs speed diagram, if the damping distribution is narrower than normal (large forces at low speeds), the damper is stuck; if the distribution is flatter than normal (about constant damping independently of the speed), an oil leakage can be the cause. Fourier analysis can be applied for natural frequency analysis, which can be exploited to understand if something is not working, by observing if all the peaks are positioned at the expected frequency.

**Differentials and gearbox** The speed of each wheel can be measured independently, to derive the amount of differential opening and check if everything is working properly for what concerns the torque vectoring technique and to correctly set the differentials pre-load.
The gearbox analysis is instead performed by mapping the gears usage and engine speed in lap, comparing different settings.

**Brakes** Their temperature is an important parameter to be measured, especially for what concerns the difference between the axles: a too large discrepancy can be detrimental for the car performance; also, front and rear pressures are analysed, to catch potential leakages or issues by observing deviations from standard operating conditions.

**Aerodynamics** Strain gauges are positioned on the pushrods, to measure the loads and derive aero-coefficients, which, together with the laser measured ride heights, compose the aero-maps. Comparisons between calculated and measured data is then performed to see if correlation is verified, if a downforce loss is happening and/or if something needs to be changed in the aerodynamic setup.
In top level competitions only, like in F1, more precise tools are available: pressure sensors under the wings, the floor and other crucial parts, are adopted to map the generated downforce everywhere on the car instantaneously, giving extremely precisely located information for instance about the laptime loss due to damage, important to make fast strategic decisions about continuing to race, making a pit-stop to replace the broken component, or retiring the car.

Figure 14.3: Example of pressure sensor positioning

# 15 Sem.07 - Ducati Motorsport

## 15.1 Introduction

The seminar by Ducati Motorsport was focused on the differences of the motorcycle world with respect to the cars-related Motorsport.
The first main difference regards the stability of the system: unlike four-wheels vehicles, which are stable up to a certain point, bikes are naturally unstable systems, in which the roll equilibrium is the most important one to be ensured, with extreme roll angle values. The intrinsic dynamics are also totally different: the COG continuously moves due to the driver movements, camber changes a lot, and so on.

Ducati Motorsport spends its main efforts in the MotoGP category. For the best results, both simulations and experiments need to be developed correctly and many efforts are applied to both.

## 15.2 Simulations

Simulation are typically solved with multi-body systems, with both rigid and flexible bodies, connected through joints. The EOMs are thus found (ODE) and solved with Newton-Euler or Lagrange approach.

Many aspect are taken into account when performing simulations, such as:

- **Engine**: different engine power maps and their impacts on lap time.
- **Vehicle**: COG position optimization with respect to laptime.
- **Aerodynamics**: lift optimization and drag reduction.
- **Electronics**: tuning and control strategies
- **Track**: rider commands optimization

For a model to be reliable, the following characteristics are required:

- Accurate CAD models.
- Accurate mass distribution, both through overall COG position (also measured for validation) and single components COG and inertia tensor.
- Characterization of purchased components (such as springs and dampers).
- Aerodynamic field of forces around the bike, obtained both by wind tunnel and CFD simulations. In the tunnel the straight position analysis is relatively easy, while it is more difficult with leaning rider, which goes out of the bike fairings, with consequent area increasing and total changes in aerodynamic characteristics.
- Tyre characterization, from a mix of tests, indoor measurements and on track measurements.

The most challenging aspect of the simulations is the driver capability of reproducing the ideal manoeuvres: as already stated, the motorcycle is an extremely unstable vehicle, the rider is required to move a lot and so the dynamics continuously and completely change; which makes the laps to be difficult both to simulate and reproduce in real life. Many resources are thrown into developing a realistic simulation driver, often done by feeding speeds profiles to machine learning algorithms.

Another important aspect are the controlling techniques. One of the main is the ECU (Engine Control Unit), which performs tasks like anti-wheelie control, to optimize the straight line acceleration; another important one is the Traction Control, useful to optimize acceleration at corner exit, more difficult task with respect to anti-wheelie, having the former more inputs and variable than the latter, which instead relies manly on 2D dynamics.
To correctly set the control parameters a numerical approach is followed: many simulations are computed by changing the PID parameters, to finally find the best compromise.

A lot of dynamics, which ask for accurate simulations, also lay in the final chain transmission: both reliability and performance are strongly accounted for, to correctly manage typical effects like the teeth jumps that can happen when the transmission centre distance reduces due to the suspension stroke variation.

Clearly, an important simulation to be correctly performed is the laptime one, which is computed by feeding the solver with the race line (possibly from GPS) and the vehicle model. After the simulation, the post processing phase comes into play: if models are accurate, good reliability from simulations is generally achieved, which is fundamental to give trustable answers on what and how actually changes by modifying some parameters.

Note that the simulations are quasi-static: components like dampers cannot be optimized because they are strongly related to dynamic effects. Everything what is characterized by oscillation frequencies above 1 $Hz$ is filtered out, such as suspension small movements, bump absorption, and similar.
The quasi-static approach is then mainly adopted to decide where to concentrate the team efforts and expenses to achieve the best possible results.

The strong difference in laptime which is experienced between MotoGP and F1 mainly comes from the smaller vehicle surface: the motorcycles has a much smaller drag coefficient, but have at the same time less available surfaces to produce downforce, while in cars the unavoidable large surfaces are exploited as well as possible to compensate the high drag with extreme downforce. Wings in bikes produce some downforce mainly to avoid wheelie, but its order of magnitude is much smaller.


Figure 15.1: Ducati MotoGP

## 15.3 Experimental approach

After simulations, the real life comes into play. During a track session, you measure everything you can, trying to estimate as many quantities as possible. Is not matter of measure, but mostly to extract relevant quantities from the acquired measurements.

There are two kinds of measurements, namely the ones that are performed during every session and the ones which the team acquires in a limited number of sessions during the year and in pre-/post-season tests. The reason behind this is that those measurements require heavy, expensive or delicate instruments.

The sensors that are always used are:

- IMU (Inertial Measurement Unit), sometimes two to be sure
- Wheels angular speed sensors
- Brake pressure sensors
- Steering angle sensor (positioned inside the steering damper)
- Engine sensors (like for throttle, gear, revs, torque, and so on)

The sensors occasionally used are instead:

- Speed and distance optical sensors (lasers for real speed and ride height)
- Strain gauges for forces and moments (instrumented wheels)
- Particular accelerometers
- Position sensors like GPS
- Special video cameras

After measurements, the kinematic quantities are reversely derived from matrices: you go back from results to causes through the vehicle dynamics (e.g. from speeds and loads to characteristics).

Some examples of quantities derived from measurements are:

- **Roll angle**: from numerical time integration of IMU data; particular attention must be paid to the noise produced by the IMU, which when integrated can generate large cumulated errors (filters are used to mitigate this effect).
- **Pitch angle**: derived from all the other quantities; the resulting pitch angle is the one which correctly closes the kinematic chain.
- **Longitudinal/lateral accelerations**: measured by the accelerometers; due to the large vehicle angles (roll and pitch in particular), to be properly projected in the direction of interest.

The typical workflow is: occasional (very accurate) measurements are exploited to directly derive the quantities of interest, which are stored as reference values; this quantities are then used for comparison with the estimated values (less precise), which comes from the always acquired measurements, to check their accuracy and reliability. Accurate sensors are thus used in rare occasions to generate models, which are then used to try to find ways to exploit the standard sensors to estimate the same quantities in all the sessions.

Some quantities are thus derived from the acquisitions. One of them is the vehicle handling, which is again much different from the cars' one. In this case, it is not simply related to how quickly the vehicle rotates (yaw rate), but also to the roll rate.
Some differences are also experienced in the G-G diagram, which is not limited by the friction only, but also by the flipping and the wheeling phenomena. The diagram shape is less rounded than the cars' one, with a heart-like shape: here the aerodynamic drag works against the traction by enhancing the wheeling effect and in favour of the braking phase, by reducing the flipping phenomena. Wings produce downforce mainly at the front, to reduce wheeling.


Figure 15.2: Ducati MotoGP wheelie

A difficult quantity to be accurately estimated is the actual speed: GPS is not accurate, during wheelie the front tyre rotates freely and tends to slow down, the rear tyre is the traction one, thus experience slips up to 15 % and you don't know the exact amount. The final value is obtained by cross-information, from the comparison of all the different available estimates, like front wheel speed and acceleration integration.
Also the longitudinal slip and slip angles are important quantities to be derived, whose estimation techniques are tuned mostly from optical sensors dataset.
Always remember that motorcycles are less neutral vehicles than road cars: strong oversteer is always present both in corners and in strong braking, with slip angles difference between front and rear up to quantities in the order of 10°. Typically, good performance is reached by letting the rear tyre slip a lot: riders like that, but attention must be paid in not exceeding with to much slip, which is detrimental for tyre wear and long run pace.

As already stated, the driver position affects a lot the bike dynamics. Cameras acquisitions and machine learning techniques are used to record the driver movements. In particular, a segmentation model of the rider body parts (head, trunk, legs, harms, ...) is trained to make it capable of tracing the position of each. Efforts are then spent trying to correlate driver position with indoor COG measurements. No physical models are used for this task, but numerical models and machine learning only.

# 16 Sem.08 - Audi Motorsport

## 16.1 Introduction

The seminar held by Audi Motorsport gave us lots of food for thought regarding the whole Motorsport world, with some particular insights on the off-road categories, such as rally raid; stressing the need of designing racing cars in the simplest way possible, to achieve results with limited time and resources.

As a reference, the Audi RS Q e-tron, which took part in the *W2RC* championship from 2022 to 2024, was presented and some characteristics were highlighted. For instance:

- **Propulsion system:** a full hybrid in series, consisting in an ICE recharging a battery that subsequently feed an electric motor which drives the wheels.
- **Battery:** High voltage line run at $800V$ to power the wheels, with some consequent safety-related issues, for both drivers and mechanics, to be considered.
- **Regenerative braking:** to be set depending on the situation. For example: in the morning at the beginning of a stage, the tyres are cold, therefore low regeneration can be set to use more the disk brakes which help in heating up the rims and so the tyres from the inside; when instead the available energy is critical as in a very long stage, higher regeneration can be chosen to improve the mileage of the car. It was also possible to use fully regenerative braking, to recover as much energy as possible, unlike in production vehicles, where, due to safety concerns, some limits on the maximum regenerative power must be respected and the disk braking is used in heavy decelerations.
- **Differentials:** one physical limited slip differential per axle and a third virtual in the centre, which sets the front/rear torque repartition mainly depending on speed and on weight distribution.
- **MGUs:** three in total, one per axle plus one for current generation by the Internal Combustion Engine to recharge the battery; the three machines are exactly the same, mainly to reduce the spare parts to ease logistics and assembly (the same reasoning is done for the upper wishbones, which are equal left to right, front to rear).
- Some technologies are shared by other categories: the MGU is the same adopted in Formula E, the ICE is the same adopted in DTM; this mainly for design and production costs efficiency.

Figure 16.1: Audi RS Q e-tron at the Dakar

Then, the focus was moved on the key aspects of the car, especially those ones which resulted to be critical in the 2023 Rally Dakar Edition and that have been improved for the 2024 one; namely, in order of importance:

1. **Safety:** related to driver, co-driver and the whole team. Unfortunately, many accidents occurred in 2023, so in 2024 many improvements were developed, based on those accidents. A new crash box was added to mitigate the risk and the consequences of pitching over accidents. At the Dakar, cars take on very high jumps and experience consequent very hard landing, with high vertical accelerations ($\approx 30g$ in more serious accidents). Therefore, some technologies also used in helicopters, were applied to the cars, such as seat foam, to reduce possible damages to the driver and co-driver spine.

2. **Reliability**: one of main problems in 2023 was VCU (Vehicle Control Unit) load, which caused the software to crash many times; the VCU load was reduced from about 95 % to about 90 %, to improve reliability.

3. **Comfort**: unlike other Motorsport categories, in Rally Raid also comfort has great importance; in those very long stages cars continuously jump and vibrate, so at the end of the day driver and co-driver can experience fatigue and exhaustion, and also develop some more serious health problems, such as the micro-traumas deriving from hand shakes, that can represent a problem also in the future. Visibility also plays a more central role than in other categories: studies are carried out to avoid dirt sticking to the windscreen.

4. **Performance**: as for Motorsport in general, performance is always a key aspect; Audi for the 2024 season focused on weight reduction (which is always a great source of performance increase), but also power and traction performance have been improved in 2024, with an improved engine and differential.

5. **Assembly time**: the team has everyday some hard work to be accomplished: early wake up, with many steps on schedule before the race, in the warm-up phase; then, during the race itself, the whole team moves to the following bivouac, waiting the car for evening check-ups and possible repairs to be done. A fast and simple disassembly-reassembly procedure is a key aspect for a proper team work.

Figure 16.2: Result of a pitching-over crash during Dakar 2023

## 16.2 Think simple

A strong remark was done, during the whole meeting, on the importance of thinking simple: when the things becomes complicated, nothing is better than a simple thinking mentality to solve them. Some key examples were presented in that sense.

- **Suspension design**: always design first the rear suspension and then the front one; only a strong rear suspension allows for a strong front design. For instance, when analysing the steering induced by lateral acceleration, the induced steering angle at the rear depends only by the lateral acceleration, while the induced steering at the front depends both on the lateral acceleration and on the induced steering on the rear itself. In an oversteering scenario, the axle to lose grip first is the rear one, and this will be felt directly on the steering wheel. When the rear tyres do not longer generate lateral force, they do not have an induced steering angle, which results in a drop in the steering wheel torque necessary to turn the front wheels.
- **Cornering stiffness**: it is of main importance on soft soils; many factor contribute in the overall stiffness, such as by acting on bump steer, toe variation with lateral force, self-aligning moment and the suspension's vertical travel. Supposing to have a target in cornering stiffness, it can be stated that about the 60 % of it comes from the tyres, and the remaining 40 % from the suspension system: toe-in variation enhances stability, which allows for a stronger rear, but in turns it also increases the self-aligning moment, which always goes against cornering stiffness as it induces toe-out; the bump steer is always to be minimized, especially in off-road where cars bump a lot; the steering lever arms on the wheel is to be maximized to ease the wheel rotation, but its position strongly depends on the presence of other elements in that region.
- **Aero-balance**: if the weight distribution loads more the rear axle (for instance), this will cause a less stable rear axle and a more stable front one; for this reason, the aerodynamic balance will be generally moved towards the rear, to increase the stability margin more on the weaker axle, especially at high speeds, where control is crucial.
- **Differentials**: they are an important element to be properly set, in terms of opening or closure of the Limited Slip Differential. Two contributions must be considered when analysing the speed difference between inner and outer wheel: the kinematic one coming from the shorter radius of the inner wheel (lower rotational speed) and the elastic one coming from the lower vertical load on the same inner wheel (lower grip, larger slip so larger rotational speed). The sum of these contributions generate an initial phase (at lower lateral accelerations) where the outer wheel rotates faster, and a second phase where the increased acceleration generates a sufficient load transfer to make the inner wheel rotate faster. The first is called stability phase (the differential will slow down the outer wheel, creating a torque vectoring moment against the car yaw rate), the second is called agility phase (where the differential will instead slow down the inner wheel, creating a torque vectoring moment enhancing the car rotation). The switching point, in terms of lateral accelerations, where the behaviour changes from stability to agility is defined as the point where the difference in rotational speed between the outer and inner wheel $\omega_2(a_y) - \omega_1(a_y)$ changes sign. Longitudinal acceleration makes the switching point to move towards lower $a_y$ values (more agility), the lifting-off causes the opposite (more stability). As a general rule of thumb, the differential is set to provide more locking while braking, to gain stability; during the turn an open differential is preferred, creating rotation to turn more easily; then at the corner exit, a semi-locked differential is preferred, to generate traction, while avoiding instability such as oversteering.

- **Roll centre**: it always moves during roll motion, how it moves depends on the suspension's kinematics; two peculiar cases were analysed:
  - If it just travels vertically, ending up precisely on the ground, (but still in the middle of the axis), the outer wheel will travel in bound, while the inner wheel will go in rebound, but the distance travelled will be exactly the same, therefore the axis will basically not move vertically, producing a null heave motion on that axle when turning.
  - If it goes near (or exactly below, in the extreme case) the outer wheel while turning, the outer wheel will not travel vertically, while the inner wheel will instead travel double the distance in bound (compared to the previous case). This will instead result in the body rising around that axle, creating unwanted motions that can hinder the vehicle's performance and drivability. At the same time also the bump steering, due to kinematics characteristics are affected: since the suspension does not compress nor extend, things like induced toe-in in compression will not be actively felt by the driver.

  Moreover, the roll centre height is important: rising its location means having a wheel speed largely pointing outside in bump, creating a lateral force which, in rough terrains, can cause excessive oscillations at the axle especially when the roll natural frequency is excited (typically about $2-2.5\ Hz$).

## 16.3 Anecdotes

Finally, some real-life experience tips were shared with us.

- In the control systems strategy, drivers generally prefer feedforward controls with respect to feedback ones, because the former always produce the same car behaviour, while the latter is more unpredictable. A clear example can be the traction control.
- The proper correlation must always be found between simulations, models, measurements and subjective evaluation: the physics are always the same, thus if some mismatch is shown, it means that for sure something went wrong in somewhere.
- In off-road, angles such as toe and camber as less relevant for performance, much more relevant are instead the vertical dynamics, vertical loads and their transfer, as statistically the car is $\approx 80\%$ of the time running on a rough straight line.

# 17 Sem.09 - Mechatronics in racing - Marelli Motorsport

## 17.1 Introduction

Especially in the modern era, electronics and mechatronics have gained great importance also in Motorsport: for this reason, Marelli Motorsport dedicated a whole seminar on the topic.

Many components can be classified as electronic or mechatronic parts, such as inverters, DC-DC converters, mini cameras and electric motors. In general, the main problems in Motorsport devices are temperatures, vibrations and water/dust.
Typically, an electronic component (ECU) is composed by a series of PCBs (Printed Circuit Boards) which carry out different jobs, an aluminium box, obtained by machining (possibily with cooling system) to contain and protect the PCBs, covers and connectors to transfer data, typically using cables.

**PCBs:** in Marelli they are developed by both electronics and mechanical departments. This joint team is organized to develop electronic components, but at the same time giving strength and stiffness to the structure, tackling also cooling problems and other mechanical issues. They can be made from insulated metal, which is good for thermal energy evacuation, adopted for instance in DC-DC converters, or from FR4 (Flame Resistant glass fibre). Usually the PCB have a thickness between $1.6mm$ and $4mm$. All the required components can be soldered on the board (SMT, Surface Mounted Technology) or inserted in holes (THT, Through Hole Technology), to form the final PCBA (Printed Circuit Board Assembly).

**Connectors:** three types exist, fulfilling different roles

- Board-To-Board connectors, called "Tiger eye", which must always ensure the correct link also when undergoing strong vibrations.
- External connectors, such as "Deutsch/Souriau" connectors, , usually derived from military industry, which remain operational even under extreme temperatures and vibrations and with very high currents.
- Power connectors, namely "Amphenol" connectors, able to withstand up to $200A$ of current.

Figure 17.1: ECU produced by Marelli, used in MotoGP

## 17.2 Temperature management

The main limit in terms of temperatures is the upper boundary set by the junction (i.e. the maximum temperature of the semiconductor material).

The temperature can be managed in two main ways: by conduction, via TIM (Thermal Interface Material), which can be thermal grease or thermal pads, or by convection via cooling fluid, typically water or glycol. Thermal grease is suitable for small layers only (few tenths of millimeters) and has the disadvantage of moving with vibrations, creating voids. Thermal pads are instead more similar to a foam: they are silicon based, can have larger thickness (up to 8 mm) and have the advantage of remaining perfectly still with vibrations, offering also a bit of damping. Also graphite-based TIMs exist, but are less used due to electrical conductivity. Moreover, the flexibility of the TIM needs to be checked: too hard materials can produce stress concentrations and create cracks on the board.

For thermal computations (both by-hand and with software), the electrical analogy, together with Fourier's first law are used: thermal powers are calculated through potentials and all the resistances in the contact faces and of the materials, up to the case. CFD analysis are also carried out to study active cooling via fluid, if needed.

Thermal tests are also performed, such as using thermal chambers (similar to ovens) to make tests on the life of the components. Some examples are: Storage and Shipping Temperature Exposure, in which the device is kept switched-off at a temperature of $-40\ °C$ for 24 $h$; High Temperature Operating Endurance, to check the device life through an accelerated test; Power Thermal Cycle Endurance, which is similar to a fatigue life test but with a thermal cycles accelerated test; Thermal Shock Test, to test the device resistance to rapid temperatures changes.

## 17.3 Vibrations management

Vibrations are a major concern for mechatronics, being responsible of fatigue failures, wear, tear and breakage of components. Shapes and materials can be optimized to reduce its impact; vibration analyses are made, such as natural frequencies studies, and vibration tests too.

Different classes of vibration can be identified depending on the specific Motorsport category and position inside the car. Also the instrumentation installed on the ground (such as antennas) is subjected to vibrations, thus requires studies and tests, too.
The two main types of vibrations are random vibrations and shock vibrations, the former coming mainly from the road irregularity, with frequencies in the range $10-2000\ Hz$ and RMS amplitudes of $6-19\ g$, the latter coming from kerbs and other obstacles.

## 17.4 Water and dust

An Index of Protection can be defined (IP classification) in terms of waterproof and dustproof properties, the first number is the level protection against dust, while the second against water.
As an example, in IP67 (the most common one) number 6 is standard dust protection, number 7 is against water, tested with a temporal immersion. In Motorsport, typical classifications are 65, 66 and 67.

To reach a certain IP grade, some protections need to be adopted: sealing gaskets are used on covers and box mating parts or, if no gaskets can be exploited, silicon is used instead; another possibility is potting with dedicated silicone, like a one piece cover; in addition, resins and coatings (HUMISEAL) can be placed directly on the PCB for increased safety. Note that HIMISEAL reduces the thermal conductivity, thus is avoided on the most critical parts.

A problem raising from the sealing usage, is that it prevents air to pass through it, that instead needs to be exchanged with external environment to balance the pressure increase due to temperature rising: high pressure can cause water condensation which is serious problem when working with electrical circuits; valves are thus adopted to allow the air exchange.

A final remark is made on the future challenges: power is increasing year by year and consequently conductivity must be increased and new materials, new solution, new cooling methods and so on must be intensely studied.

## 17.5 Mechanical design

The mechanical analysis mainly focused on the MGU (Motor Generator Unit), which can be subdivided in two types: -K (Kinetic) which recovers kinetic energy through a regenerative braking system, and -H (heat) which exploits the excess of energy coming from the turbine to charge a battery to then give a boost to the compressor to reduce the so called "turbo-lag" effect.

Just as a reference, the F1 MGU-K in the years were developed following the rules with the following powers: 60 $kW$ (KERS), 90 $kW$, 120 $kW$, 350 $kW$ (2026).
Also in FE the motor unit has been developed (at beginning only by McLaren with fixed design, then extended to free design): nowadays 2 units are used, one at front, with fixed design and used only as regenerative braking, and one at rear with free design to both power the rear wheels and regenerating; in the next generation of FE cars, the front motor will give positive power too.
Also in the later generation of WEC cars hybrid systems has been adopted.

Front electric powertrain is now under study for F1 season 2026, characterized by 350 kW, 300 Nm and 25000 rpm.

Typically three-phase brushless motors are adopted.
The typical workflow stars by the customer giving some required profiles of speed and torque at the wheels, based on simulations, which need to be reproduced on multiple laps. From here, all the other characteristics must be managed, such as maximum voltage, temperatures, cooling media, inverter maximum current and switching frequency, with the final goal of minimize losses and mass. For that, many simulations are performed, using optimization algorithms, up to reaching a series of Pareto optimal solutions: many solutions are then given to the customer, which makes the final choice, depending on what is better to be minimized for him.

One very important aspect in electric motors is cooling: dielectric oil is typically used as cooling medium because it appears to be the best solution as stator cooling flow. In the cooling design, components such as oil fan, oil reservoir, heat exchanger and oil filter need to be managed and positioned in very little space. Three types of cooling can in general be adopted:

1. Jacket + Flange water cooling: water runs outside the stator in a kind of aluminium jacket and in some flanges which cross the top and bottom parts of the motor;
2. Jacket + Flange + In-slot water cooling: the same as point 1 with the addition of some water running in the slotted portion of the stator, which is difficult to be managed in terms of leakages;
3. Oil cooling: as already stated, the best solution, being able to reduce the temperature of many degrees (about 100 $°C$ in some simulations) with respect to water cooling.

**Stator:** windings are positioned by hand, because of the too high costs and complexity in automatizing such a process, for such a low production numbers, also with designs that are changed every year, thus making it impossible to set-up automatic machines at reasonable costs. Polymeric or carbon composite materials are adopted to isolate the rotor from stator cooling fluids.

**Rotor:** the two main adopted technologies are IPM (Interior Permanent Magnets) and SPM (Surface Permanent Magnets) machines, the former being characterized by in-slots magnets, with the advantage of not requiring retention structures to keep the magnets in position but with the difficulty of properly designing the ferromagnetic portion to reach a trade-off between mechanical and magnetic requirements, the latter requiring structures to keep the magnets in contact with the rotor surface at high rotational speed (glue only is not sufficient, typically carbon fibre is used, directly wound on the surface or pre-produced and then press-fitted).
Finally, the rotor needs to be balanced, by removing material with holes creation instead of adding mass due to lack of space.


Figure 17.2: MGU produced by Marelli for F1